ICAN-6247

Application of the CA3126Q Chroma-Processing IC Using Sample-and-Hold Circuit Techniques

This Note* describes the CA3126Q monolithic integrated circuit intended for use in processing the chrominance signal in a color television receiver. In performing the functions of color subcarrier regeneration and chroma control, emphasis has been placed on utilizing all the information available in the signal so as to approach ultimate system performance capability, while at the same time substantially reducing the number of external components and adjustments.

As contrasted with prior state-of-the-art IC designs, sample-and-hold techniques are used in the phase detectors for the AFPC and the ACC-killer loops of the CA3126Q. The improved signal-to-dc unbalance attained thereby makes it possible to eliminate the adjustments conventionally used in those circuits. The only set-up adjustment is a trimmer capacitor to tune the crystal filter.

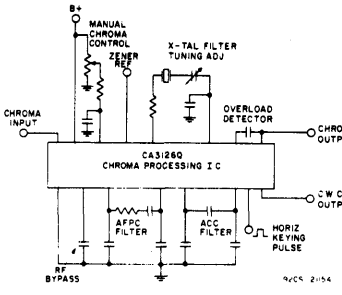


Fig. 1– Components external to the CA3126Q.

Two controls serve to adjust the chroma level: One control is automatic and functions as a supplementary ACC loop to prevent oversaturation under condition of improper transmitted burst-to-chrominance ratio, and under noisy-signal conditions. The second chroma level control is for manual adjustment of the saturation by the viewer. This control has a linear characteristic over the range of the chroma gain control.

The CA3126Q integrated circuit which performs these functions is housed in a 16-terminal package. The external components, shown in Fig. 1, are relatively few and consist of integration capacitors for the servo loops and the filter network in the VCO. Table I summarizes the performance of the circuit.

Table I – Performance Data Typical Values

Function	Nom. Supply V_{CC} = 11.2 V	Supply Var. V_{CC} ±2 V	Temp. Var. ΔT = 50°C
Oscillator Characteristic			
C. W. Carrier Ampl.	1 V_{pp}	±5%	+10%
Frequency	Nom. Sub Carrier	∓70 Hz	−70 Hz
AFPC Characteristic			
DC Loop Gain	40 Hz/deg.		
Pull-in Range	±500 Hz		
Phase Error		∓2°	−2°
Noise Bandwidth f_{NN}	100 Hz		
ACC and Killer Characteristic			
100% Input Level (Red Field)	0.25 V_{pp}		
Nominal Output with Overload Detector	0.5 V_{pp}	±2.5%	−5%
Nominal Output without Overload Detector	2.7 V_{pp}	±10%	−5%
ACC–3 dB Point	20% E_{IN}		
Killer Threshold	5% E_{IN}		
Diff. Phase Error Over Entire ACC Range	1°		
Manual Control Characteristic			
Chroma Output Linearly Proportional to Control Bias			
Diff. Phase Shift with Bias Var.	2°		

*This Note, revised by Wayne Austin (RCA Solid State Division) was originally prepared by L. A. Harwood (Consumer Electronics Division) for publication in the IEEE TRANSACTIONS ON BROADCAST AND TV RECEIVERS, May 1973, Vol. BTR-19, No. 2.

MAJOR FUNCTIONS

The signal flow and organization of the CA3126Q are shown in block form in Fig. 2. The composite chroma signal is applied to the first chroma amplifier. The output from this stage proceeds along three paths. The first path leads to the doubly-balanced wide-band AFPC detector. Here the burst signal is compared with the reference carrier to produce the required error signal for synchronization. Two sample-and-hold circuits serve to achieve high detection efficiency and bias stability. One sample-and-hold circuit samples the detected signal during the horizontal keying interval and stores the peak error signal in a filter capacitor. A second similar circuit provides an accurate reference potential as described later. The bias stability of this system is sufficient to eliminate the need for the adjustments required in conventional circuit design.

The detected and filtered burst signal controls the frequency and phase of a voltage-controlled oscillator (VCO) by operating on an electronic phase-shifter. The VCO consists of an amplifier-limiter followed by the electronic phase-shifter. A crystal filter located between the output of the phase-shifter and the input of the amplifier-limiter closes the loop of the VCO. The filtered oscillator signal is amplified to produce the required reference carriers for the AFPC and ACC synchronous detectors. The required quadrature relationship is obtained by $+\pi/4$ and $-\pi/4$ radian integrated phase-shift networks.

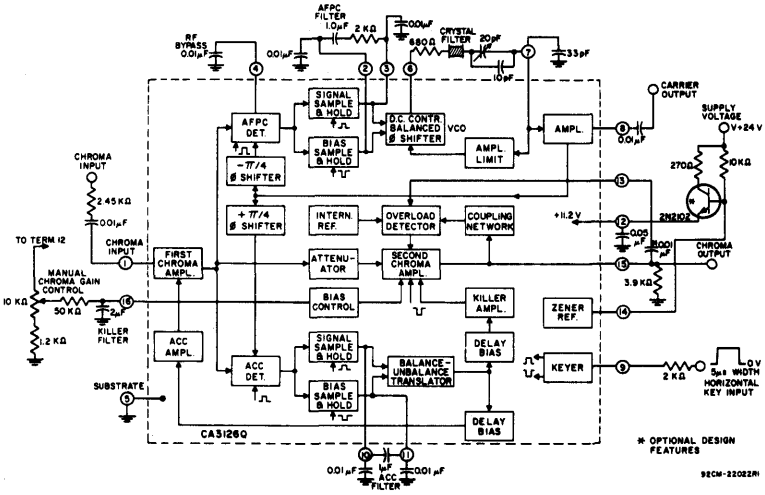


Fig. 2– Signal flow and organization of the CA3126Q.

The ACC-killer detector is similar in structure to the AFPC detector, and is also driven from the first chroma amplifier stage. It detects synchronously the in-phase component of the burst signal and produces a pulse signal proportional in amplitude to the level of the burst signal. The resulting control signal passes through a sampling circuit, as described above, and is applied to the killer and ACC amplifiers. The action of both amplifiers is delayed so that the unkill action takes place prior to ACC and the latter is fully activated upon reaching the predetermined burst level. The ACC amplifier controls the gain of the first chroma amplifier so as to maintain the burst signal constant while the killer amplifier enables the output stage in the presence of the burst signal.

The signal level to the second chroma amplifier is reduced to one fourth of the available signal level to allow for the extremes of the chroma signal excursions. A horizontal rate keyer operating on this stage removes the burst signal so that the output stage is activated only during the horizontal scanning interval. A saturation control, available for front panel control, allows a continuous gain adjustment of this amplifier. A desirable feature of this control is the linear correspondence between the control bias and the chroma output signal. The chroma maximum level corresponds to the maximum bias potential without a dead spot at the extreme of the control range. A threshold type overload detector monitors the output signal and maintains the output from the second chroma amplifier below an arbitrarily set level. This prevents the overload of the picture tube usually experienced on noisy or excessively large chroma signals. The required keying signals for the various functions are generated by two cascaded keyer stages where either polarity pulses are generated.

REGENERATION OF THE SUBCARRIER

The regeneration of the subcarrier is performed in the circuit shown in Fig. 3. This section consists of a synchronous phase detector, the sample-and-hold circuits, and a voltage-controlled oscillator. Several keying circuits serve to maintain the operation in proper time sequence.

The Phase Detector

The phase detector is formed by transistors Q_{51}, Q_{52} and Q_5 to Q_{10}. The composite chroma signal amplified by the first chroma amplifier is applied to transistors Q_7 and Q_8 and the reference carrier is applied to transistors Q_9 and Q_{52}. The product of the two signals is developed across the load resistor R_{13}. Transistors Q_5 and Q_6, triggered by a horizontal rate keyer circuit, operate on the phase detector so as to allow detection of the burst signal only. The current compensation of transistors Q_7 and Q_8 by the gating transistors Q_5 and Q_6 and the absence of filtering at the output of the detector results in transient-free switching of the phase detector. In the absence of chrominance, the potential across the load resistor R_{13} remains constant regardless of the keying. In the presence of the chrominance signal, the phase detector produces two time-spaced outputs: one during the horizontal scanning interval corresponding to the quiescent potential, the second during the horizontal keying interval representing the detected burst. Thus, the detected burst can be measured relative to the quiescent potential rather than to an arbitrary reference. This results in excellent stability for temperature and supply variations.

Sample-and-Hold Circuits

As previously stated, the sample-and-hold circuits shown in Fig. 3 allow efficient utilization of the detected error signal and provide a reliable reference potential. During the sampling interval, the detected pulse signal available at the detector load resistor R_{13} is translated to the AFPC filter capacitor of terminal 2 via transistors Q_{53} and Q_{54}. Q_{53} serves to isolate the

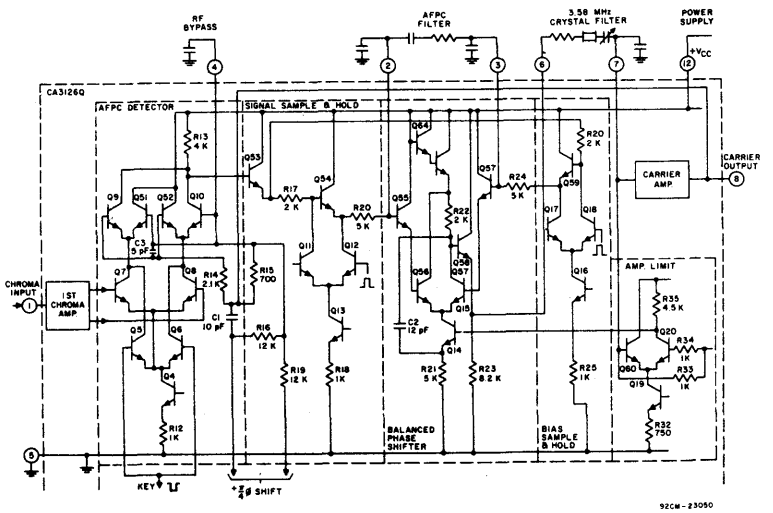


Fig. 3– Subcarrier regeneration circuit.

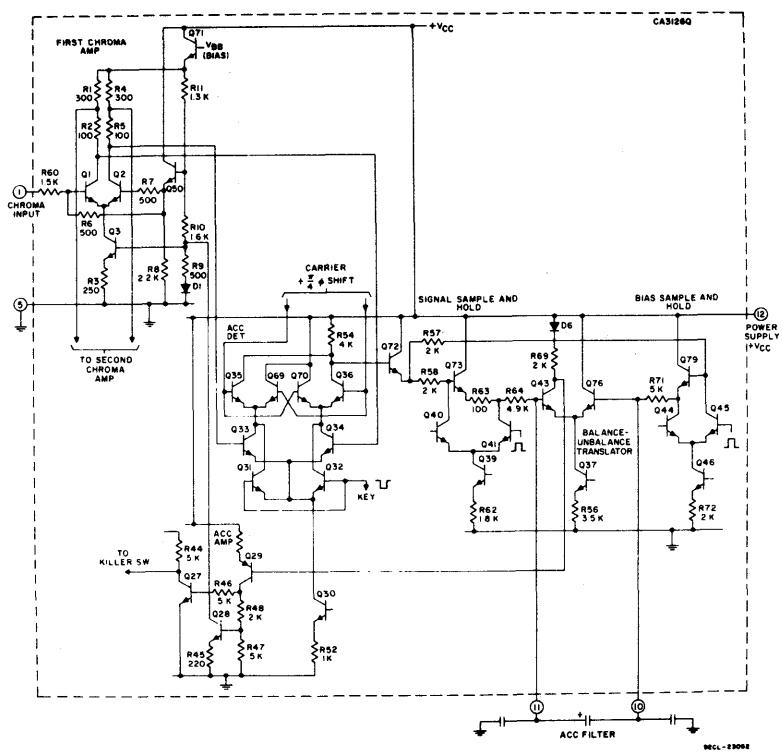


Fig. 4– The first chroma amplifier and the ACC servo loop.

detector from the switching pulses generated in the sampling circuits. The sample-and-hold action is accomplished by controlling the conduction current in transistor Q_{54} thus alternating the charge path during those intervals. During the sampling interval, transistor Q_{54} conducts and its emitter exhibits a relatively low impedance in comparison with the value of the integrated charging resistor R_{20}. The detected or sampled signal is stored in the AFPC filter capacitor which, with R_{20}, determines the time constant during this time interval. During the hold period, transistor Q_{54} is off and the filter time constant is several orders of magnitude larger than previously. The discharge of the filter capacitor is reduced to very small base bias currents only and little of the stored information is lost.

The "on" and "off" condition of the transistor Q_{54} is determined by the state of the transistor-pair Q_{11} and Q_{12}. During the "on" (sampling) interval, a signal from the horizontal rate keyer disables transistor Q_{11} and the collector current of the transistor Q_{12} maintains the transistor Q_{54} in the "on" condition. During the "off" (hold) period, transistors Q_{11} and Q_{12} change their states and the transistor Q_{54} is "off".

The bias sample-and-hold circuit, similar in structure to the above-described circuit, consists of the sampling switch Q_{59} and the transistor-pair Q_{17} and Q_{18}. This circuit, also activated by a signal from a horizontal rate keyer, samples the quiescent potential of the phase detector. The two signals, the error and the bias, processed by the sampling circuits, are stored in filter capacitors, and are applied to opposite terminals of a differential phase control. The phase control circuit synchronizes the reference carrier produced by the VCO.

Depending on the free-running frequency of the VCO, the detected signal is in the form of positive or negative going pulse trains which are then stored in a filter capacitor. The sampling switch has equal drive capabilities for both polarities of the signal; a requirement of particular importance in the presence of noise signals. Non-linear operation of the detector and sampling circuit would produce a rectified dc component causing an erroneous detuning of the VCO.

The VCO Loop

The amplification and amplitude limiting of the oscillator signal takes place in the amplifier-limiter formed by the transistor-pair Q_{60} and Q_{20}. The output from Q_{20} is fed to the dc controlled phase-shifter and returns to the amplifier through a crystal filter. The amplifier operates in a non-inverting mode, hence, the total phase shift through the phase-shifter plus crystal filter must be a multiple of 2π radians. The crystal filter is tuned to the subcarrier frequency and the filter band-width is determined by a resistor in series with the crystal. The DC controlled phase-shifter has a phase range of approximately $\pm \frac{\pi}{4}$ radians, and a phase change activated by a control signal results in a corresponding oscillator frequency change.

In the phase-shifter, the oscillator signal available at the collector of Q_{20} is applied to the base of Q_{14} from which it proceeds along two paths. An integrated capacitor C_2 couples this signal from the emitter of Q_{14} to the collector load of Q_{15} and, at this point, the signal is phase-shifted by approximately $\pi/4$ radians. In the second path, the signal arriving at the collector of Q_{15} passes through a current splitter formed by the transistor-pair Q_{56}, Q_{15}. This signal is reduced to a level determined by the control voltage at the bases of transistors Q_{56} and Q_{15}. At one extreme, transistor Q_{15} is OFF and the signal at the collector of Q_{15} arrives through the capacitor C_2 only. Conversely, with transistor Q_{15} ON, and Q_{56} OFF, the signal arriving through the transistor Q_{15} is phase-oriented so that the resultant signal has a phase of $+3/4\ \pi$ radians. The phase-control is linear throughout most of the control range.

A buffer amplifier is used to supply the CW carrier required for the demodulators, and the carrier is available at terminal 8. Internally, the buffer amplifier supplies the two synchronous detectors. Two R-C phase-shifters fed from the buffer amplifier provide the required phase orientation. A low-pass R_{14}-C_3 filter shifts the carrier to the AFPC detector by $-\pi/4$ while a high-pass filter provides a $+\pi/4$ oriented carrier for the ACC-killer detector.

AMPLITUDE CONTROL OF THE CHROMINANCE SIGNAL

Two cascaded amplifier stages serve to process the chroma signal and several signals are developed to control the gain of each stage.

First Chroma Amplifier and ACC Servo Loop

The first chroma amplifier, shown in Fig. 4, is controlled by the burst responsive ACC-killer detector only. The amplifier formed by the transistor-pair Q_1, Q_2 is driven single-ended by the applied composite chroma signal. The amplified output from this stage drives differentially the synchronous ACC-killer detector. The gain of the first amplifier is a function of the dc emitter current supplied by the constant current source Q_3. This current source is biased to provide a nominal current and, hence, a nominal gain in the first amplifier stage. The bias of the current source is reduced in response to a detected burst signal and the gain of the first stage diminishes correspondingly.

The ACC-killer detector is similar in structure to the AFPC detector. However, the CW carrier applied to the detector is in phase with the burst signal. The detected burst signal is processed by a sampling circuit in the same manner as previously described in connection with the AFPC circuit. The signal sampling consists of the transistor follower Q_{73} and the keyed transistor-pair Q_{40}, Q_{41}. Resistor R_{63} serves to produce an intentional dc offset across the inputs of the differential pair Q_{43}, Q_{76}. The detected ACC signal is unipolar with respect to the reference potential; thus, the dc offset extends the linear operating range of the amplifier Q_{43}, Q_{76}. The bias sampling circuit consisting of transistors Q_{79}, Q_{44}, Q_{45} applies the quiescent bias to the base of transistor Q_{76}. In the absence of a burst signal, the dc offset maintains transistor Q_{43} in the OFF condition and the following p-n-p transistor, Q_{29}, is also disabled. Thus, the ACC amplifier Q_{28} is non-conducting and the current source Q_3 provides the maximum current to the input stage.

The OFF state of transistor Q_{29} renders the killer amplifier (transistor Q_{27}) inoperative, a condition required to disable the second chroma stage.

Upon amplification of the burst signal in the first chroma amplifier, the detected burst signal increases proportionately to the amplitude of the input signal and combines differentially with the previously described bias signal in the collector load of transistor Q_{43}. Prior to it, the detected and bias signals are smoothed by the ACC filter capacitors. The linear operation of the chroma amplifier, the detector, and the amplifier which follows the sampling circuits is maintained to a signal level sufficient to enable the transistor Q_{28}. This potential, approximately 0.7 V, establishes the delay of the ACC characteristic as shown in Fig. 5. The chroma (burst) signal at the output of the first stage remains essentially constant with further increase of the input signal. The increasing dc potential at the collector of Q_{29} also activates the killer-amplifier Q_{27}. In order to maintain a predictable killer threshold, this action is referenced to the delay point of the ACC. As previously stated, the ACC begins to function at a signal level at which the dc potential across resistor R_{47} reaches 0.7 V. The killer threshold is lower than that of the ACC action and is determined by the voltage drop across resistors R_{48} and R_{47}. Thus, the two threshold signals are predictably established by the ratio $R_{47}/(R_{47} + R_{48})$.

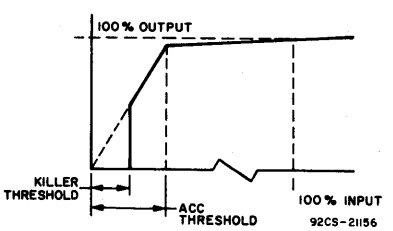


Fig. 5– Normalized ACC characteristic.

The Second Chroma Amplifier

The operation of the second chroma amplifier is controlled simultaneously by several signals. As described previously, they are: a customer-operated chroma gain (saturation) control, the killer detector signal, the overload detector, and the keyer.

The amplifier circuit shown in Fig. 6 is formed by the transistor-pair Q_{65}, Q_{24} and is driven differentially by the first chroma amplifier. The signal level to this stage is reduced by means of a resistive voltage divider. The amplifier Q_{65},Q_{24} is interrupted during the horizontal keying interval by the transistor-pair Q_{66}, Q_{23} to remove the burst information from the composite signal. The gating transistors Q_{66} and Q_{23} are connected so that their emitters and collectors are in parallel with the respective emitters and collectors of transistors Q_{65} and Q_{24}. The resulting collector current compensation maintains the quiescent output potential regardless of the keying operation.

The gain of the second chroma amplifier is adjusted by varying the current in the transistor Q_{25}. A resistive divider R_{41},R_{42} fed from a follower stage Q_{67} provides the bias potential to the base of the transistor Q_{25} and the voltage drop across resistor R_{40} determines the current flowing from the collector of Q_{25} to the emitters of Q_{65} and Q_{24}. The diode D_3 compensates the base to emitter potential of the transistor Q_{25}.

Since the bias resistors R_{40}, R_{41}, and R_{47}, and also the amplifier load resistor R_{43}, are located on the same IC chip, the resistance ratio of these components is accurately controlled. Thus, the gain of the second chroma amplifier determined by these components is very predictable, and is a function of the bias potential applied to the base of transistor Q_{67} only.

The interfacing of IC's with external control circuits usually presents problems due to the large tolerances associated with both components. The circuit used here overcomes these difficulties. The transistor follower Q_{67} exhibits negligible loading on the bias set by the manual chroma control. Thus, the gain of the second chroma amplifier is uniquely determined by the rotation of the gain control potentiometer and is relatively independent of its resistance value.

The killer operation is also performed on the second chroma amplifier. The amplified output from the ACC-killer detector is applied to the killer switch Q_{20}. In the presence of a burst signal, transistor Q_{20} is off and the chroma amplifier remains undisturbed. In the absence of a burst signal, the collector current in Q_{20} reduces the potential on the base of the transistor Q_{67} so as to cut off the second chroma amplifier.

The Overload Detector

The ACC and the manually operated saturation control provide the essential means to maintain the proper chrominance level to the picture tube. Under certain conditions, however, the presence of the ACC is detrimental. As previously stated, the ACC servo loop maintains a constant output level of the burst signal regardless of the chroma information. Transmitter variations in burst-to-chroma ratios are improperly corrected by the ACC action and, on signals with low burst-to-chroma ratios, the excessively amplified chroma can exceed the dynamic range of the picture tube.

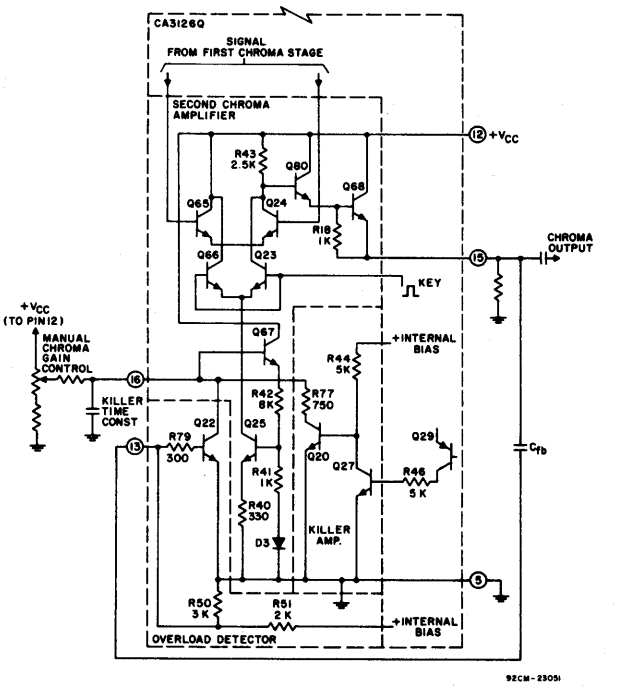


Fig. 6– Chroma output stage.

Similar overload problems are experienced when receiving weak signals. The Synchronous ACC detector produces a control signal proportional to the average value of the burst interval signal, and noise does not contribute to the output. Although this type of noise-immune detection is necessary for reliable operation of the killer circuits, it is less desirable for the ACC action because the noise-peaks plus signal tend to produce undesirable over-saturation effects.

The overload detector operating on the second chroma stage eliminates both these overload problems. The chroma signal from the output terminal of the second chroma amplifier is coupled, by means of the coupling capacitor C_{fb} to overload detector Q_{22}. Transistor Q_{22} is biased by means of an internal bias supply to 0.5 V, and remains off until its base potential is raised to approximately 0.7 V. Thus, detection takes place whenever the chroma signal plus dc bias is equal to or exceeds 0.7 V. The detected and filtered signal lowers the bias potential on the base of transistor Q_{67} and reduces the gain of the output stage.

KEYING CIRCUITS

Details of the keying circuit and of the internal bias circuits are shown in the complete diagram of the CA3126Q in Fig. 7. A positive horizontal rate keying pulse applied to terminal 9 activates the keying circuit. This circuit maintains the AFPC and ACC detectors, with the corresponding sample-and-hold circuit, in the ON position during the keying interval, and disables the chroma output stage at the same time.

CONCLUSION

The new chroma processing circuit improves the performance of a color television receiver. The use of synchronous detection and sampling results in excellent signal stability and fewer external components and adjustments. An overload detector prevents over-saturation of the picture tube, and the improved manual control simplifies the adjustment of the chroma level.

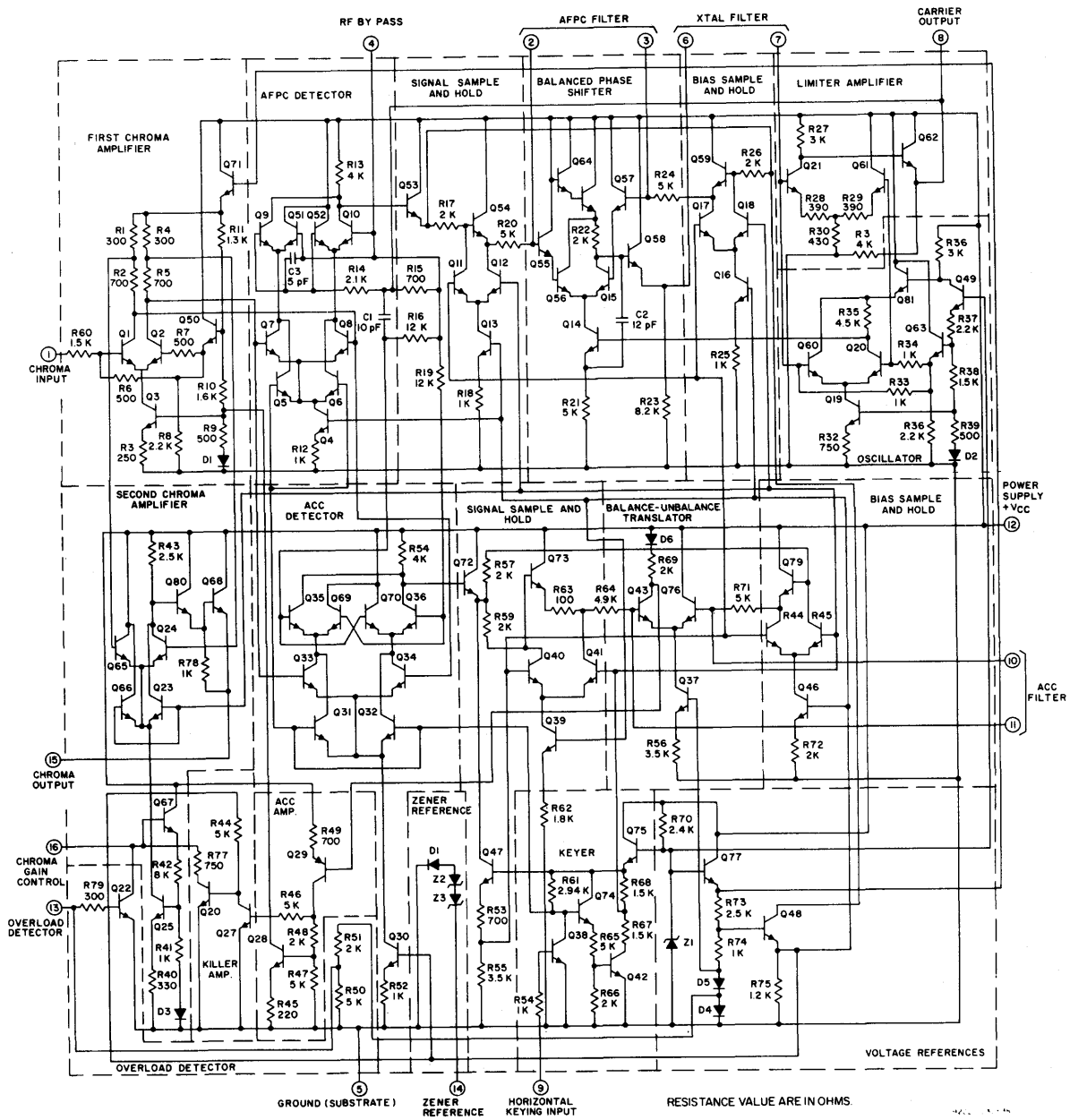


Fig. 7– Complete circuit diagram showing details of the keying circuit and internal bias circuits.

ICAN-6257

Application of the CA3089E FM-IF Subsystem

by L. S. Baar

The RCA-CA3089E, shown in Fig. 1, is an FM-IF subsystem intended for use in FM receiver applications. In addition to the amplifier-limiter and quadrature detector sections, the CA3089E provides such auxiliary functions as mute, AFC output, tuning-meter output, and delayed rf-AGC.[1] This Note briefly describes each circuit section and discusses practical aspects of designing with this device.

Circuit Description

The three-stage direct-coupled amplifier-limiter uses a cascode input stage to reduce input noise and provide better stability. The peak-to-peak swing of approximately 300 millivolts is developed across the 390-ohm resistor, R31, at pin 8. The operating-point stability is provided by dc feedback to the input stage. The input voltage for an output 3 dB below limiting is typically 12 microvolts rms.

The detector is a doubly balanced circuit driven symmetrically by the output of the if amplifier. The voltage at pin 8 is coupled through a reactance to the tuned circuit at pin 9. The detector output is taken from both sides and combined differentially to produce an audio output and automatic-frequency-control voltage. The audio output may be attenuated by a current driving pin 5. The current is normally provided by the mute drive, which reduces the level by more than 50 dB. Fig. 2 shows the detector and audio-AFC translator circuits redrawn to illustrate the balanced circuitry. The audio output is developed across a 5,000-ohm resistor, R49, Fig. 1. The AFC output can be used either as a current or voltage source.

The meter output and rf-AGC circuits are driven by three level detectors which detect the output levels of each of the if amplifier stages. The tuning-meter circuit sums these levels and provides a voltage which is a function of the input signal. The rf-AGC circuit is driven by the level detector connected to the output of the first amplifier stage, which provides the delay. The mute logic output is developed from the output of the third limiter. With a large signal, the if envelope is detected, and drives the mute logic voltage low. As the signal-to-noise ratio deteriorates, "holes" are created in the envelope; these "holes" are detected, and provide the voltage to drive pin 5.

The bias supply maintains the device current drain virtually constant from a supply voltage of 16 volts to approximately 8 volts over a temperature range of −40°C to +100°C while maintaining the performance of the device virtually unchanged. The typical curves in Figs. 3 through 7 illustrate these characteristics. A reference voltage brought out to pin 10 may be used in conjunction with the AFC, if desired.

Stability Considerations

Because the CA3089E is a very high gain device, the external circuit must be laid out carefully to eliminate or reduce any feedback path.[2] Fig. 8 shows a 10.7-MHz printed-circuit-board layout of the circuit in Fig. 9(a). The ground-plane layout was devised to prevent large rf currents at the output terminals from returning to the input grounds. Bypass-capacitor grounds also were selected to achieve the same purpose. It is recommended that bypass capacitors be placed on terminals of the auxiliary functions since most of them are connected to rectifier circuits which are not completely filtered within the device. Capacitors of the disc ceramic type with a 0.01- to 0.02-microfarad value are usually good bypass capacitors at 10.7 MHz. Larger values may exhibit a self-resonance below 10.7 MHz, and actually

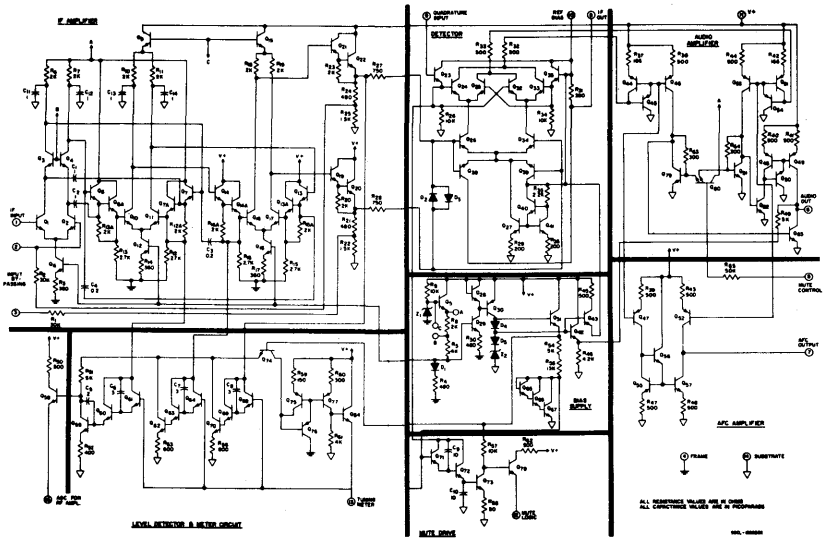


Fig. 1– Schematic diagram of the CA3089E.

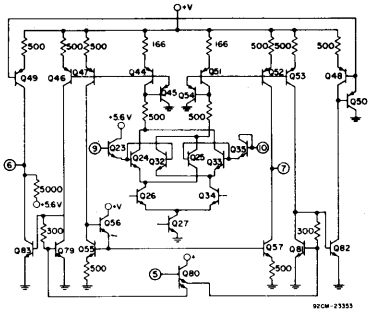


Fig. 2– Detector, audio, and AFC circuits.

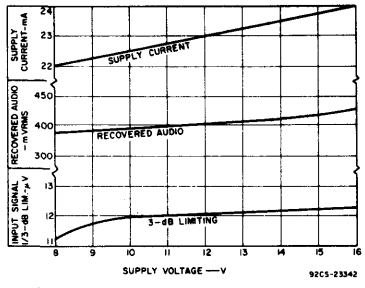


Fig. 3– Supply current, recovered audio, and input limiting as a function of supply voltage.

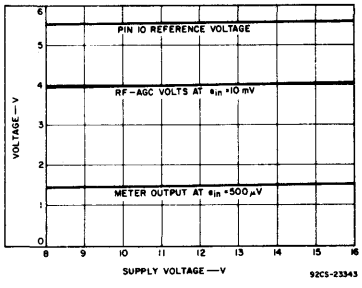


Fig. 4– Reference voltage, rf-AGC, and meter output as a function of supply voltage.

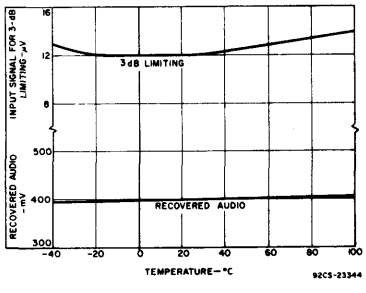


Fig. 5– Input limiting and recovered audio as a function of temperature.

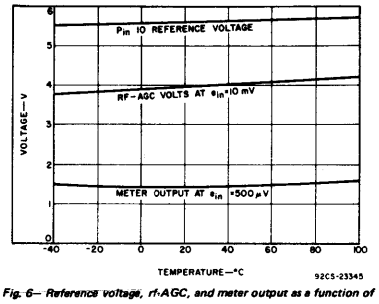


Fig. 6– Reference voltage, rf-AGC, and meter output as a function of temperature.

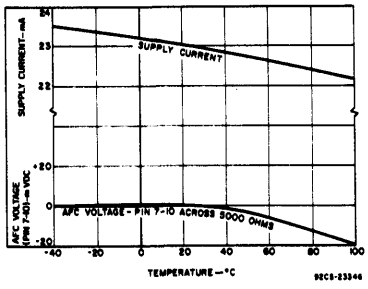
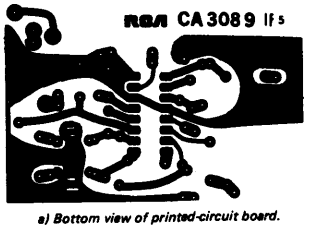


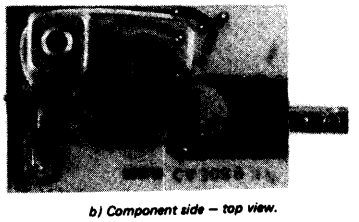
Fig. 7– Supply current and AFC voltage as a function of temperature.

exhibit inductive reactance at their terminals. The nominal input impedance of the CA3089E is approximately 9,000 ohms, and it is not recommended that an impedance match be attempted. Most commercial receivers use ceramic-filter frequency-selective elements that normally have source impedances of 500 ohms or less. When these filters are properly terminated with loading resistors, the typical source impedance is further decreased to 250 ohms or less. Higher levels of source impedance are possible with very careful circuit layout; however, the maintenance of stability could be difficult.

The CA3089E has a frequency response that is typically flat to 20 MHz; consequently, the device can provide useful gain well above that frequency. If the device is used at lower frequencies, the larger-value bypass capacitors required may not be adequate to bypass the higher frequencies. Double bypassing with lower-value capacitors can overcome such a problem. Another means of alleviating the problem is to externally reduce the frequency response by using a small capacitance across the output load of the device.



a) Bottom view of printed-circuit board.

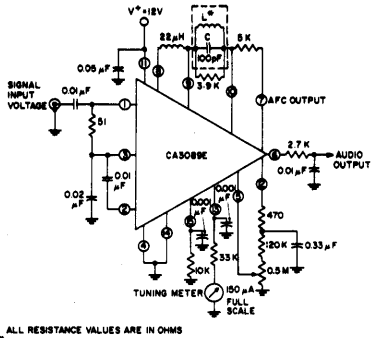


b) Component side – top view.

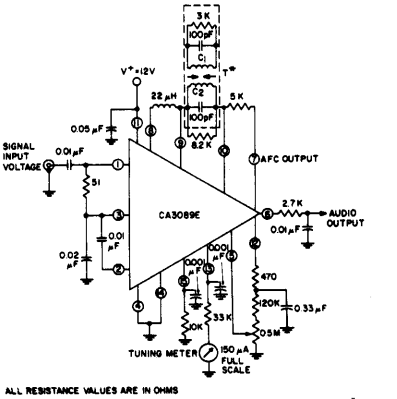
Fig. 8.– CA3089E and outboard components mounted on a printed circuit board.

Quadrature-Detector Circuits

The quadrature-detector tuned circuit is connected between pins 9 and 10. The signal voltage at pin 8 is normally coupled to pin 9 through a choke. The circuit values for the detector network are determined by several factors, the primary one being distortion at a particular level of recovered audio. Distortion is determined by the phase linearity of the quadrature network and is not influenced by the device unless excessive, recovered audio overdrives the audio circuit. With a single tuned network, the phase linearity improves as the bandwidth increases; however, recovered audio decreases. A satisfactory compromise for most FM-receiver applications is reflected in the circuit of Fig. 9(a). This circuit typically provides 400 millivolts rms of recovered audio with less than 0.5-percent distortion. Because a double-tuned circuit has better phase linearity over a wider bandwidth, distortion figures of less than 0.1-percent are attainable with the network used in the circuit of Fig. 9(b). Proper alignment and coupling adjustment of the double-tuned circuit are most easily accomplished while viewing the resulting S curve. Initial adjustment of the primary tuning slug to the proper crossover is made with the secondary slug removed. The secondary tuning slug is then



(a)



(b)

Fig. 9– (a) Test circuit for the CA3089E using a single-tuned detector coil, (b) test circuit for the CA3089E using a double-tuned detector coil.

adjusted until a slight "ripple" is observed moving along the S curve. If the ripple is excessive (enough to distort the S curve) the coupling is too tight. If no ripple is observed, the coupling is too loose. As the ripple moves through the crossover point, it will be observed that the S curve becomes more linear near the center frequency. Slight readjustment of both slugs may be necessary for final alignment. The best performance can then be achieved by slight adjustment while measuring distortion. The coupling may be varied by either moving the coils or by changing the value of the secondary load resistor.

Various circuit values can be used to obtain the same recovered audio, but the basic conditions of circuit bandwidth and phase linearity must be maintained. The detector circuit also sets up conditions which are required for proper operation of the mute circuit. The rf voltage on pin 9 must be held at approximately 175 millivolts rms, ±25 millivolts. The reason for this requirement is discussed subsequently in connection with the mute logic circuit. The approximate voltage at pin 9 is determined from the equivalent circuit shown in Fig. 10.

The peak-to-peak voltage on pin 9 is:

$$|V_9| \cong |V_8| \frac{R1}{\omega L_{ch}}$$

where R1 is the total parallel resistance and V8 is approximately 300 millivolts, peak-to-peak.

The Q of the tuned circuit between pins 9 and 10 may be affected by the effective Q of the choke between pins 8 and 9 and the series resistor R31 in the CA3089E. All of the above factors should be considered in selecting circuit values. Table 1 lists some typical combinations of component values under various conditions.

A choke is normally selected to equalize delays in the signal path and in the limiter–quadrature path. It also reduces the if harmonic content across the quadrature circuit. In some cases, such as in narrow-band applications, it may become necessary to use a capacitor as the coupling component where large values of inductance with high Q's are difficult to obtain. If a capacitor is used, the phase of the recovered audio and AFC voltage will be reversed, some asymmetry of the S curve may result, and the distortion may be adversely affected to a small degree.

As indicated above, the inductance between pins 8 and 9 tends to equalize delays in the detector signal paths. The matching of elements of the IC in the balanced detector

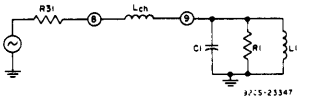


Fig. 10– Equivalent circuit used to determine approximate voltage on pin 9 of the CA3089E in Fig. 9.

circuit results in an AFC output with a very small offset when referred to the voltage at pin 10. For most applications, the inherent offset variation is well within tolerances, and does not affect circuit performance. In some narrow-band applications, however, the offset becomes more critical because of the very narrow bandwidth. In such situations, the combination of normal production variations of the device and the external circuit components results in receiver detuning when the AFC loop is closed. This detuning results in an increased distortion of the recovered audio. This distortion can be corrected with the addition of a variable capacitor from pin 8 to ground to provide phase compensation. The capacitor can be adjusted to provide zero AFC offset with minimum distortion. Generally, the offset is in one direction for a given set of conditions. The addition of a fixed capacitor will minimize variations sufficiently to satisfy many applications. A value of 5 picofarads is an effective value for the circuit of Fig. 9(a) with the recommended PC-board layout. Conversely, the offset created by using a capacitor between pins 8 and 9, as mentioned earlier, may be compensated by placing an inductance between pins 8 and 10.

Audio and AFC Circuits

The audio and AFC circuits are very similar, and both develop the same audio signal at their respective output terminals. The audio output voltage on pin 6 is developed across an internal, nominal, 5,000-ohm resistor (R49) connected to the 5.6-volt reference. In addition, the audio signal level can be attenuated by providing a direct current into pin 5 without any shift in its dc level. The audio output,

TABLE I – FIG. 10 COMPONENT VALUES AND CHARACTERISTICS AS A FUNCTION OF FREQUENCY

Freq. (Hz)	L_1 (H)	QL_1	C_1 (pF)	R1 (ohms)	X (pin 8 to pin 9)	Deviation (kHz)	Recovered Audio (mV)
10.7M	2.2μ	75	100	3900	22μH	±75	400
10.7M	2.2μ	120	100		120μH	±5	280
10.7M	2.2μ	120	100		1.3pF	±5	290
455k	0.1m	65	1000	68k	1 mH	±5	400

as shown in Fig. 11, is uniform to a frequency of more than 1 MHz when measured in the circuit shown.

The AFC output at pin 7 is a current source and, if terminated with 5,000 ohms, will provide an audio output identical to that at pin 6. The AFC output may be referenced to a wide range of voltages, from near ground potential to near supply voltage. However, because of the balanced circuit configuration, the best AFC sensitivity and offset will occur at, or near, the 5.6-volt internal reference. An AFC voltage developed across a load tied back to pin 10 is recommended. As a consequence of this connection, variations in the AFC voltage as a function of operating voltage and temperature are minimized because the voltage on pin 7 tends to follow changes in the reference voltage.

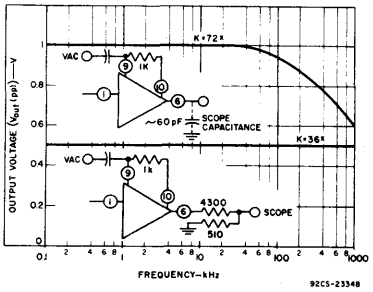


Fig. 11—Audio response at pin 6.

Mute Circuit

The signal to the mute logic circuit is taken from the emitter follower connected to pin 9. This signal drives a peak detector followed by an inverter such that the mute logic developed is zero volts with an input signal level sufficient to develop a fully limited output signal. As the input signal level is reduced below limiting, noise becomes significant, and creates "holes" in the if envelope. The detected drive voltage decreases and causes the mute logic voltage at pin 12 to increase. This voltage, in turn, is fed to pin 5 to provide the current to attenuate the audio. If the if level at pin 9 is too high, the "holes" created by the noise are insufficient to drive the mute logic voltage high enough to attenuate the audio. If the pin 9 voltage is too low, the mute drive voltage never reaches zero, and the external mute-threshold control behaves like a volume control. It is for this reason that the mute logic circuit requirements influence the selection of detector circuit values.

Another condition affecting proper mute performance is excessive gain in the tuner or preceding if stages. High gain ahead of the CA3089E under a condition of low signal-to-noise ratio results in the noise being clipped by the limiting amplifiers. The clipping has the effect of reducing the number of "holes" in the if envelope, and limits the mute drive voltage at pin 12 to values insufficient to attenuate the audio. If high gain is a system requirement, an externally derived mute logic voltage is necessary to drive pin 5.

The external circuit on pin 12 in Fig. 9(a) serves to filter the output, and provides a variable potential for mute-threshold adjustment. The 470-ohm resistor in series with pin 12 reduces the effective Q of the filter capacitor and prevents the circuit from setting up on noise current transients as the mute circuit begins to function. The voltage divider, composed of the 500 kilohm potentiometer and 120 kilohm resistor, controls the threshold point. These values are suggested ones, and may be altered to suit the user. Curve A in Fig. 12 shows the change in audio output level as a function of input signal with the mute-threshold control circuit (also shown in Fig. 12) at its maximum-voltage setting. Because of the more shallow slope and the larger circuit time constant involved, a "soft" mute action results.

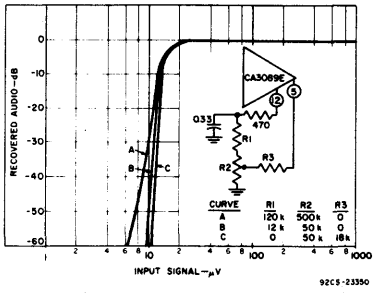


Fig. 12—Mute curves.

Curves B and C illustrate the change in the curves resulting from adjustment of the values of the threshold control circuit. These latter circuits provide a faster acting mute. The fixed resistor, R1, in addition to controlling the slope of the mute characteristic, limits the voltage appearing at pin 5. The use of this resistor is recommended to prevent a latch-up at the attenuating circuit, which, if it occurs, maintains the circuit in muted condition until the supply voltage is removed.

The curves in Fig. 12 show that the muting action cannot be initiated under any condition until some noise is present in the output signal. In this respect, the mute performance of the CA3089E differs from that of some other systems which are activated by signal level. Such systems can be adjusted to allow noise-free signals to be processed further. When the CA3089E circuit operates under small-signal conditions, noise may be audible before muting action occurs. The threshold-level adjustment only permits more or less noise to appear at the output; a listener can use the control to adjust the interstation hiss to the level of his preference.

Tuning Meter and RF-AGC Circuit

The tuning-meter circuit sums the output of three peak detectors connected to successive stages of the if amplifier-limiter. These circuits detect not only the carrier, but also harmonics developed when successive stages go into limiting and eventually form a square wave. The result is a logarithmic dc output as a function of input signal, as shown by the curve in Fig. 13. The circuit developing the delayed

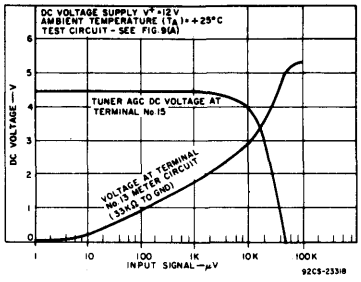


Fig. 13—Tuner AGC and tuning-meter output as a function of input signal voltage.

rf-AGC voltage is driven by the level detector connected to the first if stage. As a result, no output is detected until the input signal is large enough to drive the peak detector; the input signal is large enough to drive the peak detector; the signal is a delayed AGC action. The curve of rf-AGC voltage as a function of input signal is also shown in Fig. 13.

IF Amplifier/Detector System and Stereo Decoder

Fig. 14 shows the circuit diagram of a complete FM-if detector system driving a stereo decoder. Using the selectivity of two ceramic filters, the CA3089E in conjunction with the CA3090AQ stereo decoder provides the basic signal processing between the tuner output and the audio amplifiers. The gain of the silicon n-p-n bipolar-transistor stage is adjusted to make up the losses of the two filters. In addition to driving a tuning meter, the voltage at pin 13 of the CA3089E may be used to drive a "stereo defeat" circuit in the CA3090AQ, thereby holding the decoder in a monaural condition to improve the signal-to-noise ratio under weak signal conditions. A suggested PC-board pattern and parts layout are shown in Fig. 15.

Operation at Frequencies Other Than 10.7 MHz

Because the CA3089E was designed for use in FM broadcast receivers, its circuits are optimized for use at 10.7 MHz. Nevertheless, the device performs equally well over a wide range of frequencies both above and below 10.7 MHz. The if amplifier response is essentially flat from dc to more than 20 MHz. The operation of the detector circuit is dependent only on the external components. The operation of the auxiliary sections–rf-AGC, meter output, and mute logic – depend on internal peak detectors, and, as a consequence, their performance at lower frequencies is limited. The internal capacitors were optimized for 10.7 MHz operation, and are too small to operate effectively at lower frequencies. The detector efficiencies begin to deteriorate at about 2 MHz, and the detectors are essentially unusable at

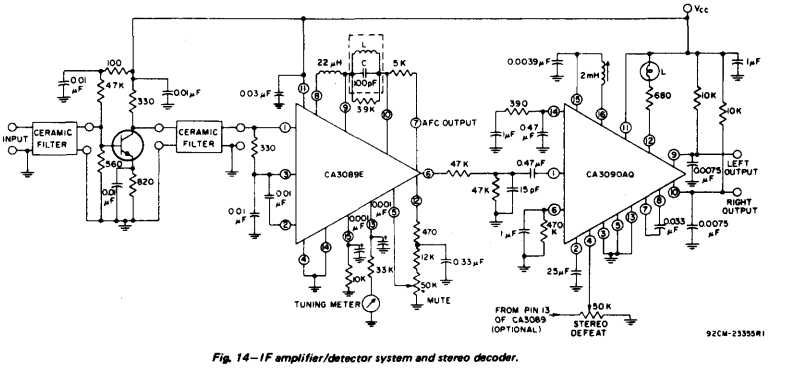


Fig. 14—IF amplifier/detector system and stereo decoder.

455 kHz without the use of external circuitry. The rf-AGC and mute logic circuits do not develop sufficient dc voltage

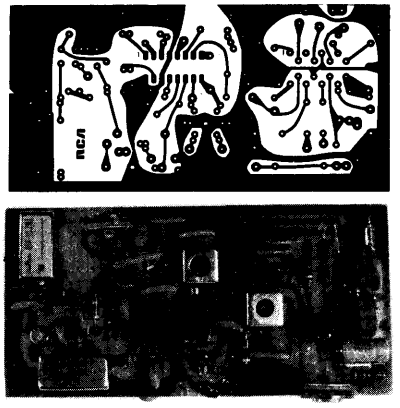


Fig. 15—Suggested PC-board pattern and parts layout for the circuit of Fig. 14.

to perform their functions, and the meter output signal loses its logarithmic characteristic and exhibits peaks and valleys as input signal is increased. Operation of the rf-AGC and mute logic circuits may be enhanced by the addition of a dc amplifier and inverter to each circuit. A simple example using a CA3096E IC transistor array is shown in Fig. 16.[3]

The CA3089E may be used effectively in narrow-band communication receivers. In double-conversion receivers, some of the functions of the CA3089E are negated at a 455-kHz intermediate frequency. However, if a 10.7-MHz intermediate frequency is used, all of the auxiliary features

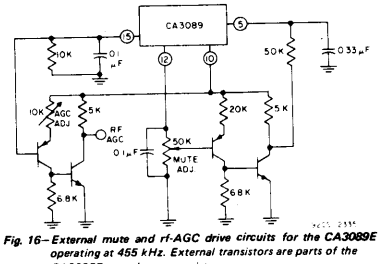


Fig. 16—External mute and rf-AGC drive circuits for the CA3089E operating at 455 kHz. External transistors are parts of the CA3096E n-p-n/p-n-p transistor array.

may be used, but another set of problems is encountered. The small deviation signals encountered in narrow-band systems require the use of high-Q circuits in the quadrature detector, as indicated in Table I. However, variations in external-component parameters with temperature changes may cause the tuned frequency of the detector to drift out of the if pass band. Normally temperature-compensated components are necessary. The CA3089E, operating in conjunction with an inexpensive operational transconductance amplifier,[4,5] provides means of locking the tuned circuit to the incoming frequency. Fig. 17 shows the block diagram of such a system. The AFC output voltage developed across the resistor between pins 7 and 10 is amplified by the op-amp and drives a varactor to maintain the tuned frequency on the incoming-signal frequency.

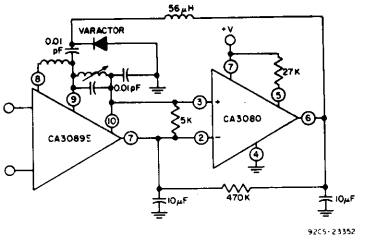


Fig. 17—Detector frequency-stabilization circuit.

The CA3089E may also be used as the core of an ultra-linear FM generator; Fig. 18 shows the circuit. The carrier is generated by the CA3089E with the introduction of feedback from the output terminal, pin 8. The carrier is modulated by the varactor connected across the tuned circuit at the input of the CA3089E. The varactor is driven by the output of the differential amplifier, A1, using a CA3028 IC.[6,7] This differential-amplifier stage is driven at one of its input terminals by the audio modulating signal. Negative feedback of the audio signal is provided by driving the other differential-amplifier input from the recovered audio output of the CA3089E at pin 6. The detector circuit uses a double-tuned transformer to produce audio with very little distortion at pin 6. This feedback technique results in a very low distortion modulation. The rf output of the CA3089E at pin 8 is essentially a square wave, and is fed to a tuned-amplifier stage to buffer the signal and restore the sine-wave-shaped rf output signal.

Acknowledgments

The author thanks Jack Craft for his aid and suggestions in many discussions and Frank Curley for his aid in circuit construction and collection of data.

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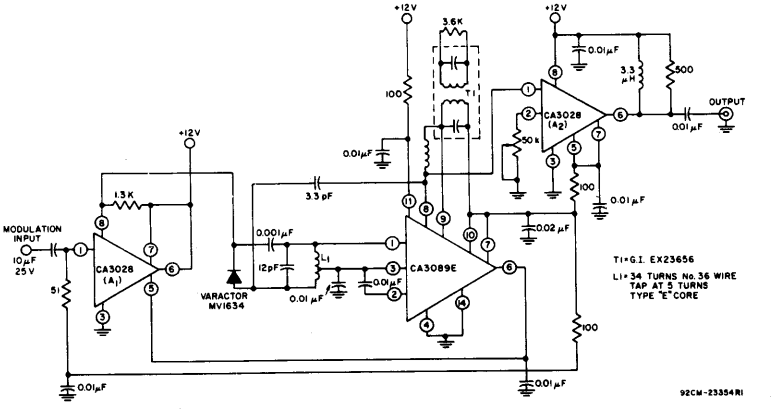


Fig. 18—FM generator circuit.

ICAN-6303

A Single IC for the Complete PIX-IF-System in TV Receivers

The RCA-CA3068 linear integrated circuit is a PIX-IF-subsystem in a shielded, quad-formed, dual-in-line, 20-lead, plastic package. This package contains all the active devices and most of the passive elements necessary for a high performance, PIX-if-system for a TV receiver. **This Note* describes** the receiver functions performed by the CA3068, and its application to color and monochrome TV receivers.

Specifically, the receiver functions performed by the CA3068 are:

Video if amplification
Linear video detection
Noise-limited amplification of detected video
Keyed age, with noise-immunity circuits
AGC delay for tuner rf stage
Buffered output signal to drive Automatic-Fine-Tuning (AFT) circuits
Amplification of intercarrier frequencies
Sound-carrier detection
Sound-carrier amplification
Zener reference diode for voltage regulation.

The only external components required for the operation of this if subsystem are bandwidth shaping networks, biasing networks, and a power supply. A functional block diagram of the signal portion of a typical color-TV receiver is shown in Fig. 1. A detailed block diagram of the CA3068, together with its peripheral tuned-circuits, is provided in Fig. 2.

This Note contains a detailed description of circuit functions within the integrated circuit, together with examples of the use of the CA3068 in PIX-IF amplifier PC-boards for color and monochrome TV.

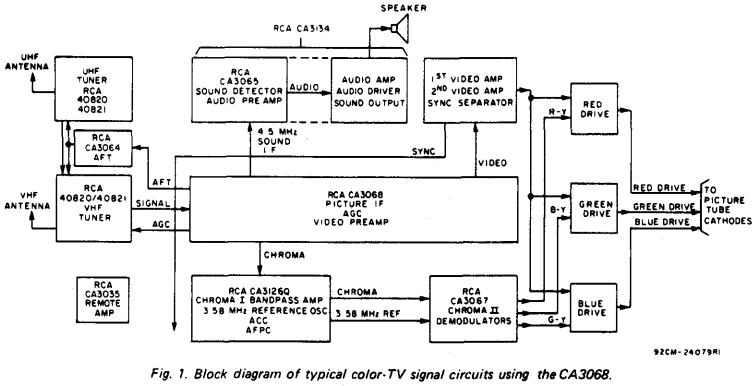


Fig. 1. Block diagram of typical color-TV signal circuits using the CA3068.

GENERAL DESCRIPTION OF CIRCUIT FUNCTIONS IN THE CA3068

As shown in the block diagram of Fig. 2, the if signal from the tuner is applied to the input (terminal 6) of the cascode if amplifier. Output from the cascode amplifier is then coupled to a wideband amplifier at terminal 13 through the interstage transformer (T2). Under maximum-gain conditions, the over-all gain of the CA3068 is typically 75 dB at PIX-IF frequencies. This signal is then applied to a linear video detector whose output signal is fed to a video amplifier having a gain of 12 dB.

Bandpass shaping is accomplished by means of tuned-circuits preceding the input stage (terminal 6) and at the interstage circuit comprising input and output terminations via terminals 9, 12 and 13, as shown in Fig. 2. Terminal 16 is tied in at this point to provide loop bias for the input stages of the amplifiers connected to terminals 12 and 13. The agc voltage developed within the CA3068 is applied to its input stage by an external path from terminal 4 to terminal 6 through the input circuitry, as shown in Fig. 2. The developed agc is gated by a keying pulse applied to terminal 3 from the horizontal sweep circuit of the TV receiver. Delayed agc for the rf amplifier in the tuner is obtained from terminal 7; the delay is variable by adjustment of a resistance (25 kilohms) in series with the supply to terminal 8.

The zener reference voltage for the power-supply regulating pass-transistor is developed at terminal 18 when this terminal is connected to a voltage supply through a current-limiting resistor. This resistor value should be selected to provide a quiescent current into the zener of 0.5 to 1.5 milliamperes (excluding the base current for the pass transistor).

Terminal 15 is the dc input terminal that provides power for most of the CA3068 and should be connected to the 11.2-volt regulated supply as shown in Fig. 2. The CA3068 package has a 20-lead configuration with 18 active terminals. Terminals 11 and 20 have been omitted from the package; their corresponding leads are internally connected to the shield. Terminals 1, 5, and 10 are grounding terminals. In addition, terminal 17 is at ground potential. Additional information relative to dc grounding is given in the section concerning if design.

DETAILED CIRCUIT FUNCTIONS

Fig. 3 is a schematic diagram of the CA3068. The diagram is partitioned to facilitate the explanation of the circuit configuration and its functions.

The cascode input amplifier (first if) is a unique circuit designed for dual-mode operation. At low-level input signals, the buffer stages formed by Q3 and Q4 drive the base of the cascode-if amplifier composed of Q7 and Q6. Negative-going agc applied to Q3 (through an external connection to terminal 6) increases in proportion to the increase of the input signal level. After approximately 40 dB of gain reduction is reached in this operational mode, Q7 is cut off, and its function is assumed by Q5. Emitter degeneration in Q5 increases the dynamic input range of the cascode amplifier sufficiently to cope with the higher range of input signal level. The point at which Q5 assumes the input amplifier function is sensed by Q11. It should be understood that transistors Q11, Q4, and Q7 approach cut-off at essentially the same signal level. As Q11 approaches cut-off, it draws less shunting current from terminal 8, and base current drive to Q8 is increased. The point at which sufficient base current is available to drive Q8 into conduction is determined by an external-delay agc potentiometer connected in series with the V+ supply-lead and terminal 8. As Q11 cuts-off, the voltage increases at terminal 8, and current flowing into terminal 8 is diverted to the base of Q8. When Q8 starts to conduct, it turns on Q9 and Q10, thereby causing the open-circuit voltage at terminal 7 to drop and produce a negative-going agc voltage for the rf stage of the tuner. Q8 is also part of the if-agc feedback loop, and provides an increase in agc loop-gain. This increase compensates for the decrease in agc loop-gain that occurs when the cascode if amplifier is transitioned to its modified cut-off characteristic. After tuner gain reduction has reached its maximum, an additional 10 dB of gain reduction can be obtained in the cascode-amplifier under this modified cut-off condition.

This reverse-agc system is used for the cascode input stages because the stability achieved under maximum-gain conditions is maintained throughout the range of agc functioning.

The wideband if amplifier consists of transistors Q12, Q13, Q14 and Q15. Q12 serves as a buffer stage between the interstage tuned-circuits and the automatic-fine-tuning (AFT) output-signal terminal. The actual if signal amplification takes place in Q13, Q14 and Q15, which effectively serve the function of second and third PIX-IF stages. Transistor Q15 is the driving source to Q17, the video detector. This driving source impedance is approximately 500 ohms as a result of the degenerative feedback loop through Q16. The feedback network also extends the 3-dB-down frequency response to beyond 70 MHz. It is this low detector-driving-point impedance and the absence of a tuned-circuit at this interstage point that contribute to the superior performance of the detector system. In most conventional detection systems, the detector is driven from a high-impedance source involving a double-tuned interstage transformer with unequal primary and secondary Q's. In such a system, variations in detector impedance (caused by normal video excursions) can produce significant phase shifts that adversely affect color fidelity. In the CA3068, the untuned, low-impedance detector drive circuit produces a nearly optimum condition for the detector circuit.

The detector circuit consists of transistor Q17 and its biasing network Q18, Q19 and R20. Q18 is biased to the same potentials as Q17 because the bases are tied together through the resistance element of the low-pass filter that consists of R19 and C3. R20 and C4 form a conventional peak detector in which the time constants are selected for optimum detector efficiency and desired video bandwidth. This system detects chroma subcarrier without introducing differential phase errors as a function of the video signal, and detects the video signals with a minimum of amplitude distortion. The low signal-level requirements for the detector, the absence of tuned-circuits in the detector drive circuit, and the low source impedance for the detector, all contribute to the superior detector performance.

The video detector is direct-coupled to the video amplifier. Consequently, a dc input voltage above the level of one V_{be} (0.7 volt) drop at the input to Q23 determines the condition for white level (dc) at the output (terminal 19). It is necessary, therefore, to bias Q23 to the threshold of conduction in the absence of detected video. This function is accomplished by the differential-amplifier circuit arrangement consisting of transistors Q20 and Q21. In the absence of signal, the dc potentials at the emitters of Q20 and Q21 are identical. The current through Q20 must equal the current through Q21 because R25 is similar in value to (R27 + R28). This current also flows through D3 (which has the same geometry as Q22). Consequently, Q22 carries all the current supplied by Q20, and no current is available for the base of Q23, so that Q23 is held on the edge of conduction. When an rf carrier is present, the current in Q20 increases in direct proportion to the carrier level; however, the current in Q21 remains fixed. When the current increases in Q20, this increase can only flow to the base of Q23. Since the current in Q23 is directly proportional to its base current flow, a corresponding increase in current through Q20 as a result of rf carrier detection produces a video output at terminal 19.

As the video carrier signal increases, the dc level at the base of Q23 increases, and there is an accompanying decrease in the dc level at the base of Q25 and, consequently, at terminal 19. With a sufficiently strong rf signal, the current through Q23 and R29 increases such that the base voltage of Q25 is driven toward dc ground. The "bottoming" level at terminal 19 under nominal signal conditions is locked to about 0.8-volt as a result of the high loop gain of the agc system. Any further increase in the signal, after "bottoming" is reached, will be clipped. This operational feature serves as a highly effective mechanism to limit impulse noise.

When a signal is present at the input, the composite video signal at the base of Q25 appears at terminal 19 through the Darlington connection to the emitter of Q26. The sync tips in

*This Note, revised by Maurice Caputo (Solid State Division), was orginally prepared by S. Reich and R. T. Peterson.

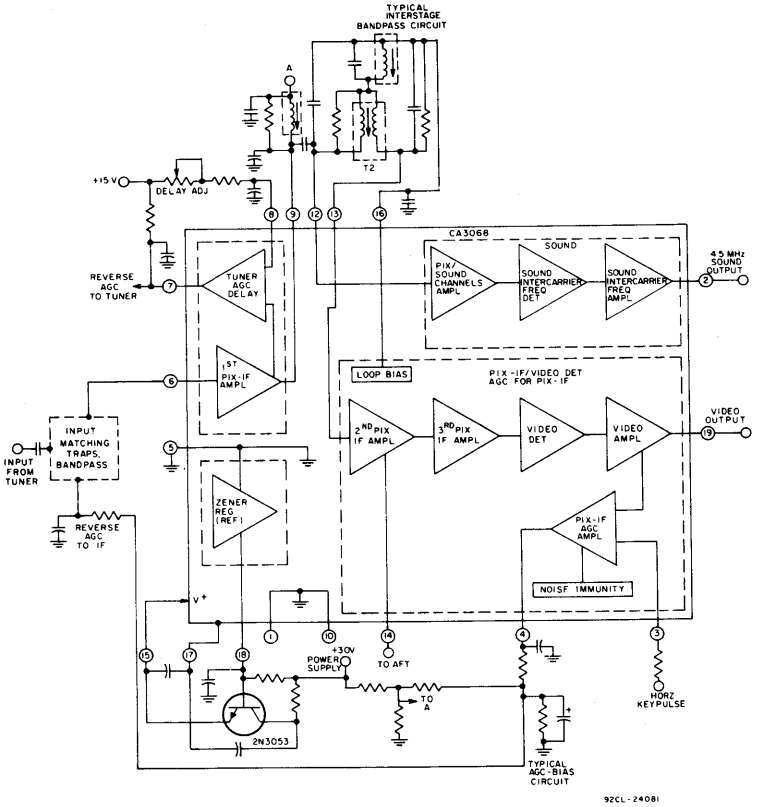


Fig. 2. Detailed block diagram of the CA3068 together with its peripheral tuned circuits.

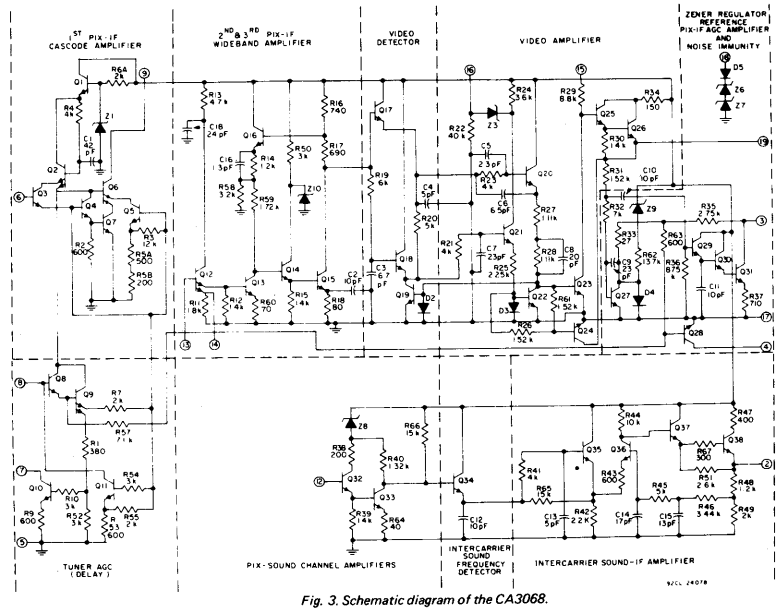


Fig. 3. Schematic diagram of the CA3068.

this composite waveform drive the keyed agc amplifier Q27, which in turn drives Q28. Without a video rf signal there is no video signal output, and Q27 conducts during the keying intervals (the horizontal pulse is connected to terminal 3). As the detected signal level increases in amplitude and the output voltage at terminal 19 approaches its typical operational level of 7 volts peak-to-peak, the peak potential at the base of Q27 begins to fall below 0.8 volt. Under these conditions, the keying current formerly channeled through Q27 is diverted through diode D4. As the signal level rises even higher, a greater portion of the Q27 collector current is diverted through D4, and the base current to Q28 is proportionately increased. A 10-microfarad capacitor is normally connected between terminal and ground and is, by this connection, put in shunt with Q28. The charge on this external capacitor is maintained through a bleeder resistor to V+. As the base current to Q28 increases, Q28 discharges the capacitor at a rate that is proportional to the base current of Q28. Integration of the total charge on the capacitor over the keying interval yields a dc level (agc voltage) that is inversely proportional to the incoming signal level; i.e., agc voltage approaches zero as the signal increases.

Any high-performance agc system must have noise-immunity characteristics in order to avoid the establishment of false agc levels. AGC voltage developed from random noise can produce "wash-out", "blank raster" and/or a momentary "loss of sync". The CA3068 is designed with an improved noise-immunity circuit that essentially removes the keying current during periods of high noise input. The active devices responsible for providing protection against this deleterious effect of the impulse noise are the "noise detector", Q29, and the "noise clamp" Q31, which is driven by Q30. Impulse noise is channeled through the high-pass filter network consisting of C10 and R36 to the detector input Q29. Q29 and C11 comprise a conventional peak detector. The dc level across C11, which is proportional to the level of impulse noise, turn on Q30 and Q31, thereby clamping the keying supply voltage (terminal 3) to ground. In actual operation, the terminal-3 supply has a series resistance that is large enough to limit the peak current into the zener diode (Z5) to approximately 0.8 milliampere. When Q31 conducts, it shunts this current to ground.

The sound-if-channel and PIX-IF-channel signals whose "carrier" frequencies are 41.25 MHz and 45.75 MHz, respectively, are applied to terminal 12. Q32 functions as a buffer between the interstage-tuned-circuits associated with terminal 12 and the PIX/sound-channels amplifier, Q33. The intercarrier frequency (the difference frequency between the PIX and sound "carrier" frequencies) is detected by the peak detector Q34 and C12. The resultant 4.5 MHz FM sound-intercarrier signal is fed to transistor Q35. This transistor and Q36 form a differential pair that provides an amplified intercarrier sound-if signal to the base of Q37. A feedback system through the RC networks in the Darlington emitter-follower output of Q37 provides bandpass shaping in the region of 4.5 MHz while maintaining a low dc gain. The low level of dc gain is desirable because the circuit receives its bias in an open-loop manner from terminal 16. The bandpass of this amplifier system is fairly broad, and even though it is optimized for 4.5 MHz operation, there is relatively high output at other intercarrier frequencies, as shown in the curve in Fig. 4.

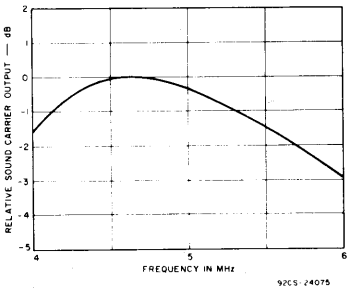
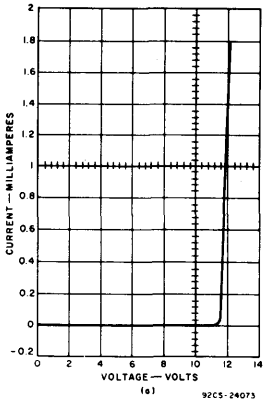


Fig. 4. Relative sound-carrier output as a function of frequency.

The internal zener reference-diode consists of the series diode arrangement shown connected between terminal 18 and the substrate in Fig. 3. A regulator-circuit configuration showing the

pass transistor interconnected with the reference diode is given as part of the color and monochrome if amplifier circuits that are discussed in the following paragraphs. Similarly, the regulation curves shown in Figs. 5 (a) and 5 (b) are discussed below in more detail. It should be noted that (with a heat sink for the 2N3053 and a lower value for the resistor in series with the collector) the regulated voltage from this supply may be used to provide power to other circuits in addition to the CA3068.

The distribution of tuned circuits around the CA3068 amplifier circuit is a matter of preference of the circuit designer. In general, a total of five tuned circuits will be required subsequent to the mixer for proper selectivity and bandpass shaping. In addition, at least one 47.25 MHz adjacent sound-channel and one 41.25 MHz sound-channel trap will be required. The systems to be discussed in this Note are designed to be driven from a single tuned circuit connected to the mixer output. In addition, both the color and monochrome if systems described subsequently utilize tuned circuits at the input and output to the cascode amplifier. The second transformer is used to couple



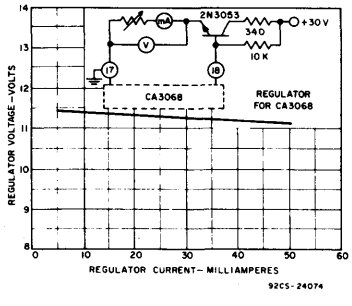


Fig. 5. Regulated supply of the CA3068: (a) voltage-versus-current for zener reference diode of CA3068; (b) voltage-versus-current for regulated supply of CA3068.

output from the cascode if amplifier to the wideband if amplifier (i.e., the output from terminal 9 to input terminal 13 for the PIX-channel and input terminal 12 for the sound-channel). All of the if transformers are synchronously tuned.

PIX–IF SYSTEM DESIGNS

The use of the CA3068 in the two major categories of PIX-IF application, PIX-IF for color-TV receivers and PIX-IF for monochrome receivers, is described below. To illustrate the use of the CA3068 in a tuner requiring "reverse" agc action, the rf-stage of the tuner employed in the PIX-IF for color-TV receiver contains a MOSFET. In contrast, the rf-stage of the tuner employed in the PIX-IF for the monochrome receiver makes use of a bipolar transistor in a "forward" agc arrangement.

COLOR TV

A block diagram of a color-TV receiver is shown in Fig. 1. In the design to be described, the input to the if system is intended to be coupled through a 50-ohm cable from the TV mixer; the mixer employs a single tuned-output coil having an impedance transformed down to 50 ohms. The if input circuit drives a cascode if amplifier with a gain capability of 35 dB. The input impedance to the cascode if amplifier is greater than 4000-ohms at minimum signal levels and increases with agc action. The source impedance as seen by the CA3068 should be approximately 500 ohms to dominate the input-circuit conductance node. Similarly, the output impedance of the cascode amplifier should be loaded by a tuned circuit with an impedance of approximately 3000 ohms to dominate the output node. The if amplifier stability is then unaffected by the IC impedance variations, but is a function of the feedback component. This feedback component consists of coupling within the IC packaging, PC-board stray capacitances, and PC-board common impedances. It can be shown that with the maximum device feedback capacitance the amplifier is stable. For example, with circuit bias conditions of I9 = 2 mA, Y21 = 50 mmhos, and C_{fb} (max) = 0.005pF, the maximum usable gain (MUG) is 42 dB (which allows for a 20 percent skew factor). The fact that this value of MUG is greater than the actual circuit gain (35 dB) substantiates the stability.

Although these calculations show the device to be stable, it must be recognized that poorly controlled external feedback mechanisms may raise the level of feedback in a high-gain, physically small rf amplifier so as to produce instabilities. For this reason, the PC-board layout is extremely important, and any high-gain if amplifier design should include a board layout.

As mentioned previously, the interstage transformer should load the cascode amplifier with approximately 3000 ohms, and should provide a 500-ohm source impedance to input terminal 13 (the wideband if amplifier section). The impedance at terminal 13 is approximately 5000 ohms. The driving-point impedance to sound-if terminal 12 should be about 1000 ohms, this terminal looks into a 5000-ohm input circuit. The 41.25 MHz trap is a rejection filter for the video amplifier and allows the carrier to pass into the sound system.

The circuit design in Fig. 6 shows a typical cable-link circuit which includes a 47.25 MHz bridged "T" adjacent-sound-channel trap at the input circuit. It also includes a 39.75-MHz trap for an adjacent video carrier for operation in CATV systems or in areas where adjacent channels are available. This trap may consist of a 39.75-MHz bridge "T" connected in parallel directly across the 47.25-MHz trap. Both traps provide the additional selectivity necessary for attenuation of the undesired frequencies by more that 40 dB.

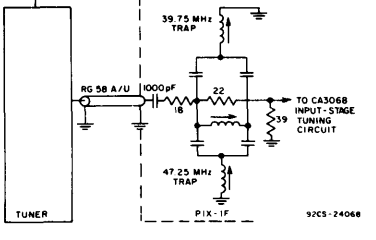


Fig. 6. Schematic diagram of a typical tuner-to-PIX-IF link circuit.

The second and third PIX-IF amplifier stages provide two extra stages of gain (approximately 40 dB). The stages present a very low driving-point impedance to the linear detector, as described earlier. The detected signal then undergoes an additional 12 dB of video amplification. The video output at terminal 19 is nominally 7 volts (peak-to-peak). AGC is developed when the input signal reaches and exceeds the magnitude necessary to produce this video output level. Fig. 7 (a) shows the developed agc bias (terminal-4 voltage) as a function of signal level at terminal 6. Fig. 7 (b) shows the delayed agc voltage at terminal 7 (for application to the tuner) with R1 adjusted so that this delay-bias is generated whenever the input signal at terminal 6 exceeds 8 millivolts.

An output signal is available at terminal 14 to drive an automatic-fine-tuning (AFT) subsystem-IC, such as the RCA CA3064. This connection is a buffered output from an emitter follower as described earlier. The level of signal at 45.75 MHz to drive the AFT circuit is nominally 15 millivolts.

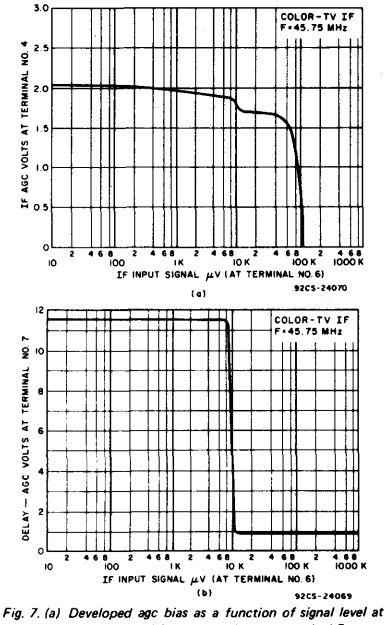


Fig. 7. (a) Developed agc bias as a function of signal level at terminal 7; (b) delayed agc voltage at terminal 7.

Fig. 8 shows the RCA CA3068 coupled to a tuner that uses an RCA type 40820 MOSFET in the rf-amplifier stage. AGC voltages are applied (shown in Fig. 8) to optimize over-all TV-receiver performance, so that, when maximum receiver sensitivity is required (such as during the reception of weak signals from the antenna) the tuner will operate at optimum noise factor and maximum gain. As the input signal level increases, it is still desirable to operate the rf stage at optimum signal-to-noise ratio until the signal level is of sufficient magnitude to override any tuner noise degradation brought about by the application of agc. Therefore, the gain-reduction voltage to the tuner should be delayed until the signal level builds up. Fig. 7 (b) shows that this agc is delayed until the if signal level reaches an 8-millivolt level. Then the tuner-gain-reduction mode is initiated. After the tuner gain reduction is expended, at least another 10-dB gain reduction is still available in the cascode portion of the if amplifier.

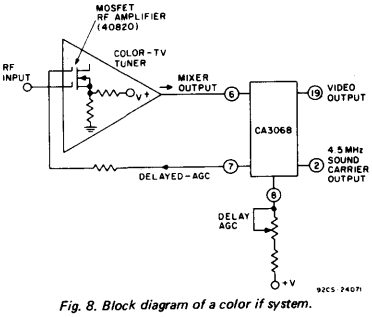


Fig. 8. Block diagram of a color if system.

As shown in Fig. 2, the agc system is, for the most part, self-contained. An optimized agc response characteristic can be achieved by use of a high-quality tantalum 10-microfarad capac-

itor connected between terminal 4 and ground. An RC decoupling network smooths the agc ripple associated with the charge and discharge of the 10-microfarad capacitor at the horizontal-oscillator frequency rate. The agc system is normally keyed from the horizontal-output circuit in the TV system. This keying pulse should be applied to terminal 3. The magnitude of the pulse should be sufficient to supply a nominal peak current value of 0.8 milliampere into terminal 3. The value of the series resistor R_s associated with terminal 3 may be computed as follows: During the conduction period (with keying applied), the constant-voltage components within the integrated circuit account for:

$V_k = 8.2$ V

(It is assumed that I3 = 0.8 mA)

If the keying-pulse magnitude, V_p, is 15 V, then:

$$I3 = 0.8 \text{ mA} = \frac{15 - V_k}{R_s} = \frac{(15 - 8.2)\text{ V}}{R_s}$$

$R_s = 8.5$ kilohms

The sound output is derived from terminal 2 at a level compatible with the input requirements of a TV-sound-if-subsystem IC, such as the RCA CA3065. There is also a dc component of approximately 6.7 volts present at terminal 2. Coupling networks to subsequent circuits must contain a suitable dc-blocking capacitor.

Small chokes located in the sound and video outputs (terminals 2 and 19) should be self-resonating at the intermediate frequencies to prevent if leakage into subsequent stages.

The CA3068 if subsystem has an internal zener reference-diode that permits operation of the subsystem with an external voltage-regulator pass transistor. A suggested circuit arrangement is shown as part of the over-all if schematic diagram in Fig. 5 (b). The voltage-regulator pass-transistor has a nominal output voltage of 11.2 volts. Bypassing of the V+ supply with reference to the if subsystem is important, and the suggested arrangement shown in the application circuit (Fig. 10) should be used. Specifically, terminal 15 should be bypassed to terminal 17 on the CA3068. Even though terminal 17 is at dc ground potential, it should not be tied to ground but rather should be bypassed in the manner shown to avoid mutual impedance coupling within the CA3068.

MONOCHROME TV

The delayed-agc circuits used in the CA3068 were originally intended to control a MOSFET in the rf-stage of the TV tuner. This arrangement permits direct application of the delayed-agc voltage from the CA3068 to the tuner. In monochrome receivers, however, it is common practice to employ a bipolar transistor in the rf-stage of the tuner, and a circuit with a "forward"-agc characteristic is required to control the rf-stage. This characteristic is easily established by means of an inverter network utilizing a p-n-p transistor, as shown in the circuit of Fig. 9.

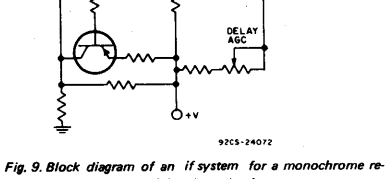


Fig. 9. Block diagram of an if system for a monochrome receiver showing peripheral agc circuit.

As the input signal level increases, the forward-agc delay voltage is developed at the tuner when the voltage at terminal 7 of the CA3068 decreases. The agc voltage applied to the rf-stage of the tuner (Fig. 9) is derived from the collector of the p-n-p transistor. As the delay-agc voltage is generated at terminal 7 of the CA3068, the base of the p-n-p inverter is driven into conduction, which causes more current to flow through the collector circuit, so that a positive (or forward) agc potential is generated for the bipolar transistor in the tuner.

TV RECEIVER PIX–IF CIRCUIT APPLICATIONS

In this section, the application of the CA3068 integrated circuit in a color and a monochrome TV receiver is described. The circuits shown were constructed on single-sided copper PC boards.

As previously noted, because of the high gain encountered in PIX-IF design, positive feedback must be avoided if the amplifier is to remain free of spurious oscillation. To this end, the optimization of printed board layout and component placement is essential. The proper choice of bypassing components and signal-path layout is necessary to avoid feedback through ground loops.

IF CIRCUIT FOR COLOR TV RECEIVER

The schematic diagram of an if system for a color-TV receiver is shown in Fig. 10. A parts list and illustrations showing the PC-board component layout (top view) and the actual printed circuit (bottom view of board) are shown in Appendix A. Since most current color-TV receivers employ automatic-fine-tuning (AFT) systems, an AFT system using the CA3064 has also been included on the same board; Fig. 10 includes the AFT circuit.

The if-response is determined by the triple-tuned circuit, which consists of three traps: two preceding the IC and an interstage double-tuner circuit with one trap. In the tripled-tuned circuit, the two bridge-T traps are used to provide attenuation of the adjacent-channel picture carrier (frequency 39.75 MHz) and adjacent-channel sound carrier frequency (47.25 MHz). A

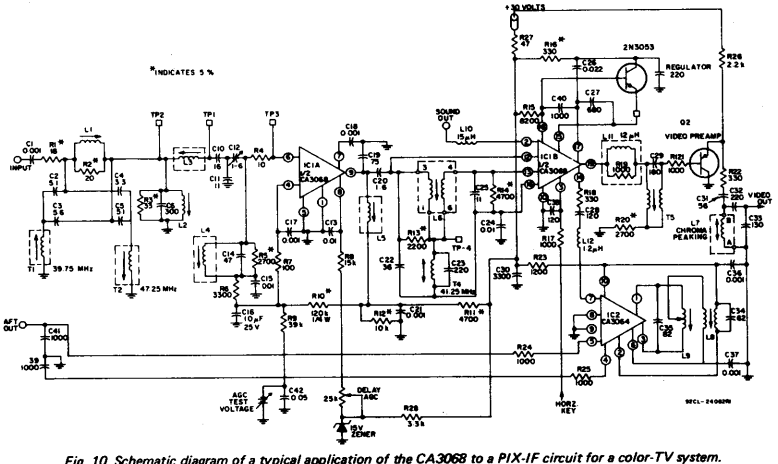


Fig. 10. Schematic diagram of a typical application of the CA3068 to a PIX-IF circuit for a color-TV system. A template of the printed circuit board used to construct this circuit, a diagram of the position of all components on the board, a block diagram of the location of major components on the board, and a circuit parts list are given in Appendix A.

common bridge impedance consisting of parallel-connected L1 and R2 is used. Adjustment of L1 for best null of the 47.25 MHz trap assures the desired 60-dB minimum attenuation.

The triple-tuned circuit provides, at center frequency, a source resistance to the IC of 800 ohms and a voltage gain of three from the input to pin 6 of the IC. The first section of the triple-tuned circuit consists of L2 and C6. Capacitor C6 is in parallel resonance with coil L2 at 44 MHz. The third section of the triple-tuned circuit consists of coil L4 and capacitor C14. Coupling and voltage-gain from L2 to L4 are provided by the second section, coil L3 and capacitors C10, C11, and C12. The inductive reactance of L3 is made 75 times larger than that of L2 to provide a high degree of tuned-circuit isolation for ease of alignment.

The circuit provides protection against interference resulting from a strong rf signal which might inadvertently be introduced between the tuner and the if stage. Parasitic resonance and couplings have been minimized to maintain a high degree of attenuation at frequencies remote from the if-resonance frequency.

The interstage double-tuned bandpass circuit, with a bifilar T-trap at 41.25 MHz, is similar to that commonly used in the third stage of color-TV receivers. The sound and picture carriers are present at the input (terminal 12) to the 4.5 MHz sound-if detector circuit. Trapping action removes the 41.25 MHz sound carrier at terminal 13 to prevent a difference-frequency beat of 0.92 MHz with the chroma subcarrier at 42.17 MHz. The picture carrier and chroma subcarrier entering terminal 13 are amplified, detected, and additionally amplified as detected video signal. If the sound carrier is not attenuated by the 41.25 MHz trap, the carrier will be detected as a large 4.5 MHz difference-signal in the video output. A 4.5 MHz trap (T5) is included to prevent interference of a residual 4.5 MHz intercarrier signal in the chroma and luminance circuits.

The chroma peaking circuit compensates for the slope of the video response, as shown in Figs. 11 (a), 11 (b) and 11 (c). The actual slope and shape of the video response between 3.08 MHz and 4.08 MHz will vary because of normal component tolerance. The chroma-peaking coil, L7, has two cores, one to adjust inductance to center the response at 3.58 MHz, and the other to adjust chroma output level and bandwidth. The latter core controls circuit Q with little effect on over-all inductance.

Photographs of the detected sweep-response characteristics are shown in Fig. 12. The sweep-response of Fig. 12 (f) shows the interstage alignment from TP3 (of Fig. 10) to terminal 9 of the CA3068. The sweep-response curves in Figs. 12 (a) through 12 (e) show 60 dB of agc range from a level of 100 microvolts (Fig. 12 (e)) to 100 millivolts (Fig. 12 (a)).

The alignment procedure for the color-TV PIX-IF system using the CA3068, Fig. 10, is given in Appendix A.

IF CIRCUIT FOR MONOCHROME TV RECEIVER

The schematic diagram for a PIX-IF system for a monochrome TV system that employs the CA3068 is shown in Fig. 13. A PC-board component-layout diagram (top view), the actual printed circuit (bottom view of board), and a circuit parts list are shown in Appendix B. A sound-if system using the CA3065 has been included to show the simplicity with which it can be used in conjunction with the CA3068.

The selectivity is provided in two sections, an input single-tuned circuit with trap, and a double-tuned interstage circuit. The resistive pad, R1, R2, and R3 of Fig. 13, is used to terminate the link-cable and isolate cable effects from the high-Q input circuit. The bridge-T trap-circuit is used to give maximum attenuation to the adjacent-channel sound carrier. Precision components (R2, C1, C2) achieve a good null at 47.25 MHz

without the need for additional components. The circuit Q is controlled by R11 and the resistive input network to yield a 3-dB bandwidth of 3 MHz centered at 44.5 MHz. The "T"-equivalent circuit is used for interstage coupling to realize a miniature, precision, double-tuned transformer. The mutual coupling element, L5, is an air-core, spring-winding coil which is actually calibrated by physical dimensions. If necessary, this coil may be "knifed" to provide a simple and effective coupling adjustment. The circuit Q's are each set at 21, and are controlled by R17 and R18, which also feed bias for the broadband amplifier and sound channels, respectively. The picture-carrier at 45.75 MHz is set at 50 percent to yield proper reception of the vestigial sideband. The color subcarrier at 42.17 MHz is placed comparatively low on the response curve, since the resulting beat with the 41.25 MHz is placed at greater than 5 percent but less than 10 percent to produce an adequate sound-if intercarrier signal at 4.5 MHz, and yet maintain low intermodulation. Typical over-all sensitivity of the if circuit is approximately 150 microvolts for full video output.

Interference from the 45-MHz high-level signals and harmonics is prevented by care in passing and filtering. A 12-microhenry choke (L4 of Fig. 13), self-resonant at the fourth harmonic, is used in the video output lead; the sound output contains a ferrite bead. The B+ supply must be bypassed to provide a low-impedance source for the video driver stages and to provide high-frequency filtering. The 1-microhenry choke (L3 of Fig. 13) is made very lossy to prevent resonance with C8. The ferrite bead and C9 provide high-frequency filtering for harmonics of the 45-MHz signal.

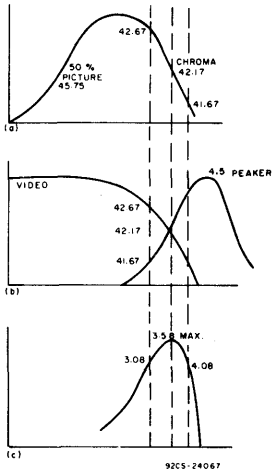


Fig. 11. (a) Over-all if response, (b) video and peaking circuit response, (c) chroma response for the circuit of Fig. 10 (frequency values in MHz).

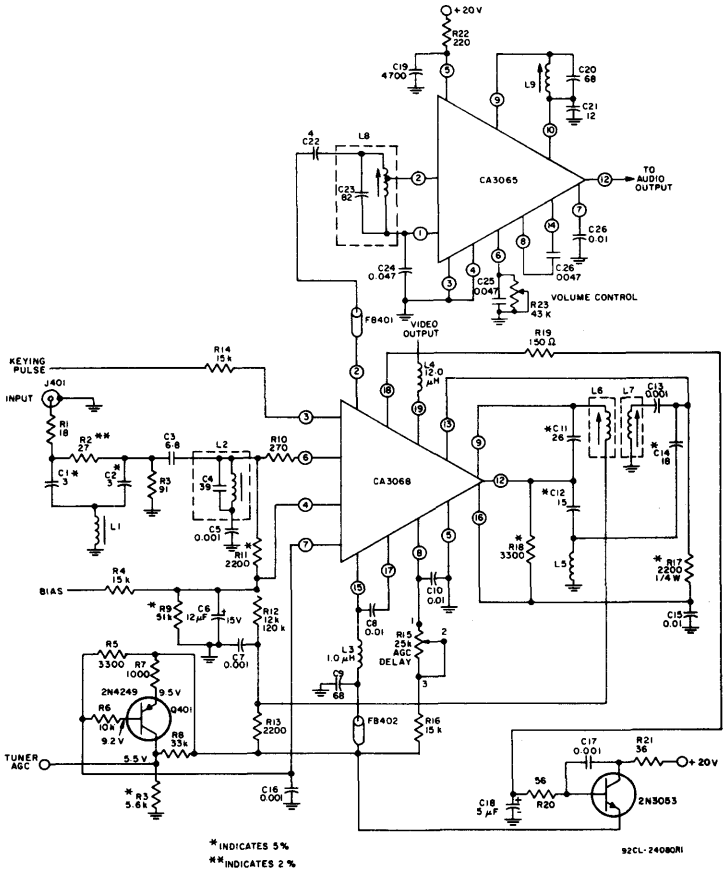


Fig. 13. Schematic diagram of a typical application of the CA3068 to a PIX-IF circuit for a monochrome-TV system. A template of the printed-circuit board used to construct this circuit, a diagram of the position of all components on the board, and a circuit parts list are given in Appendix B.

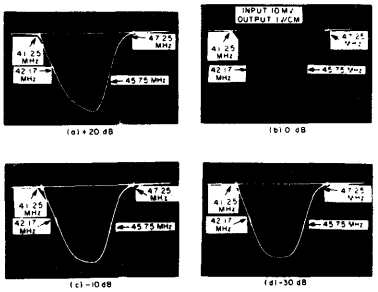


Fig. 12. Detected sweep-response characteristics for the circuit of Fig. 10.

Typical sweep-response characteristics are shown in Fig. 14. The alignment instructions for the monochrome, PIX-IF circuit are given in Appendix B.

SUMMARY

A complete if subsystem has been described for use in both color and monochrome TV receivers. The only signal inputs required by the CA3068 are if signals from the tuner and a keying pulse from the horizontal circuitry. The CA3068 provides all outputs needed to drive the video output stage, delay line, sync-separator circuitry, RCA CA3065 sound if subsystem, RCA CA3064 AFT subsystem, and delayed-age voltage for the rf stage in the tuner. Additionally, circuits for noise immunity and signal overload protection are an integral part of the CA3068 design. These subsystems have typical input sensitivities of 100 microvolts for a 4-volt, peak-to-peak video output. A unique video detector arrangement provides extremely linear output throughout the 7-volt, peak-to-peak, video-output range of the system.

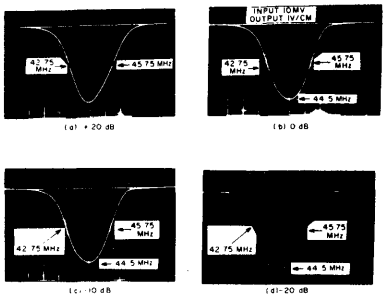


Fig. 14. Typical sweep-response characteristics for the circuit of Fig. 13.

Although this Application Note describes subsystem designs in TV receivers, the CA3068 is also applicable in AM communications systems requiring performance at frequencies within the range of 10 to 70 MHz.

REFERENCES

1. RCA Data Bulletin File No. 396 concerning the CA3064 and CA3064E, "TV Automatic Fine Tuning Circuit", or the RCA DATABOOK, 1975 Series SSD-201C.
2. RCA Data Bulletin File No. 412 concerning the CA3065, "TV IF Sound System", or the RCA DATABOOK, 1975 Series SSD-201C.
3. RCA Data Bulletin File No. 467 concerning the CA3068, "Television Video IF System", or the RCA DATABOOK, 1975 Series SSD201C.

APPENDIX A – THE COLOR CIRCUIT

ALIGNMENT PROCEDURE FOR THE COLOR CIRCUIT

Preliminary Adjustments and Calibration

1. Adjust delay-agc (noise pot) fully cw.
2. Connect supplies as indicated on schematic diagram (Fig. 10), set bias to zero.
3. Set sweep generator to 10 millivolts as indicated on Boonton 91DA meter with 56-ohm termination.

Step 1 - IF Interstage Alignment

a. Ground TP1 with short clip lead.
b. Connect sweep generator with 56-ohm termination and 1000-picofarad decoupling capacitor to TP3.
c. Connect oscilloscope to video output.
d. Adjust bias for 5-volt peak-to-peak response on oscilloscope.
e. Adjust bottom core of T4 for minimum at 41.25 MHz.
f. Adjust L5 and L6 for symmetrical response with PIX and color markers equal (Fig. 12 (a)): L5 controls markers and L6 controls tilt.
g. Adjust top and bottom cores of T4 simultaneously, top core for maximum rejection of 41.25 MHz and bottom core to maintain minimum 41.25 MHz.

Step 2 - IF Overall Alignment

a. Leave ground clip lead on TP1.
b. Remove sweep input from TP3.
c. Connect TP2 through a 1000-picofarad capacitor to TP3.
d. Connect sweep generator to input.
e. Readjust variable bias to maintain 5-volts peak-to-peak response on oscilloscope.
f. Adjust T1 for minimum 39.75 MHz.
g. Adjust T2 for minimum 47.25 MHz.
h. Adjust L2 for equal height of PIX and color markers.
i. Remove ground-clip lead from TP1 and 1000-picofarad capacitor from between TP2 and TP3.
j. Maintain 5-volts peak-to-peak response on oscilloscope by re-adjusting bias.
k. Adjust L3 and L4 simultaneously for symmetrical response with PIX and color markers equal: L4 controls markers and L3 controls tilt.
l. Adjust bandpass trimmer, C12, to place PIX and color markers at 40 percent while readjusting L3 and L4 (Fig. 12 (b)).
m. Re-adjust T1 for minimum at 39.75 MHz if necessary.
n. Re-adjust T2 for minimum at 49.25 MHz. Then adjust L2 to maximize the rejection at 47.25 MHz.

AFT Alignment

a. With oscilloscope on AFT output, adjust bias for 10-volts peak-to-peak response.
b. Adjust L8 for maximum 45.75 MHz.
c. Adjust L9 for crossover at 45.75 MHz.
d. Re-adjust L8 and L9 to obtain symmetry.
e. Adjust L8 to obtain maximum width.

Color-Circuit Parts List

Capacitors		Resistors (All values in ohms)	
C1	0.001μF	R1	18
C2	5.1pF	R2	20
C3	5.6pF	R3	33
C4	3.3pF	R4	10
C5	5.1pF	R5	2.7k
C6	300pF	R6	3.3k
C10	16pF	R7	100
C11	11pF	R8	15k
C12	1-6pF	R9	39k
C13	0.01μF	R10	120k
C14	47pF	R11	4.7k
C15	0.01μF	R12	10k
C16	10μF	R13	2.2k
C17	0.001μF	R14	4.7k
C18	0.001μF	R15	8.2k
C19	7.5pF	R16	330
C20	1.6pF	R17	1k
C21	0.001μF	R18	330
C22	3.6pF	R19	1k
C23	220pF	R20	2.7k
C24	0.01μF	R21	1k
C25	11pF	R22	330
C26	0.022μF	R23	1.2k
C27	680pF	R24	1k
C28	120pF	R25	1k
C29	180pF	R26	2.2k
C30	0.022μF	R27	47
C31	56pF	R28	3.3k
C32	220pF	R29	25k
C33	130pF		
C34	62pF		
C35	82pF		
C36	0.001μF		
C40	1000pF		
C41	1000pF		
C42	1000pF		

Inductors	RCA Stock No.
L1	132159
L2	132161
L3	132839
L4	132658
L5	137126
L6	132146
T1	132839
T2	132157
T4	132150
T5	132135

APPENDIX B – THE MONOCHROME CIRCUIT

ALIGNMENT PROCEDURE FOR THE MONOCHROME-CIRCUIT

Step 1 -

1. Connect +20 volts to appropriate points on board.
2. Connect sweep generator to input
3. Connect dc bias voltage to appropriate point on board.
4. Adjust sweep generator for 10-millivolt input.
5. Adjust bias voltage for 5-volt, peak-to-peak output.

Step 2 -

1. Adjust LT for minimum response at 47.25 MHz.
2. Adjust L2 for maximum at 44.5 MHz.
3. Adjust L6, L7 for bandpass shown in Fig. 14 (b). The curve should have 3-MHz bandwidth centered at 44.5 MHz.

Monochrome-Circuit Parts List

Capacitors	
C1	3.0pF
C2	3.0pF
C3	6.8pF
C4	3.9pF
C5	0.001μF
C6	12μF
C7	0.001μF
C8	0.001μF
C9	6.8pF
C10	0.01μF
C11	20pF
C12	15pF
C13	0.001μF
C14	18pF
C15	0.01μF
C16	0.001μF
C17	0.001μF
C18	5μF
C19	4700pF
C20	68pF
C21	12pF
C22	4pF
C23	82pF
C24	0.047μF
C25	0.047μF
C26	0.01μF
C27	0.047μF

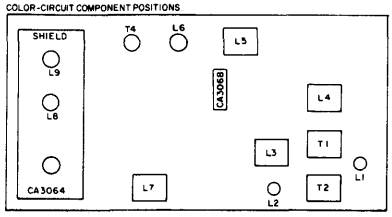
Inductors	RCA Stock No.
L1	131655
L2	133463
L3	1.0μH
L4	12.0μH
L5	134754*
L6	131465
L7	133546
L8	130120
L9	130121

*(9 turns No. 23 wire; use 1/2 W resistor to form coil)

Resistors (All values in ohms)

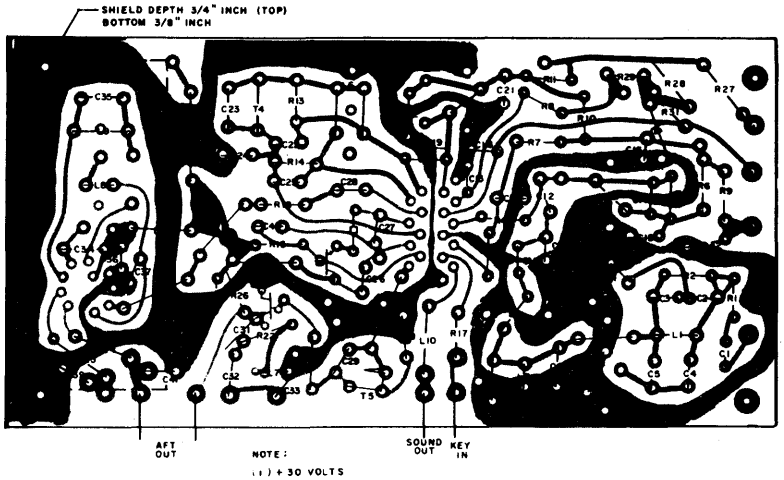
R1	18
R2	27
R3	91
R4	15k
R5	3.3k
R6	10k
R7	1.0k
R8	33k
R9	51k
R10	270
R11	2.2k
R12	120k
R13	2.2k
R14	15k
R15	25k
R16	8.2k
R17	2.2k
R18	3.3k
R19	150
R20	56
R21	36
R22	220
R23	5.6k

COLOR-CIRCUIT COMPONENT POSITIONS



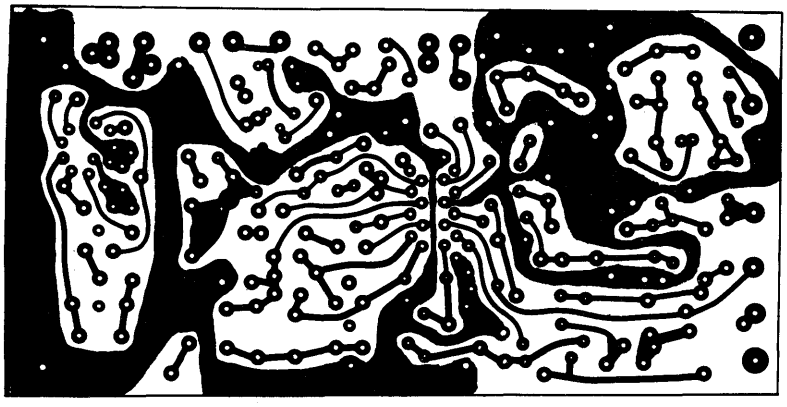
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THE COLOR CIRCUIT

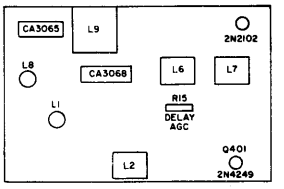


NOTE:
(1) + 30 VOLTS
(2) TUNER AGC
(3) CONTROL AGC
(4) GND
(5) INPUT

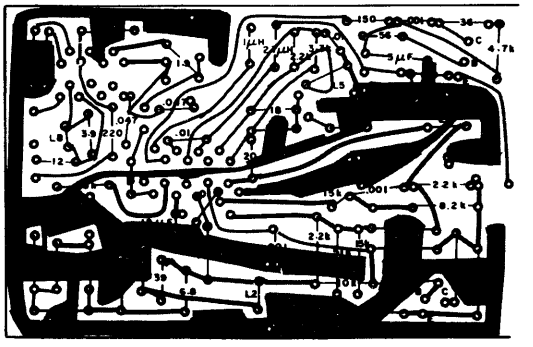
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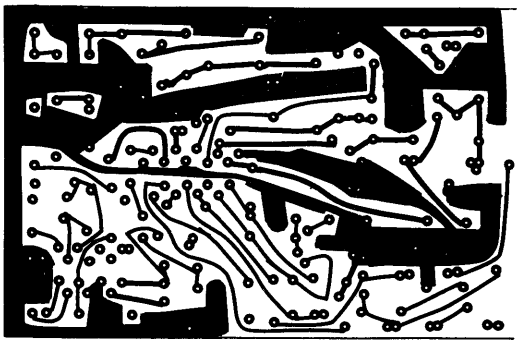


THE MONOCHROME CIRCUIT



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ICAN-6668

Applications of the CA3080 and CA3080A High-Performance Operational Transconductance Amplifiers

by H. A. Wittlinger

The CA3080 and CA3080A are similar in generic form to conventional operational amplifiers, but differ sufficiently to justify an explanation of their unique characteristics. This new class of operational amplifier not only includes the usual differential input terminals, but also contains an additional control terminal which enhances the device's flexibility for use in a broad spectrum of applications. The amplifier incorporated in these devices is referred to as an Operational Transconductance Amplifier (OTA), because its output signal is best described in terms of the output-current that it can supply. (Transconductance $g_m = \frac{\Delta i_{out}}{\Delta e_{in}}$). The amplifier's output-current is proportional to the voltage difference at its differential input terminals.

This Note describes the operation of the OTA and features various circuits using the OTA. For example, communications and industrial applications including modulators, multiplexers, sample-and-hold-circuits, gain control circuits and micropower comparators are shown and discussed. In addition, circuits have been included to show the operation of the OTA being used in conjunction with RCA COS/MOS devices as post-amplifiers.

Fig. 1 shows the equivalent circuit for the OTA. The output signal is a "current" which is proportional to the transconductance (gm) of the OTA established by the amplifier bias current (I_{ABC}) and the differential input voltage. The OTA can either source or sink current at the output terminals, depending on the polarity of the input signal.

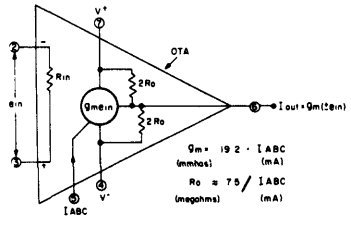


Fig. 1— Basic equivalent circuit of the OTA.

The availability of the amplifier bias current (I_{ABC}) terminal significantly increases the flexibility of the OTA and permits the circuit designer to exercise his creativity in the utilization of this device in many unique applications not possible with the conventional operational amplifier.

Circuit Description

A simplified block diagram of the OTA is shown in Fig. 2. Transistors Q1 and Q2 comprise the differential input amplifier found in most operational amplifiers, while the lettered-circles (with arrows leading either into or out of the circles) denote "current-mirrors". Fig. 3a shows the basic type of current-mirror which is comprised of two transistors,

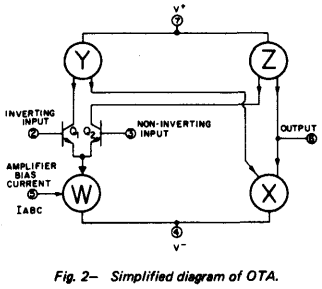


Fig. 2— Simplified diagram of OTA.

one of which is diode-connected. In a "current-mirror", with similar geometries for Q_A and Q_B, the current I' establishes a second current I whose value is essentially equal to that of I'. This basic current-mirror configuration is sensitive to the transistor beta (β). The addition of another active transistor, shown in Fig. 3b, greatly diminishes the circuit sensitivity to transistor beta (β) and increases the current-source output impedance in direct proportion to the transistor beta (β). Current-mirror W (Fig. 2) uses the configuration shown in Fig. 3a, while mirrors X, Y, and Z are basically the version shown in Fig. 3b. Mirrors Y and Z employ p-n-p transistors, as depicted by the arrows pointing outward from the mirrors. Appendix I describes "current-mirrors" in more detail.

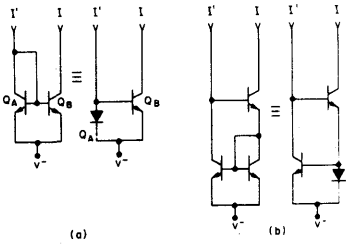


Fig. 3— Basic types of current mirrors; a) diode-connected transistor paired with transistor; b) improved version: employs an extra transistor.

Fig. 4 is the complete schematic diagram of the OTA. The OTA employs only active devices (transistors and diodes). Current applied to the amplifier-bias-current terminal, I_{ABC}, establishes the emitter current of the input differential amplifier Q1 and Q2. Hence, effective control of the differential transconductance (g_m) is achieved.

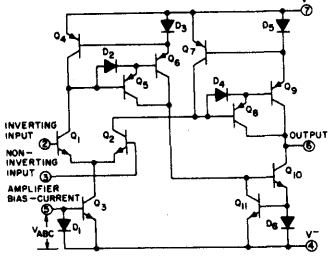


Fig. 4— Schematic diagram of OTA types CA3080 and CA3080A.

The g_m of a differential amplifier is equal to

$$\frac{qaI_C}{2KT}$$

(see Ref. 2 for derivation) where q is the charge on an electron, a is the ratio of collector current to emitter current of the differential amplifier transistors, (assumed to be 0.99 in this case), I_C is the collector current of the constant-current source (I_{ABC} in this case), K is Boltzman's constant, and T is the ambient temperature in degrees Kelvin. At room temperature, $g_m = 19.2 \times I_{ABC}$, where gm is in mmho and I_{ABC} is in milliamperes. The temperature coefficient of g_m is approximately -0.33%/°C (at room temperature).

Transistor Q3 and diode D1 (shown in Fig. 4) comprise the current mirror "W" of Fig. 2. Similarly, transistors Q7, Q8 and Q9 and diode D5 of Fig. 4 comprise the generic current mirror "Z" of Fig. 2. Darlington-connected transistors are employed in mirrors "Y" and "Z" to reduce the voltage sensitivity of the mirror, by the increase of the mirror output impedance. Transistors Q10, Q11, and diode D6 of Fig. 2 comprise the current-mirror "X" of Fig. 2. Diodes D2 and D4 are connected across the base-emitter junctions of Q5 and Q8, respectively, to improve the circuit speed. The amplifier output signal is derived from the collectors of the "Z" and "X" current-mirror of Fig. 2, providing a push-pull Class A output stage that produces full differential g_m. This circuit description applies to both the CA3080 and CA3080A. The CA3080A offers tighter control of g_m and input offset voltage, less variation of input offset voltage with variation of I_{ABC} and controlled cut-off leakage current. In the CA3080A, both the output and the input cut-off leakage resistances are greater than 1,000 MΩ.

APPLICATIONS

Multiplexing

The availability of the bias current terminal, I_{ABC}, allows the device to be gated for multiplex applications. Fig. 5 shows a simple two-channel multiplex system using two CA3080 OTA devices. The maximum level-shift from input to output is low (approximately 2mV for the CA3080A and 5mV for the CA3080). This shift is determined by the amplifier input offset voltage of the particular device used, because the open-loop gain of the system is typically 100dB when the loading on the output of the CA3080A is low. To further increase the gain and reduce the effects of loading, an additional buffer and/or gain-stage may be added. Methods will be shown to successfully perform these functions.

In this example positive and negative 5-V power-supplies were used, with the IC flip-flop powered by the positive supply. The negative supply-voltage may be increased to -15 V, with the positive-supply at 5 V to satisfy the logic supply voltage requirements. Outputs from the clocked flip-flop are applied through p-n-p transistors to gate the CA3080 amplifier-bias-current terminals. The grounded-base con-

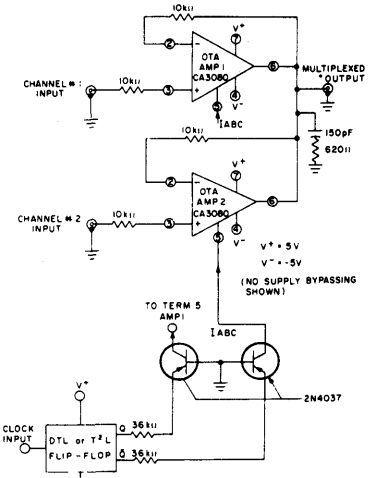


Fig. 5— Schematic diagram of OTAs in a two-channel linear time-shared multiplex circuit.

figuration is used to minimize capacitive feed-through coupling via the base-collector junction of the p-n-p transistor.

Another multiplex system using the OTA's clocked by a COS/MOS flip-flop is shown in Fig. 6. The high output voltage capability of the COS/MOS flip-flop permits the circuit to be driven directly without the need for p-n-p level-shifting transistors.

A simple RC phase-compensation network is used on the output of the OTA in the circuits shown in Figs. 5 & 6. The values of the RC-network are chosen so that $\frac{1}{2\pi RC} \cong 2MHz$.

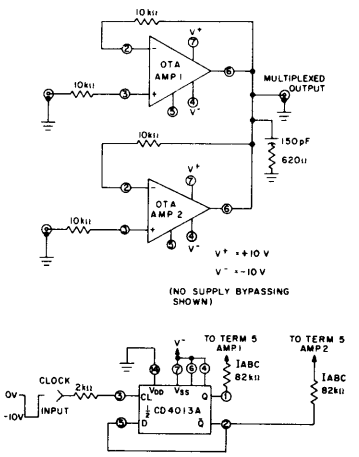


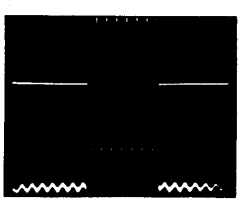
Fig. 6– Schematic diagram of a two-channel linear multiplex system using a COS/MOS flip-flop to gate two OTAs.

This RC-network is connected to the point shown because the lowest-frequency pole for the system is usually found at this point. Fig. 7 shows an oscilloscope photograph of the multiplex circuit functioning with two input signals. Fig. 8



TOP TRACE: MULTIPLEXED OUTPUT 1V/DIV & 100μsec/DIV
BOTTOM TRACE: TIME EXPANSION OF SWITCHING BETWEEN INPUTS 2V/DIV & 5 μsec/DIV

Fig. 7– Voltage waveforms for circuit of Fig. 6; top trace: multiplexed output; lower trace: time expansion of switching between inputs.



TOP TRACE: 1V/DIV & 100μsec/DIV – OUTPUT
BOTTOM TRACE: VOLTAGE EXPANSION OF OUTPUT 1mV/DIV & 100μsec/DIV ISOLATION IS IN EXCESS OF 80 db

Fig. 8– Voltage waveforms for circuit of Fig. 6; top trace: output; lower trace: voltage expansion of output; isolation in excess of 80 dB.

shows an oscilloscope photograph of the output of the multiplexer with a 6-V p-p, sine wave signal (22 kHz) applied to one amplifier and the input to the other amplifier grounded. This photograph demonstrates an isolation of at least 80 dB between channels.

Sample-and-Hold Circuits

An extension of the multiplex system application is a sample-and-hold circuit (Fig. 9), using the strobing characteristics of the OTA amplifier bias-current (ABC) terminal as a means of control. Fig. 9 shows the basic system using the CA3080A as an OTA in a simple voltage-follower configuration with the phase-compensation capacitor serving the additional function of sampled-signal storage. The major consideration for the use of this method to "hold" charge is that neither the charging amplifier nor the signal readout device significantly alter the charge stored on the capacitor. The CA3080A is a particularly suitable capacitor-charging amplifier because its output resistance is more than 1000 MΩ under cut-off conditions, and the loading on the storage capacitor during the hold-mode is minimized. An effective solution to the read-out requirement involves the use of an RCA 3N138 insulated-gate field-effect transistor (MOS/FET) in the feedback loop. This transistor has a maximum gate-leakage current of 10 picoamperes; its loading on the charge "holding" capacitor is negligible. The open-loop voltage-gain of the system (Fig. 9) is approximately 100 dB if the MOS/FET is used in the source-follower mode to the CA3080A as the input amplifier. The open-loop output impedance $(\frac{1}{g_m})$ of the 3N138 is approximately 220 Ω because its transconductance is about 4,600 μmho at an operating current of 5 mA. When the CA3080A drives the 3N138 (Fig. 9), the closed loop operational-amplifier output impedance characteristic

$$Z_{out} \cong \frac{Z_o \text{ (open-loop)}}{A \text{ (open-loop voltage-gain)}}$$

$$\cong \frac{220\ \Omega}{100dB} \cong \frac{220\ \Omega}{10^5} \cong 0.0022\ \Omega$$

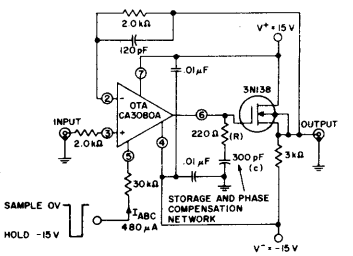
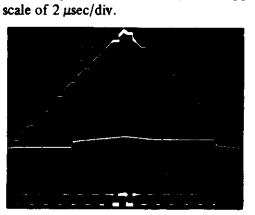


Fig. 9– Schematic diagram of OTA in a sample-and-hold circuit.

Fig. 10 shows a "sampled" triangular signal. The lower trace in the photograph is the sampling signal. When this signal goes negative, the CA3080A is cutoff and the signal is "held" on the storage capacitor, as shown by the plateaus on the triangular wave-form. The center trace is a time expansion of the top-most transition (in the upper trace) with a time scale of 2 μsec/div.



TOP TRACE: SAMPLED SIGNAL 1V/DIV & 20μsec/DIV
CENTER TRACE: TOP PORTION OF UPPER SIGNAL 1V/DIV & 2μsec/DIV
BOTTOM TRACE: SAMPLING SIGNAL 20V/DIV & 20μsec/DIV

Fig. 10– Waveforms for circuit of Fig. 9; top trace: sampled signal; center trace: top portion of upper signal; lower trace: sampling signal.

Once the signal is acquired, variation in the stored-signal level during the hold-period is of concern. This variation is primarily a function of the cutoff leakage current of the CA3080A (a maximum limit of 5 ηA), the leakage of the storage element, and other extraneous paths. These leakage currents may be either "positive" or "negative" and, consequently, the stored-signal may rise or fall during the "hold" interval. The term "tilt" is used to describe this condition. Fig. 11 shows the expected pulse "tilt" in microvolts as a function of time for various values of the compensation/storage capacitor. The horizontal axis shows three scales representing leakage currents of 50 ηA, 5 ηA, 500 pA.

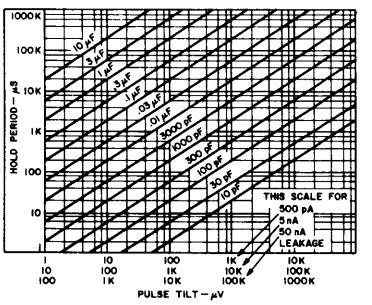
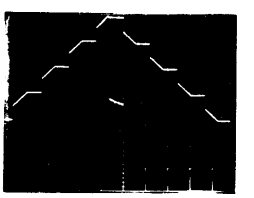


Fig. 11– Chart showing "tilt" in sample-and-hold potentials as a function of hold time with load capacitance as a parameter.

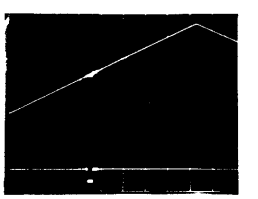
Fig. 12 shows a dual-trace photograph of a triangular signal being "sampled-and-held" for approximately 14 ms with a 300 pF storage capacitor. The center trace (expanded to 20 mV/div) shows the worst-case "tilt" for all the steps shown in the upper trace. The total equivalent leakage current in this case is only 170 pA $(I = C\frac{dv}{dt})$.



TOP TRACE SAMPLED SIGNAL 1V/DIV & 20msec/DIV
CENTER TRACE WORSE CASE TILT 20mV/DIV & 20msec/DIV

Fig. 12– Oscilloscope photo of "triangular-voltage" being sampled by circuit of Fig. 9.

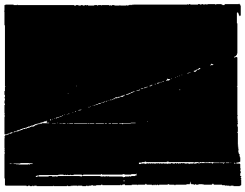
Fig. 13 is an oscilloscope photograph of a ramp voltage being sampled by the "sample-and-hold" circuit of Fig. 9. The input signal and sampled-output signal are superimposed. The lower trace shows the sampling signal. Data shown in Fig. 13 were recorded with supply voltages of ±10 V and the series input resistor at terminal 5 was 22 kΩ.



TOP TRACE: INPUT AND OUTPUT SUPERIMPOSED 1V/DIV & 2μsec/DIV
BOTTOM TRACE: SAMPLING SIGNAL 20V/DIV & 2μsec/DIV

Fig. 13– Oscilloscope photo of "ramp-voltage" being sampled by circuit of Fig. 9.

In Fig. 14, the trace of Fig. 13 has been expanded (100 mV/div and 100 ηsec/div) to show the response of the sample-and-hold circuit with respect to the sampling signal. After the sampling interval, the amplifier overshoots the signal level and settles (within the amplifier offset voltage) in approximately 1 μs. The resistor in series with the 300 pF phase-compensation capacitor was adjusted to 68 ohms for minimum recovery time.



TOP TRACE INPUT AND SAMPLED OUTPUT SUPERIMPOSED 100 mV/DIV & 100 ns/DIV
BOTTOM TRACE SAMPLING SIGNAL 20 V/DIV & 100 ns/DIV

Fig. 14– Oscilloscope photo showing response of sample-and-hold circuit (Fig. 9).

Fig. 15 shows the basic circuit of Fig. 9 implemented with an RCA 2N4037 p-n-p transistor to minimize capacitive feedthrough. Fig. 16 shows oscilloscope photographs taken with the circuit of Fig. 15 operating in the sampling mode at supply-voltage of ±15 V. The 9.1 kΩ resistor in series with the p-n-p transistor emitter establishes amplifier-bias-current (I_{ABC}) conditions similar to those used in the circuit of Fig. 9.

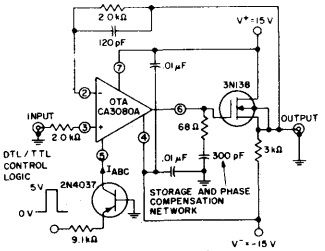
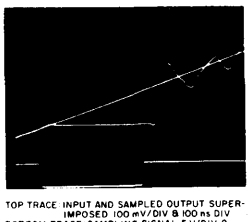


Fig. 15– Schematic diagram of the OTA in a sample-and-hold configuration (DTL/TTL control logic).

Considerations of circuit stability and signal retention require the use of the largest possible phase-compensation capacitor, compatible with the required slew rate. In most systems the capacitor is chosen for the maximum allowable "tilt" in the storage mode and the resistor is chosen so that $\frac{1}{2\pi RC} \cong 2$MHz, corresponding to the first pole in the amplifier at an output current level of 500 μA. It is frequently desirable to optimize the system response by the placement of a small variable resistor in series with the capacitor, as is shown in Figs. 9 and 15. The 120 pF capacitor shunting the 2 kΩ resistor improves the amplifier transient response.

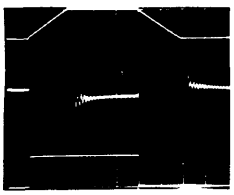


TOP TRACE INPUT AND SAMPLED OUTPUT SUPERIMPOSED 100 mV/DIV & 100 ns DIV
BOTTOM TRACE SAMPLING SIGNAL 5 V/DIV & 100 ns/DIV

Fig. 16– Oscilloscope photo for circuit of Fig. 15 operating in sampling mode.

Fig. 17 shows a multi-trace oscilloscope photograph of input and output signals for the circuit of Fig. 9, operating in the linear mode. The lower portion of the photograph shows the input signal, and the upper portion shows the output signal. The amplifier slew-rate is determined by the output current and the capacitive loading: in this case the slew rate (dV/dt) = 1.8V/μs.

The center trace in Fig. 17 shows the difference between the input and output signals as displayed on a Tektronix 7A13 differential amplifier at 2 mV/div. The output of the amplifier system settles to within 2 mV (the offset voltage specification for the CA3080A) of the input level in 1 μs after slewing.



TOP TRACE OUTPUT 5 V/DIV & 2 μsec/DIV
CENTER TRACE DIFFERENTIAL COMPARSION OF INPUT AND OUTPUT 2 mV/DIV— 0 VOLTS THROUGH CENTER— 2 μsec/DIV
BOTTOM TRACE INPUT 5 V/DIV & 2 μsec/DIV

Fig. 17– Oscilloscope photo showing circuit of Fig. 9 operating in the linear sample-mode.

Fig. 18 is a curve of slew-rate as a function of amplifier-bias-current (I_{ABC}) with various storage/compensation capacitors. The magnitude of the current being supplied to the storage/compensation capacitor is equal to the amplifier-bias-current (I_{ABC}) when the OTA is supplying its maximum output current.

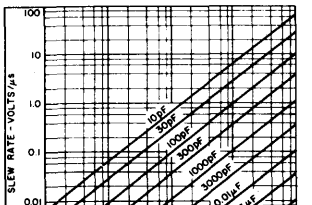


Fig. 18– Slew rate as a function of amplifier-bias-current (I_{ABC}) with phase-compensation capacitance as a parameter.

Gain Control – Amplitude Modulation

Effective gain control of a signal may be obtained by controlled variation of the amplifier-bias-current (I_{ABC}) in the OTA because its g_m is directly proportional to the amplifier-bias-current (I_{ABC}). For a specified value of amplifier-bias-current, the output current (I_O) is equal to the product of g_m and the input signal magnitude. The output voltage swing is the product of output current (I_O) and the load resistance (R_L).

Fig. 19 shows the configuration for this form of basic gain control (a modulation system). The output signal current (I_O) is equal to $-g_m V_x$; the sign of the output signal is negative because the input signal is applied to the inverting input terminal of the OTA. The transconductance of the OTA is controlled by adjustment of the amplifier bias current, I_{ABC}. In this circuit the level of the unmodulated carrier output is established by a particular amplifier-bias-current (I_{ABC}) through resistor Rm. Amplitude modulation of the carrier frequency occurs because variation of the voltage Vm forces a change in the amplifier-bias-current (I_{ABC}) supplied via resistor Rm. When Vm goes positive the bias current increases which causes a corresponding increase in the g_m of the OTA. When the Vm goes in the negative direction (toward the amplifier-bias-current terminal potential), the amplifier-bias-current decreases, and reduces the g_m of the OTA.

As discussed earlier, $g_m = 19.2 \times I_{ABC}$, where g_m is in millimhos when I_{ABC} is in milliamperes. In this case, I_{ABC} is approximately equal to:

$$\frac{Vm - (V^-)}{Rm} = I_{ABC}$$

$$(I_O) = -g_m V_x$$

$$g_m V_x = (19.2)(I_{ABC})(V_x)$$

$$I_O = \frac{-19.2\,[Vm - (V^-)]\,V_x}{Rm}$$

$$I_O = \frac{19.2\,(V_x)(V^-)}{Rm} - \frac{19.2\,(V_x)(V_m)}{Rm} \text{ (Modulation Equation).}$$

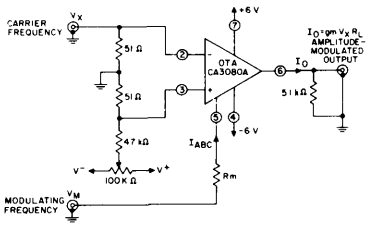


Fig. 19– Amplitude modulator circuit using the OTA.

There are two terms in the modulation equation: the first term represents the fixed carrier input, independent of Vm, and the second term represents the modulation, which either adds to or subtracts from the first term. When Vm is equal to the V- term, the output is reduced to zero.

In the preceding modulation equations the term

$$(19.2)(V_x)\ \frac{V_{ABC}}{Rm}$$

involving the amplifier-bias-current terminal voltage (V_{ABC}) (see Fig. 4 for V_{ABC}) was neglected. This term was assumed to be small because V_{ABC} is small compared with V- in the equation. If the amplifier-bias-current terminal is driven by a current-source (such as from the collector of a p-n-p transistor), the effect of V_{ABC} variation is eliminated and transferred to the involvement of the p-n-p transistor base-emitter junction characteristics. Fig. 20 shows a method of driving the amplifier-bias-current terminal to effectively remove this latter variation. If an n-p-n transistor is added to

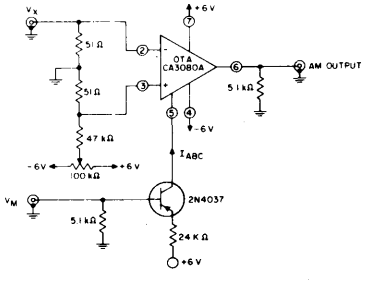


Fig. 20– Amplitude modulator using OTA controlled by p-n-p transistor.

the circuit of Fig. 20 as an emitter-follower to drive the p-n-p transistor, variations due to base-emitter characteristics are considerably reduced due to the complementary nature of the n-p-n base-emitter junctions. Moreover, the temperature coefficients of the two base-emitter junctions tend to cancel one another. Fig. 21 shows a configuration using one transistor in the RCA type CA3018A n-p-n transistor-array as

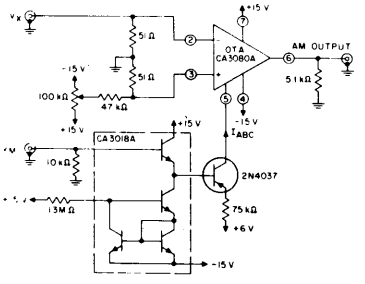


Fig. 21– Amplitude modulator using OTA controlled by p-n-p and n-p-n transistors.

an input emitter-follower, with the three remaining transistors of the transistor-array connected as a current-source for the emitter – followers. The 100-kΩ potentiometer shown in these schematics is used to null the effects of amplifier input offset voltage. This potentiometer is adjusted to set the output voltage symmetrically about zero. Figs. 22a and 22b show oscilloscope photographs of the output voltages obtained when the circuit of Fig. 19 is used as a modulator for both sinusoidal and triangular modulating signals. This method of modulation permits a range exceeding 1000:1 in the gain, and thus provides modulation of the carrier input in excess of 99%. The photo in Fig. 22c shows the excellent isolation achieved in this modulator during the "gated-off" condition.

Four-Quadrant Multipliers

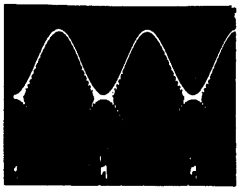
A single CA3080A is especially suited for many low-frequency, low-power four-quadrant multiplier applications. The basic multiplier circuit of Fig. 23 is particularly useful for waveform generation, doubly balanced modulation, and other signal processing applications, in portable equipment, where low-power consumption is essential and accuracy requirements are moderate. The multiplier configuration is basically an extension of the previously discussed gain-controlled configuration (Fig. 19).

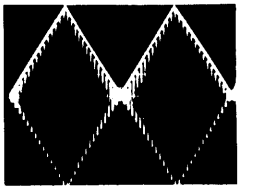
To obtain a four-quadrant multiplier, the first term of the modulation equation (which represents the fixed carrier) must be reduced to zero. This term is reduced to zero by the placement of a feedback resistor (R) between the output and the inverting input terminal of the CA3080A, with the value of the feedback resistor (R) equal to $1/g_m$. The output current is $I_O = g_m$ (-Vx) because the input is applied to the inverting terminal of the OTA. The output current due to the resistor (R) is $\frac{Vx}{R}$. Hence, the two signals cancel when R = $1/g_m$. The current for this configuration is:

$$I_O = \frac{-19.2\ Vx\ Vm}{Rm} \quad \text{and } Vm = Vy$$

The output signal for these configurations is a "current" which is best terminated by a short-circuit. This condition can be satisfied by making the load resistance for the multiplier output very small. Alternatively, the output can be applied to a current-to-voltage converter shown in Fig. 24.

In Fig. 23, the current "cancellation" in the resistor R is a direct function of the OTA differential amplifier linearity. In the following example, the signal excursion is limited to ±10 mV to preserve this linearity. Greater signal-excursions on the input terminal will result in a significant departure from linear operation (which may be entirely satisfactory in many applications).





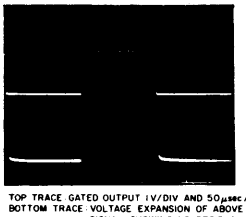


Fig. 22– a) Oscilloscope photo of amplitude modulator circuit of Fig. 15 with R_m = 40 kΩ, V^+ = 10 v and V- = -10 V. Top trace: modulation frequency input ≃ 20-V p-p; center trace: amplitude modulated output 500-mV/div.; lower trace: expanded output to show depth of modulation, 20 mV/div.; b) triangular modulation; top trace: modulation frequency input ≃ 20 V; lower trace: amplitude modulated output 500 mV/div.; c) square wave modulation, top trace: gated output 1 V/div.; lower trace: expanded scale, showing no residual (1 mV/div) and at least 80 dB of isolation at fq = 100 kHz.

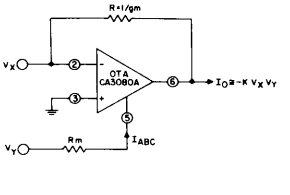


Fig. 23– Basic four quadrant analog multiplier using an OTA.

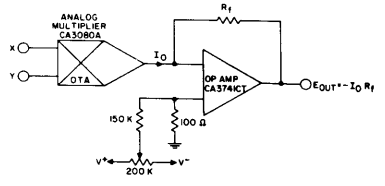


Fig. 24– OTA analog multiplier driving an op-amp that operates as a current-to-voltage converter.

Fig. 25 shows a schematic diagram of the basic multiplier with the adjustments set-up to give the multiplier an accuracy of approximately ±7 percent "full-scale". There are only three adjustments: 1) one is on the output, to

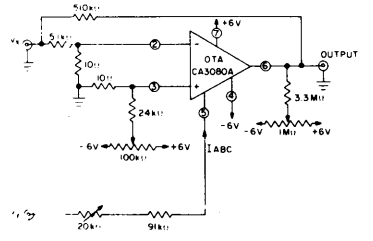


Fig. 25– Schematic diagram of analog multiplier using OTA.

compensate for slight variations in the current-transfer ratio of the current-mirrors (which would otherwise result in a symmetrical output about some current level other than zero); 2) the adjustment of the 20-kΩ potentiometer establishes the g_m of the system equal to the value of the fixed resistor shunting the system when the Y-input is zero; 3) compensates for error due to input offset voltage.

Procedure for adjustment of the circuit:

1. a) Set the 1 MΩ output-current balancing potentiometer to the center of its range
   b) Ground the X- and Y- inputs
   c) Adjust the 100 kΩ potentiometer until a zero-V reading is obtained at the output.
2. a) Ground the Y-input and apply a signal to the X-input through a low source-impedance generator. (It is essential that a low impedance source be used; this minimizes any change in the g_m balance or zero-point due to the 50-μA Y-input bias current).
   b) Adjust the 20-kΩ potentiometer in series with Y-input until a reading of zero-V is obtained at the output. This adjustment establishes the g_m of the CA3080A at the proper level to cancel the output signal. The output current is diverted through the 510-kΩ resistor.
3. a) Ground the X-input and apply a signal to the Y-input through a low source-impedance generator.
   b) Adjust the 1-MΩ resistor for an output voltage of zero-V.

There will be some interaction among the adjustments and the procedure should be repeated to optimize the circuit performance.

Fig. 26 shows the schematic of an analog multiplier circuit with a 2N4037 p-n-p transistor replacing the Y-input "current" resistor. The advantage of this system is the higher input resistance resulting from the current-gain of the p-n-p transistor. The addition of another emitter-follower preceding the p-n-p transistor (shown in Fig. 21) will further increase the current gain while markedly reducing the effect of the V_{be} temperature-dependent characteristic and the offset voltage of the two base-emitter junctions.

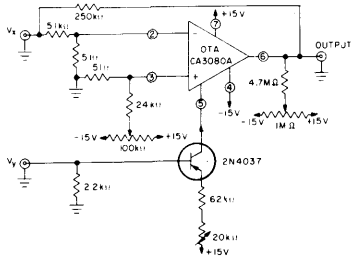
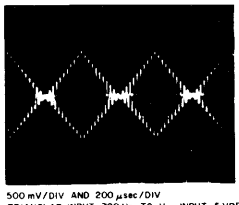
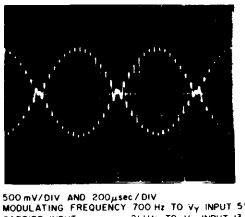


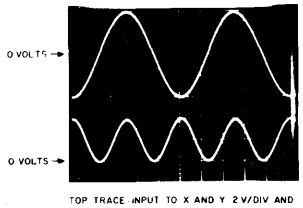
Fig. 26– Schematic diagram of analog multiplier using OTA controlled by a p-n-p transistor.



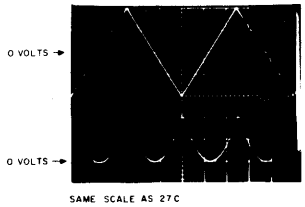
500 mV/DIV AND 200 μsec/DIV
TRIANGLAR INPUT 700 Hz TO V_Y INPUT 5 VPP
CARRIER INPUT 30 kHz TO V_X INPUT 13.5 VPP



500 mV/DIV AND 200μsec/DIV
MODULATING FREQUENCY 700 Hz TO V_Y INPUT 5 VPP
CARRIER INPUT 21 kHz TO V_X INPUT 13.5 VPP



TOP TRACE: INPUT TO X AND Y 2 V/DIV AND
1 msec/DIV – 200 Hz
BOTTOM TRACE: OUTPUT 500 mV/DIV AND
1 msec/DIV – 400 Hz



SAME SCALE AS 27C

Fig. 27– a) Waveforms observed with OTA analog multiplier used as a suppressed carrier generator; b) waveforms observed with OTA analog multiplier used in signal-squaring circuits.

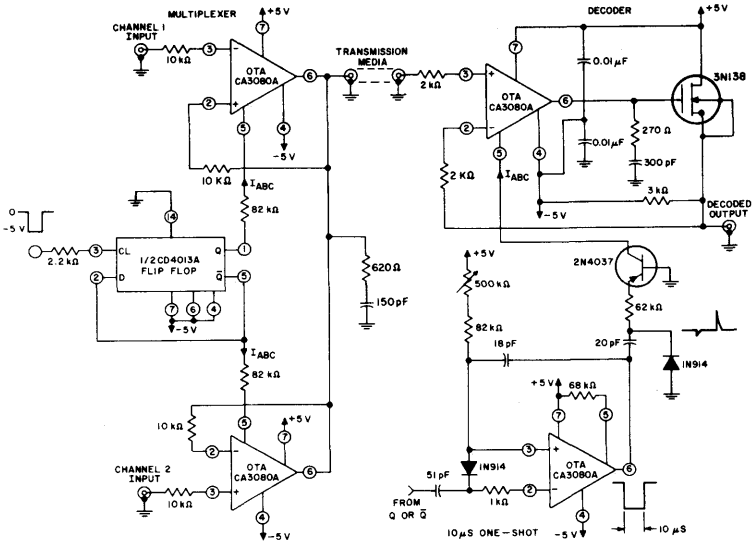


Fig. 28– Two-channel multiplexer and decoder using OTAs.

Figs. 27a and 27b show oscilloscope photographs of the output signals delivered by the circuit of Fig. 26 which is connected as a suppressed-carrier generator. Figs. 27c and 27d contain photos of the outputs obtained in signal "squaring" circuits, i.e. "squaring" sine-wave and triangular-wave inputs.

If ±15-V power supplies are used (shown in Fig. 26), both inputs can accept ±10-V input signals. Adjustment of this multiplier circuit is similar to that already described above.

The accuracy and stability of these multipliers are a direct function of the power supply-voltage stability because the Y-input is referred to the negative supply-voltage. Tracking of the positive and negative supply is also important because the balance adjustments for both the offset voltage and output current are also referenced to these supplies.

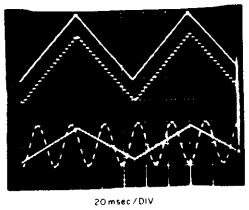
Other forms of four-quadrant multipliers using operational transconductance amplifiers have been published. (See Ref. 2.) the circuit shown in Ref. 2 tends to reduce the effects of the previously discussed g_m temperature dependency.

Linear Multiplexer – Decoder

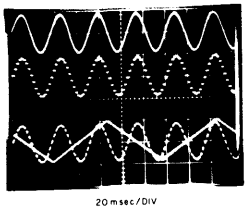
A simple, but effective system for multiplexing and decoding can be assembled with the CA3080 shown in Fig. 28. Only two channels are shown in this schematic, but the number of channels may be extended as desired. Fig. 29 shows oscilloscope photos taken during operation of the multiplexer and decoder. A CA3080 is used as a 10 μsec delay- "one-shot" multivibrator in the decoder to insure that the sample-and-hold circuit can sample only after the input signal has settled. Thus, the trailing edge of the "one-shot" output-signal is used to sample the input at the sample-and-hold circuit for approximately 1 μs. Fig. 30 shows oscilloscope photos of various waveforms observed during operation of the multiplexer/decoder circuit. Either the Q or $\overline{Q}$ output from the flip-flop may be used to trigger the 10 μsec "one-shot" to decode a signal.

High-Gain, High-Current Output Stages

In the previously discussed examples, the OTA has been buffered by a single insulated-gate field-effect-transistor (MOS/FET) shown in Fig. 9. This configuration yields a voltage gain equal to the (g_m) (R_o) product of the CA3080, which is typically 142,000 (103dB). The output voltage and current-swing of the operational amplifier formed by this configuration (Fig. 9) are limited by the 3N138 MOS/FET performance and its source-terminal load. In the positive direction, the MOS/FET may be driven into saturation; the source-load resistance and the MOS/FET characteristics become the factors limiting the output-voltage swing in the negative direction. The available negative-going load current may be kept constant by the return of the source-terminal to

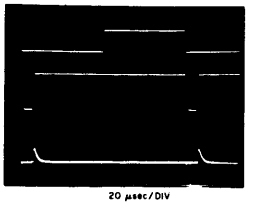


20 msec/DIV
TOP TRACE: INPUT SIGNAL (1 VOLT/DIV)
CENTER TRACE: RECOVERED OUTPUT (1 VOLT/DIV)
BOTTOM TRACE: MULTIPLEXED SIGNALS (2 VOLTS/DIV)

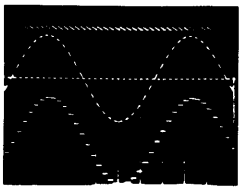


20 msec/DIV
TOP TRACE: INPUT SIGNAL (1 VOLT/DIV)
CENTER TRACE: RECOVERED OUTPUT (1 VOLT/DIV)
BOTTOM TRACE: MULTIPLEXED SIGNALS (1 VOLT/DIV)

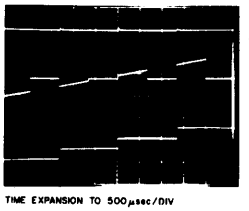
Fig. 29– Waveforms showing operation of linear multiplexer/sample-and-hold decode circuitry (Fig. 28).



20 μsec/DIV
TOP TRACE: FLIP-FLOP OUTPUT (5 VOLTS/DIV)
CENTER TRACE: "ONE-SHOT" OUTPUT (5 VOLTS/DIV)
BOTTOM TRACE: PULSE AT THE COLLECTOR OF THE 2N4037 TRANSISTOR (0.1 VOLTS/DIV)



TOP TRACE: COLLECTOR OF PNP TRANSISTOR (0.5 V/DIV)
CENTER TRACE: MULTIPLEXED OUTPUT WITH ONE CHANNEL INPUT GROUNDED (0.5 V/DIV)
LOWER TRACE: DECODED OUTPUT (0.5 V/DIV)
TIME ALL SCALES: 5 msec/DIV



TIME EXPANSION TO 500 μsec/DIV

Fig. 30– (a) Waveforms showing timing of flip-flop, delay– "one-shot" and the strobing pulse to the sample-and-hold circuit (Fig. 28): top trace: flip-flop output (5 V/div); center trace: "one-shot" output (5 V/div); lower trace: pulse at collector of 2N4037 transistor (0.1 V/div); b) Waveforms showing the decoding operation from the decoder keying pulse (top traces) to the recovered "decoded" sampled output (lower traces). 1) top trace: collector of 2N4037; center trace: multiplexed output with one channel input grounded; lower trace: decoded output; 2) Expanded scale of (1).

a constant-current transistor. Phase compensation is applied at the interface of the CA3080 and the 3N138 MOS/FET shown in Fig. 9.

Another variation of this generic form of amplifier utilizes the RCA CD4007A (COS/MOS) "inverter" as an amplifier driven by the CA3080. Each of the three "inverter"/amplifiers in the CD4007A has a typical voltage gain of 30 dB. The gain of a single COS/MOS "inverter"/amplifier coupled with the 100 dB gain of the CA3080 yields a total forward-gain of about 130 dB. Use of a two-stage COS/MOS amplifier configuration will increase the total open-loop gain of the system to about 160 dB (100,000,000). Figs. 31 through 34 show examples of these configurations. Each COS/MOS "inverter"/amplifier can sink or source a current of 6 mA (typ.). In Figs. 33 and 34, two COS/MOS "inverter"/amplifiers have been connected in parallel to provide additional output current.

The open-loop slew-rate of the circuit in Fig. 31 is approximately 65 V/μsec. When compensated for the unity-gain voltage-follower mode, the slew-rate is about 1 V/μsec (shown in Fig. 32). Even when the three "inverter"/

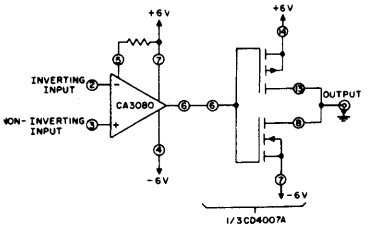


Fig. 31– Schematic diagram showing OTA driving COS/MOS Inverter/Amplifier (open-loop mode). For greater current output the two remaining amplifiers of the CD4007A may be connected in parallel with the single stage shown. Open-loop gain ≃ 130 dB.

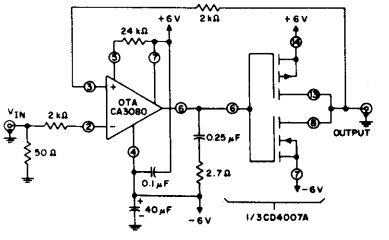


Fig. 32– Schematic diagram showing OTA driving COS/MOS Inverter/Amplifier (unity-gain closed-loop mode). For greater current output, the two remaining amplifiers of the CD4007A may be connected in parallel with the single stage shown.

amplifiers in the CD4007A are connected as shown in Fig. 33, the open-loop slew-rate remains at 65 V/μsec. A slew-rate of about 1 V/μsec is maintained with this circuit connected in the unity-gain voltage-follower mode, as shown in Fig. 34. Fig. 35 contains oscilloscope photos of input-output waveforms under small-signal and large-signal conditions for the circuits of Figs. 32 and 34. These photos illustrate the inherent stability of the OTA and COS/MOS circuits operating in concert.

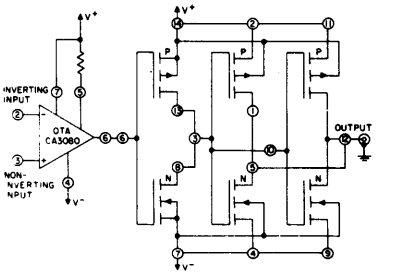


Fig. 33– Schematic diagram showing OTA driving two-stage COS/MOS Inverter/Amplifier (open-loop mode). gain ≃ 160 dB.

Precision Multistable Circuits

The micropower capabilities of the CA3080, when combined with the characteristics of the CD4007A COS/MOS "inverter"/amplifiers, are ideally suited for use in connection with precision multistable circuits. In the circuits of Figs. 31, 32, 33, and 34, for example, power-supply current drawn by the COS/MOS "inverter"/amplifier approaches zero as the output voltage swings either positive or negative, while the CA3080 current-drain remains constant.

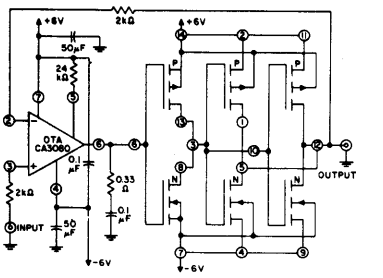
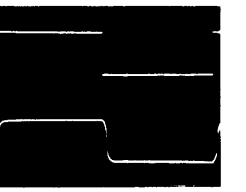


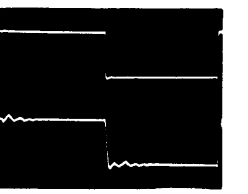
Fig. 34– Schematic diagram showing OTA driving two-stage COS/MOS Inverter/Amplifier (unity gain closed-loop mode).

(a)



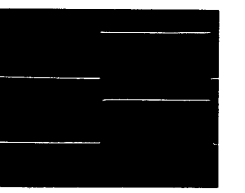
TOP TRACE: INPUT 5 V/DIV – 100 μsec/DIV
BOTTOM TRACE: OUTPUT SAME SCALE

(b)



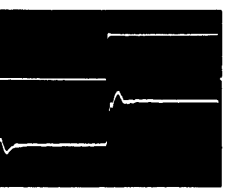
TOP TRACE: INPUT – 50 mV/DIV – 1 μsec/DIV
BOTTOM TRACE: OUTPUT – SAME SCALE

(c)



TOP TRACE: INPUT – 5 V/DIV – 100 μsec/DIV
BOTTOM TRACE: OUTPUT – SAME SCALE

(d)



TOP TRACE: INPUT – 50 mV/DIV – 1 μsec/DIV
BOTTOM TRACE: OUTPUT – SAME SCALE

Fig. 35– a) Waveforms for circuit of Fig. 32 with large signal input; b) Waveforms for circuit of Fig. 32 with small signal input; c) Waveforms for circuit of Fig. 34 with large signal input; d) Waveforms for circuit of Fig. 34 with small signal input.

Fig. 36 shows a variety of circuits that can be assembled using the CA3080 to drive one "inverter"/amplifier in the CD4007A. Precise timing and thresholds are assured by the stable characteristics of the input differential amplifier in the CA3080. Moreover, speed vs. power consumption tradeoffs may be made by adjustment of the I_{ABC} current to the CA3080. The quiescent power consumption of the circuits shown in Fig. 36 is typically 6 mW, but can be made to operate in the micropower region by suitable circuit modifications.

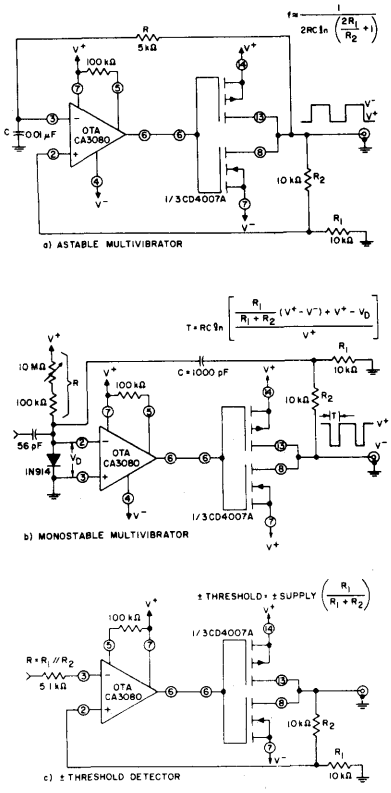


Fig. 36– Multistable circuits using the OTA and COS/MOS Inverter/Amplifiers: a) astable multivibrator; b) monostable multivibrator; c) threshold detector (plus or minus). For greater current output, the remaining amplifiers in the CD4007A may be connected in parallel with the single stage shown.

Micropower Comparator

The schematic diagram of a micropower comparator is shown in Fig. 37. Quiescent power consumption of this circuit is about 10 μW (typ). When the comparator is strobed "ON", the CA3080A becomes active and consumes 420 μW. Under these conditions, the circuit responds to a differential input signal in about 8 μsec. By suitably biasing the CA3080A, the circuit response time can be decreased to about 150 nsec., but the power consumption rises to 21 mW.

The differential amplifier input common-mode range for the circuit of Fig. 37 is -1V to +10.5 V. Voltage of the micropower comparator is typically 130 dB. For example, a 5 μV input signal will toggle the output.

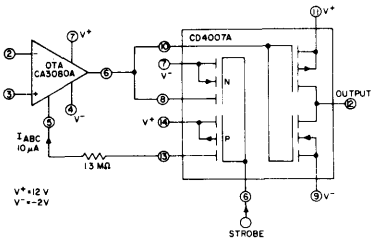


Fig. 37– Schematic diagram of micropower comparator using the CA3080A and COS/MOS CD4007A.

APPENDIX I
CURRENT MIRRORS

The basic current-mirror, described in the beginning of this note, in its rudimentary form, is a transistor with a second transistor connected as a diode. Fig. A shows this basic configuration of the current-mirror. Q2 is a diode connected transistor. Because this diode-connected transistor is not in saturation and is "active", the "diode" formed by this connection may be considered as a transistor with 100% feedback. Therefore, the base current still controls the collector current as is the case in normal transistor action, i.e., $I_C = \beta I_b$. If a current I_1 is forced into the diode-connected transistor, the base-to-emitter voltage will rise until equilibrium is reached and the total current being supplied is divided between the collector and base regions. Thus, a base-to-emitter voltage is established in Q2 such that Q2 "sinks" the applied current I_1.

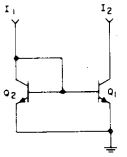


Fig. A– Diode – transistor current source.

If the base of a second transistor (Q1) is connected to the base-to-collector junction of Q2, shown in Fig. A, Q1 will also be able to "sink" a current approximately equal to that flowing in the collector lead of the diode-connected transistor Q2. This assumes that both transistors have identical characteristics, a prerequisite established by the IC fabrication technique. The difference in current between the input current (I_1) and the collector current (I_2) of transistor Q, is due to the fact that the base-current for both transistors is supplied from I_1. Fig. B shows this current division, using a unit of base current (1) to each transistor base. This base

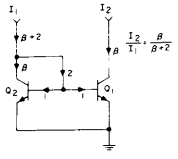


Fig. B– Diode – transistor current source. Analysis of current flow.

current causes a collector current to flow in direct proportion to the β of each transistor. The ratio of the "sinking" current I_2 to the input current I_1 is therefore equal to $\frac{I_2}{I_1} = \beta/(\beta+2)$. Thus, as β increases, the output "sinking" current (I_2) level approaches that of the input current (I_1). The curves in Fig. C show this ratio as a function of the transistor β. When the transistor β is equal to 100, for example, the difference between the two currents is only two percent.

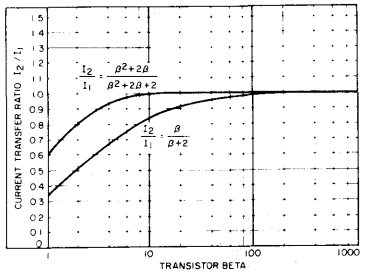


Fig. C– Current transfer ratio I_2/I_1 as a function of transistor beta.

Fig. D shows a curve-tracer photograph of characteristics for the circuit of Figs. A and B. No consideration in this discussion is given to the variation of the transistor (Q1) collector current as a function of its collector-to-emitter voltage. The output resistance characteristic of Q1 retains its similarity to that of a single transistor operating under similar conditions. An improvement in its output resistance characteristic can be made by the insertion of a diode-connected transistor in series with the emitter of Q1.

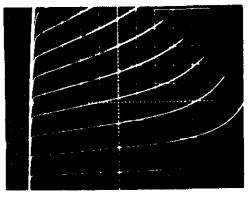


Fig. D– Photo showing results of Figs. A & B.

This diode-connected transistor (Q3 in Fig. E) may be considered as a current-sampling diode that senses the emitter-current of Q1 and adjusts the base current Q1 (via Q2) to maintain a constant-current in I_2. Because all controlling transistors are operated at relatively fixed voltages, the previously discussed effects due to voltage coefficients do not exist. The curve-tracer photograph of Fig. F shows the improved output resistance characteristics of the circuit of Fig. E. (Compare Fig. D and F).

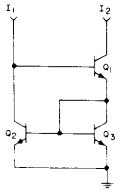


Fig. E– Diode – 2 transistor current source.

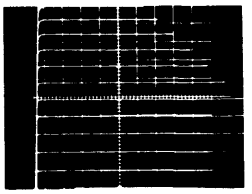


Fig. F– Photo showing results of Fig. E.

Fig. G shows the current-division within the "mirror" assuming a "unit" (1) of current in transistors (Q2 and Q3.

The resulting current-transfer ratio $I_2/I_1 = \frac{\beta^2 + 2\beta}{\beta^2 + 2\beta + 2}$; Fig. C shows this equation plotted as a function of b.ta. It is significant that the current transfer ratio (I_2/I_1) is improved by the β^2 term, and reduces the significance of the $2\beta + 2$ term in the denominator.

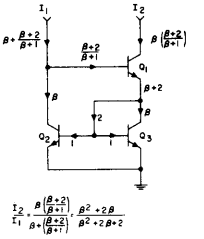


Fig. G– Current flow analysis of Fig. E.

Conclusions

The Operational Transconductance Amplifier (OTA) is a unique device with characteristics particularly suited to applications in multiplexing, amplitude modulation, analog multiplications, gain control, switching circuitry, multivibrators, comparators, and a broad spectrum of micropower circuitry. The CA3080 is ideal for use in conjunction with COS/MOS (Complementary-Symmetry MOS) IC's being operated in the linear mode.

Acknowledgements

The author is indebted to C. F. Wheatly for many helpful discussions. Valued contributions in circuit evaluation were made by A. J. Visioli Jr. and J. H. Klinger.

References

1 RCA's Linear Integrated Circuits Manual, Basic Circuits Section.
2 RCA published data for CA3060 File No. 404

Application of the CA3134G Sound IF and Output Subsystem in Television Receivers

by George M. Harayda

In the CA3134, the sound if and audio output subsystems for color or black-and-white television receivers are combined in a single monolithic integrated circuit.[1] As shown in the block diagram, Fig. 1, the CA3134 includes a multistage if amplifier-limiter, an fm detector, an electronic attenuator, and an audio power amplifier. The power amplifier is designed to drive, primarily, an 8-, 16-, or 32-ohm speaker, although, if the designer chooses, it may be used to drive 4- or 10-ohm or other sized loads. The amplifier has a typical power output of 5 watts with a 16-ohm load and a V+ of 30 volts. The consolidation of the functions mentioned into an integrated circuit minimizes the number of components and reduces the area of the printed circuit board necessary for this portion of a television receiver. This consolidation also permits a reduction in manufacturers' component inventories and simplifies field servicing.

The incorporation of additional features in the CA3134 results in an improvement in performance when the circuit is compared to a system in which a type such as the CA3065 is used with a discrete or integrated-circuit power amplifier. These additional features include a volume control with an improved taper, a provision for the optional use of an unattenuated audio output (fixed level unaffected by volume control position) (terminal 8), an additional power-supply-ripple bypass point (terminal 6), and a power amplifier with both current limiting and thermal-sensing shutdown protection.

Installation Considerations

The CA3134 employs the hermetic Gold-CHIP (G suffix) system, which is of the sealed-junction type designed to provide protection against humidity and other surface contaminants without a hermetic package enclosure. The semiconductor junctions are sealed by a silicon-nitride passivation layer. A multilayered, highly corrosion-resistant, terminal-connection system of special design is employed as the chip metalization.

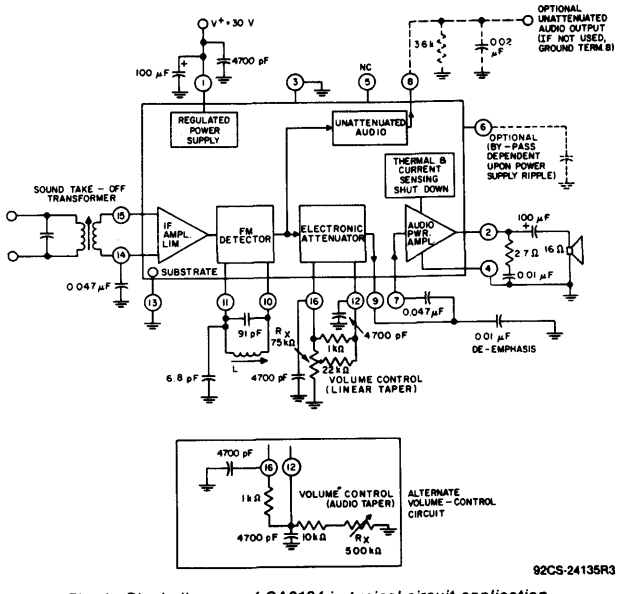


Fig. 1 - Block diagram of CA3134 in typical circuit application.

The CA3134G is encapsulated in a 16-lead plastic "power stud" package. This package has low thermal resistance, and the number of input and output terminals has not been compromised to make allowance for heat-sinking tabs. The "power stud" package lends itself to a wide variety of heat-sink methods depending on the application requirements. The CA3134GM and CA3134GQM are similar to the CA3134G except that they incorporate a tin-plated copper-strap heat sink. The CA3134GQM also has quad-formed leads.

The CA3134G is designed for applications in which it is desired to employ other than the standard heat-sinked configurations. Heat sinks similar to the type provided with the CA3134GM or GQM may also be attached to the CA3134G.[2]

Circuit Description

The CA3134 is designed primarily for use with either a single- or double-tuned sound take-off transformer (balanced input) to couple the sound if frequency (4.5 MHz) output from the video detector stage to the CA3134. This transformer serves both for impedance matching and as a bandwidth-limiting network to help reject unwanted frequencies such as intermodulation frequency products. Other undesired signals include residual am video information and sync information. Fig. 4 shows the overall circuit diagram for the CA3134.

Sound IF Amplifier-Limiter

The sound if stage amplifies the input signal until clipping eliminates am video and sync signals. In a typical TV system, the signal level available to the sound if amplifier-limiter is 35 mV rms. At this signal level, the input impedance components at terminals 14 and 15 of the CA3134 consist of a resistance, R_p, of approximately 25 kilohms in parallel with a capacitance, C_p, of approximately 3 pF. The sound if amplifier

provides enough gain to bring the input signal level to an amplitude suitable for fm detection, but not so high as to cause PC layout or coupling problems, Fig. 2.

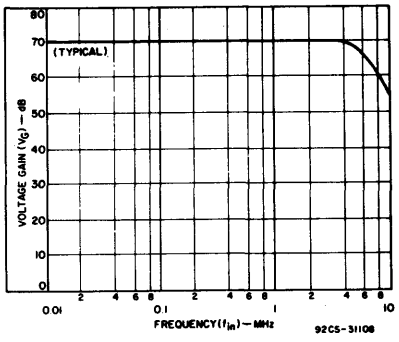


Fig. 2 - Voltage gain of if amplifier as a function of frequency.

As shown in Fig. 4, the if amplifier consists of four stages of differential amplifiers, Q15-Q16, Q19-Q20, Q23-Q24, and Q27-Q28, using resistors R13, R16, R19, and R24 as constant-current sinks; each stage is followed by emitter followers, Q17, Q21, and Q25. Because the differential amplifier functions as a limiter, am signals are eliminated and the signal into Q30 consists of constant-amplitude, frequency-modulated square waves. These square waves are shaped into approximate sine waves by Q30 and its associated RC networks to assure proper operation of the fm detector. The signal output from R31 into the base of Q41 and to terminal 10 is a constant-amplitude fm sine wave.

FM Detector

The fm sine wave at terminal 10 constitutes the input signal to the differential peak detector stage. The extracted signal contains the audio information. The detector section is formed by the differential amplifier configuration comprising transistors Q31, Q32, Q35, Q36, Q40, and Q41. Transistors Q31 and Q41 are emitter followers that operate at approximately 0.3 mA and provide high impedance at each input of the detector (terminals 10 and 11). Transistors Q32 and Q40, which operate at approximately 10 microamperes, along with the 15-picofarad capacitors C3 and C4 and the external frequency-sensitive network on terminals 10 and 11, perform peak or envelope detection. As shown in Fig. 1, this frequency-sensitive network consists of a parallel LC network in series with a 6.8-pF capacitor. The signal voltage (from Q30) is applied across the entire network connected to terminal 10. The portion of the signal from Q30 that is across the external 6.8-pF capacitor is applied to terminal 11, and the resulting difference in these signals provides the basic S curve used in the recovery of the audio signal from the fm signal.

An advantage of the differential peak detector is that it requires the alignment of only one single-tuned coil. This coil (L in Fig. 1) can be aligned by any one of the following methods (with an input terminated in 50 ohms, f_O = 4.5 MHz, f_m = 400 Hz, $\Delta f = \pm$ 25 kHz, and a voltage at terminal 15 (V_{15}) $\simeq$ 100 mV rms):

1. Tune L for maximum recovered audio. To minimize thermal effects on alignment, the volume control should be adjusted so that the maximum recovered audio level at the load is limited to a low power level (approximately 0.1 watt or less).
2. Tune L for maximum recovered audio and fine tune for minimum distortion.
3. With no rf input signal, note the dc voltage at terminal 9. Then apply a 4.5-MHz cw signal and adjust the detector coil L until the dc voltage at terminal 9 is the same as the value noted.

After aligning the differential peak detector coil, align the input transformer by reducing the fm input signal level until the recovered audio level drops approximately 3 dB. Then tune the input transformer for a maximum recovered audio level while the input level of the if amplifier-limiter is below its limiting point. Fig. 3 shows the recovered audio, am rejection, and signal-to-noise ratio for the CA3134 as a function of rf input level.

Volume Control and Electronic Attenuator

Control of the audio signal detected by the differential peak detector is accomplished by differential amplifiers Q33-Q34 and Q38-Q39.

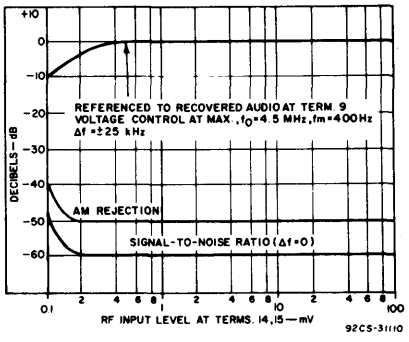


Fig. 3 - Recovered audio and signal-to-noise ratio as functions of rf input level.

The volume is controlled when the bias levels of the differential amplifiers are changed by a current flowing through an external fixed resistor between terminals 12 and 16. The amount of current flowing through this external resistor (which determines the level of recovered audio) is controlled by the position of the variable resistance (volume control) relative to ground. The voltage reference at terminal 16 is established by internal zener diode Z2, approximately 6 volts. The maximum level of recovered audio, therefore, occurs when no currents other than the base currents for Q34 and Q39 are being drawn from the zener diode through the external resistor. When the volume control is adjusted for the minimum level of recovered audio, the current drawn from terminal 16 should be limited to less than 1 milliampere.

This method of controlling the recovered audio has a very predictable volume-control taper, which can be modified to suit the designer's preference by changing the external component values. In addition, it allows for either a one- or two-wire volume control. The one-wire volume control (Fig. 4, alternate volume-control circuit) requires only one wire from the printed circuit board to the external volume control, but requires that the value of the variable resistor be large (approximately 500 kilohms) and that the resistor have an audio taper to assure an acceptable change of audio output level with a linear change (rotation) of the volume-control. The two-wire volume control allows the use of a volume control having a lower value of resistance and a linear taper.

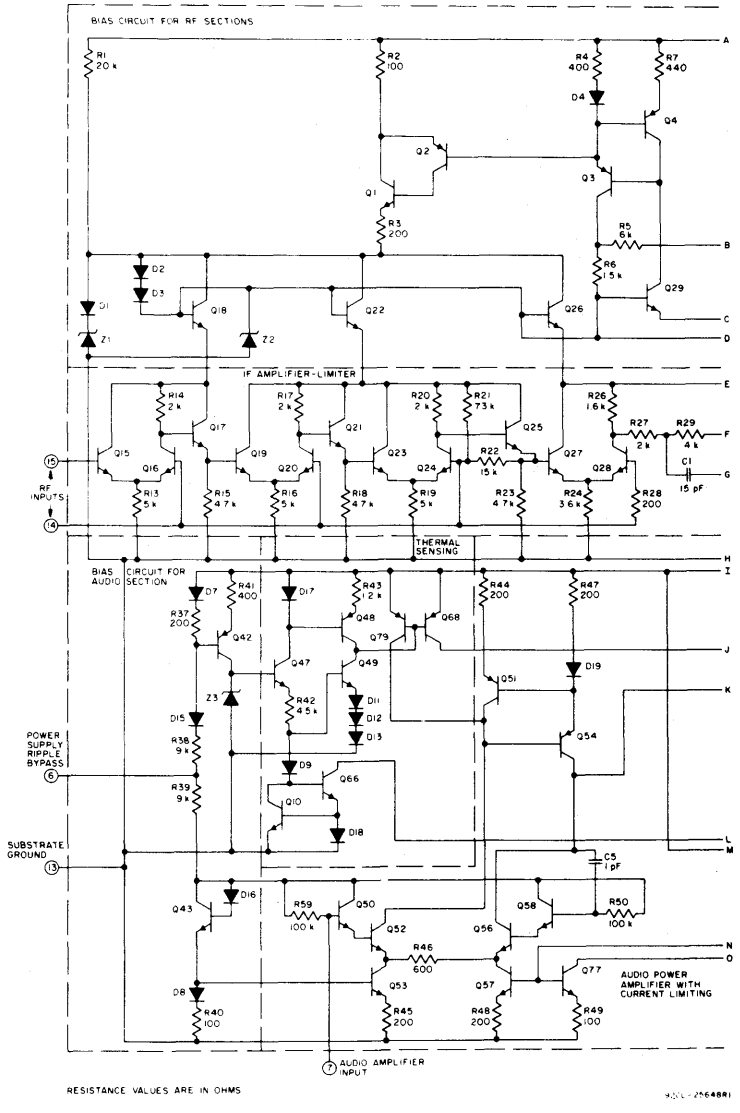


Fig. 4 - Schematic diagram of CA3134.

The output impedance of the electronic attenuator (terminal 9) is typically 7.5 kilohms (R5 + R6). A tone control may be inserted between the output of the electronic attenuator and the input of the audio power amplifier (terminal 7).

Unattenuated Audio Output

The operation of Q38 and Q39 is duplicated by Q33 and Q34 as the volume control is varied. The currents from Q33 and Q34 are combined by the current-mirror configuration produced by D5, Q5, Q6, and Q8, Q12. When an external resistor is placed between terminal 8 and ground, the current from this current-mirror configuration produces a recovered audio voltage at a fixed level independent of volume-control position. This output may be used to mute the sound in the event the broadcasting station loses its sound carrier or broadcast signal or to allow for the direct recording of the audio portion of a program.

Audio Power Amplifier

The audio power amplifier is a quasi-complementary class AB type with a typical voltage gain of 35dB. Internal feedback eliminates the need for external feedback compo-

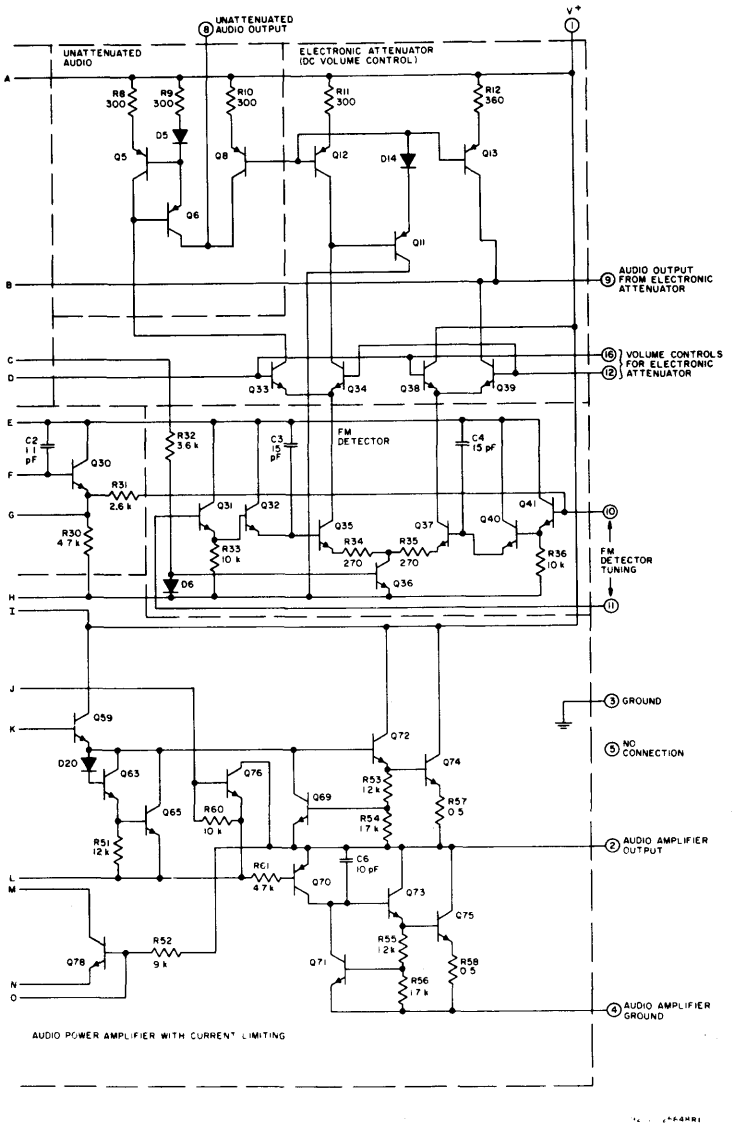


Fig. 4 – Schematic diagram of CA3134.

nents, especially costly electrolytic capacitors. The input impedance (at terminal 7) is typically 100 kilohms (R59). Fig. 5 shows the frequency response of the audio power amplifier and Fig. 6 its efficiency. Both current limiting and thermal shutdown protection are provided. Current limiting is accomplished by limiting the drive to the output transistors from the driver transistors Q72 and Q73. The limiting drive is determined by the feedback from R53, R54, and Q69 to Q72, and R55, R56, and Q71 to Q73. When the peak output current exceeds approximately 0.8 ampere, the voltage developed across the emitters of Q72 and Q73 will cause Q69 and Q71, respectively, to conduct, thereby limiting the drive to the output transistors Q74, Q75.

When the chip temperature exceeds 150°C, the thermal-sensing portion of the CA3134 begins to shut down the power amplifier by removing the bias from the power amplifier driver stages. The temperature at which the thermal shutdown circuitry is activated is determined by the relative areas of D9, Q66, and D18 and those of Q49, D11, D12, and D13. When Q49 conducts, transistors Q79, Q68, and Q76 are in turn biased into

conduction and remove bias from the amplifier driver stages. Because the drive is not removed symmetrically, the signal is distorted and gives an indication that the unit is in a fault condition.

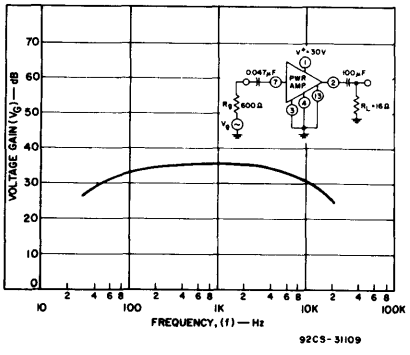


Fig. 5 - Voltage gain of audio power amplifier as a function of frequency.

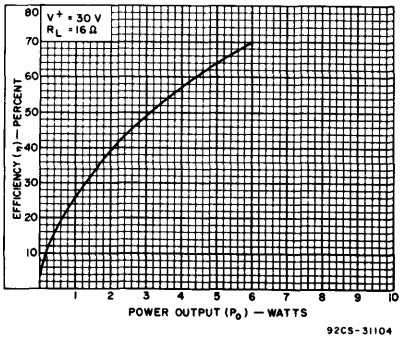


Fig. 6 - Efficiency of audio power amplifier as a function of output power.

Application

For the required power output from the CA3134 (Fig. 7), the speaker impedance must be such that its current drain will be both within the capability of the power supply and less than the current-limiting level of the CA3134. To decouple the CA3134 from the power supply and provide a means for preventing excessive drive to the speaker, a series resistor should be placed between the power supply and terminal 1. The value of this resistor depends upon the required power output level for the worst-case power supply voltage condition. This resistor also reduces the amount of power dissipated in the CA3134.

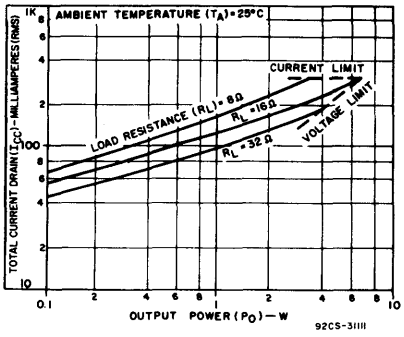


Fig. 7 - Total supply current as a function of output power.

The maximum power dissipation (Figs. 8, 9, 10, and 11) together with the anticipated maximum ambient temperature (Fig. 12) determines the required junction-to-ambient thermal resistance necessary to assure that the maximum chip temperature is lower than the rated junction temperature of 150°C. The overall thermal resistance can be lowered by careful PC board layout. As much of the copper area as possible should be exposed, and coil shields (input transformer and detector tuned circuit) should be used to help radiate heat.

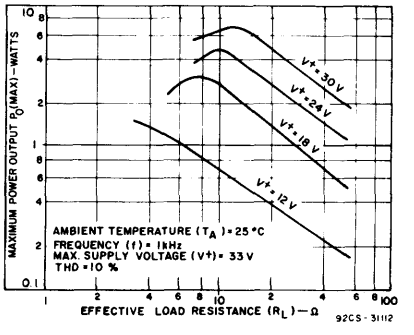


Fig. 8 - Maximum output power as a function of effective load resistance.

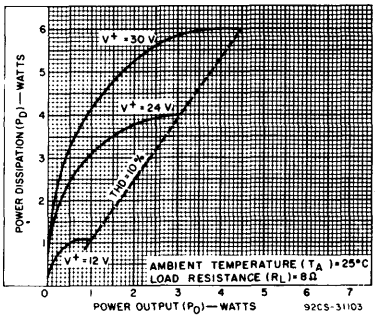


Fig. 9 - Power dissipation as a function of output power at R_L = 8 ohms.

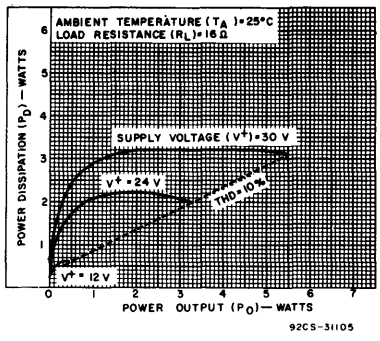


Fig. 10 - Power dissipation as a function of output power at R_L = 16 ohms.

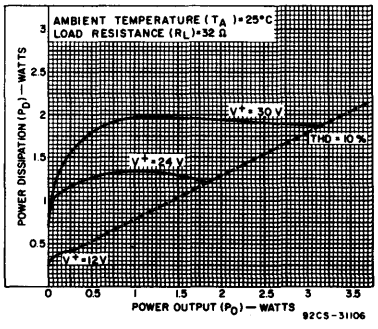


Fig. 11 - Power dissipation as a function of output power at R_L = 32 ohms.

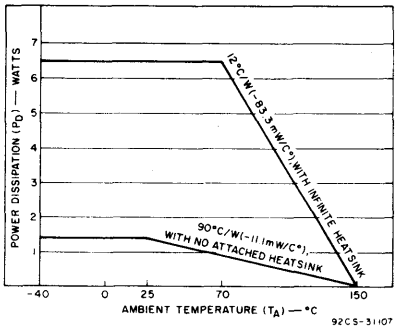


Fig. 12 - Maximum power dissipation as a function of ambient temperature.

Acknowledgment

The author is indebted to both Jack Craft for many helpful discussions and Wayne Austin for his suggestions in the preparation of this Note. The contributions of H. Chinery in the electrical characterization of the CA3134 and Ralph Thompson in the mechanical characterization of the stud package are acknowledged.

References

1. RCA Solid State Data Sheet for CA3134 types, "TV Sound IF and Audio Output Systems," File No. 1097.

2. Method of attaching heat sinks similar to the type provided with the CA3134GM or GQM to the CA3134G. First apply a non-conductive epoxy (Uniset structural adhesive or equivalent) to the top side of the plastic package. Then apply a conductive epoxy (Dupont 5504A or equivalent) in the hole of the heat sink and around the stud projecting from the plastic package. To assure good thermal conduction, use sufficient conductive epoxy to allow the excess to be forced through the hole when the heat sink is fitted over the stud. Stress applied to the stud should be limited to less than 3 in-lbs (approximately 0.35 newton-meter), 15 lbs (approximately 65 newtons) of tension, and 100 lbs (approximately 445 newtons) of compression.

Measurement of Burst ("Popcorn") Noise in Linear Integrated Circuits

by T. J. Robe

The advent in recent years of very high-gain operational amplifiers operating in the 1/f noise-frequency spectrum has placed emphasis on the need for very low-noise devices. This need is particularly true for operational amplifiers which have either low-offset characteristics and/or offset-null capability.

The traditional methods used to select such devices involve the measurement of either spot or wideband (≈ 10 kHz) noise figures in the 1/f frequency range (10 Hz to 10 kHz) at various source resistances. This type of measurement, however, only provides an indication of the average noise power at the measurement frequency and does not reveal the burst ("popcorn") noise characteristics of the Device Under Test (DUT). The metering circuits cannot respond fast enough to measure the effects of burst-noise. Fig. 1a shows a photograph of typical burst-noise as a function of time for an operational amplifier having poor burst-noise characteristics. This photo illustrates burst-noise which is characterized by random abrupt output voltage-level changes that persist for periods from approximately 1/2 millisecond to several seconds. Additionally, the random rate at which the bursts occur ranges from approximately several hundred per second to less than one per minute. Furthermore, these rates are not necessarily repetitive and predictable. Consequently, the nature of burst-noise prevents its measurement by means of the standard averaging techniques. Instead, a technique to detect individual bursts must be used and a DUT must be under observation for a period in the order of 10 seconds to one minute. Fig. 1b shows a photo of the output of a virtually burst noise-free operational amplifier, the RCA-CA6741T.

Test Configuration

Some of the major questions relevant to the type of test required are:

1. What characteristics of the burst-noise should be detected?
2. What test-circuit configuration is most suitable to detect these characteristics?

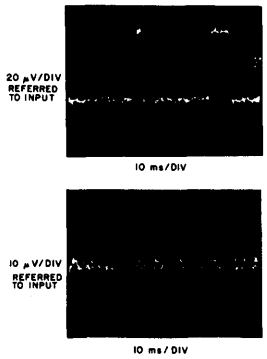


Fig. 1— (a) Photo of output waveforms for amplifier with poor burst-noise characteristics; (b) photo of output waveform for the RCA-CA6741T.

3. What are the "Pass-Fail" criteria?

There are three major characteristics of the noise burst which have an impact on the suitability of a device from the standpoint of applications: burst amplitude, duration, and rate of occurrence. Of these, burst-amplitude and rate of occurrence are of primary interest to potential users of a particular device. Long duration bursts (of sufficient amplitude) seriously degrade the performance of dc amplifiers; however, suitable devices could be selected by the rejection of any unit which produced even one burst during some prescribed test period. Therefore, an absolute measurement of burst duration is not a prime necessity.

The rate of occurrence, on the other hand, as measured by the burst-count in a given test period could conceivably be considered as a variable of prime importance in the selection process. For instance, a burst-rate of 100 per second is clearly objectionable in almost any low-level low-frequency application, whereas the occurrence of only one low-amplitude burst in a one-minute period might be quite acceptable. Consequently, it is desirable to include flexibility in the testing system so that "Pass-Fail" criteria can be established on the basis of burst-noise count in some prescribed period of time. The test equipment described herein detects total noise (1/f noise plus burst noise) bursts with amplitudes above a preset threshold level during a given test period and allows acceptance or rejection on the basis of the number of noise voltage excursions beyond the threshold level, in the selected test period.

Another factor to be considered is the bandwidth of the test system. Excessive bandwidth allows the normal "white" noise of the terminating resistors and the DUT to obscure burst-noise occurrences and does not realistically simulate the low-frequency applications in which burst-noise is particularly objectionable. On the other hand, a test circuit having excessively narrow bandwidth prevents detection of the shorter-duration bursts (≈ 1/2 ms) even if their amplitude is relatively high. A suitable compromise is chosen in which the system rise time permits a burst of "minimum" duration to reach essentially its full amplitude. Because the rise time and bandwidth of an amplifier are related by the equation:

$$BW \approx \frac{0.4}{t_r}$$

the minimum bandwidth to detect a 0.5 ms burst is approximately:

$$BW_{min} = \frac{0.4}{(0.5)(10^{-3})} = 0.8 \text{ kHz}.$$

Consequently, a 1 kHz bandwidth has been selected as a reasonable one for a burst-noise test system and, therefore, prescribes the need for a low-pass filter in the system.

The test requirements outlined above can be implemented with the following circuit elements shown in the block diagram of Fig. 2a. Fig. 2b shows the complete system schematic:

1. A fixed high-gain amplifier incorporating the DUT as the first stage to amplify the microvolt-level burst to an easily detectable level (this should be a burst noise-free unit);
2. A low-pass filter to limit the test bandwidth to approximately 1 kHz,
3. A comparator to produce a fast-rise high-level single-polarity output pulse whenever an input burst-noise pulse (of either polarity) exceeds a preset (but adjustable) threshold level;
4. A counter to tally the number of pulses emanating from the comparator during the test period: a single decade counter is adequate.
5. A latch circuit which trips to the "latched" state when the count exceeds a preselected number (e.g. 1 to n). The latch circuit, if tripped, energizes an indicator lamp.
6. A timer to control the period over which the counter is enabled. It should incorporate the capability to reset both the counter and the latch circuit at the beginning of each test period.
7. Power supplies for the DUT and other auxiliary circuits.

Test Conditions

Some of the conditions which affect the burst-noise performance of the DUT include bias-level, source resistance (Rs), and ambient temperature (T_A).

The quiescent operating conditions in operational amplifiers are normally set by the magnitude of the positive and negative supplies. Many of the newer Op-Amp types, however, have bias-terminals into which fixed currents can be injected to set their performance characteristics. The RCA-CA3060, CA3080, and CA3080A Operational Transconductance Amplifiers (OTA's) and; the RCA-CA3078 and CA3078A Micropower Op-Amps are examples of such devices. For best low-frequency and burst-noise performance, these amplifiers should be operated at the lowest bias

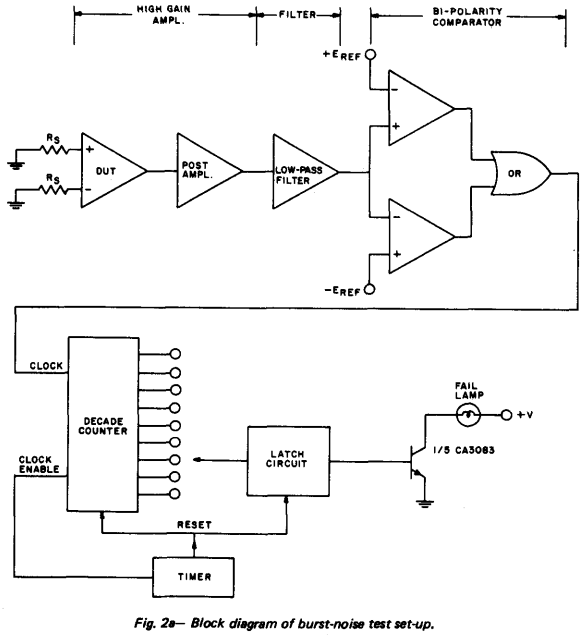


Fig. 2a— Block diagram of burst-noise test set-up.

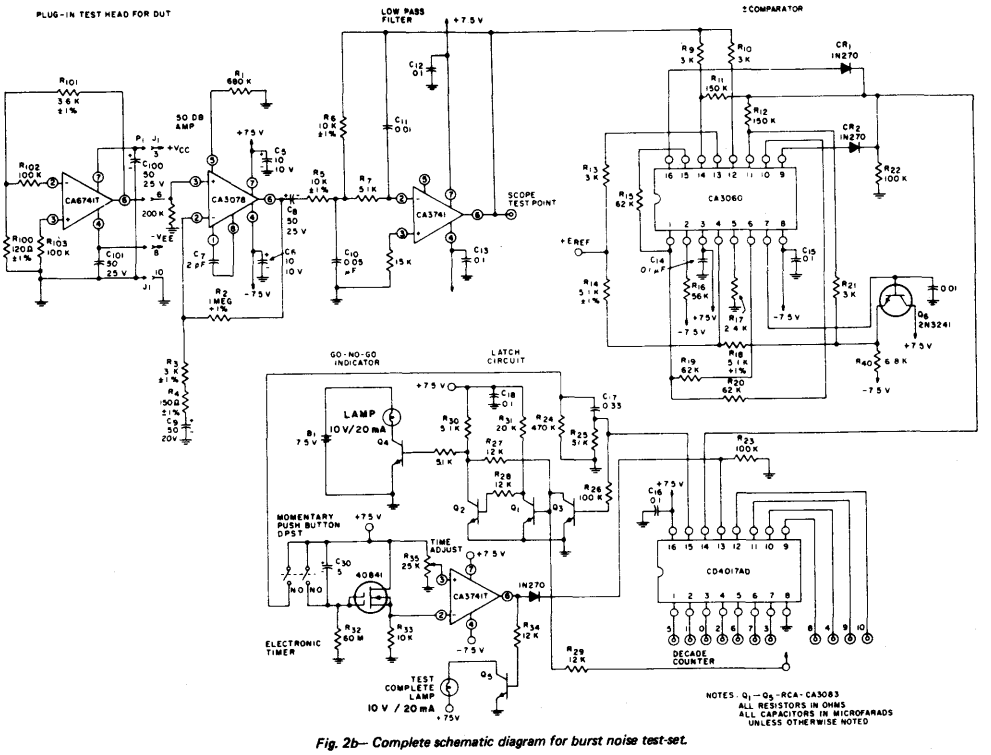


Fig. 2b– Complete schematic diagram for burst noise test-set.

currents consistent with the gain-bandwidth requirements of the particular application.

In the test for burst noise, the source resistance (Rs) seen by the input terminals of the DUT, is a key test parameter. Burst noise causes effects which are equivalent to a spurious current-source at the device input and, therefore, burst-noise current generates an equivalent input noise-voltage in proportion to the magnitude of the source resistance through which it flows. Accordingly, to increase the sensitivity of the test system, it is desirable to use the highest source resistance consistent with the input offset-current of the DUT. For example, an Op-Amp which has 0.1 μA input offset current could realistically be tested with source-resistance in the order of 100KΩ (10 mV input offset), whereas a 1 MΩ source-resistance (100 mV input offset) could cause excessive offset in the output. For 741 type Op-Amps a 100kΩ resistance is recommended.

Burst-noise generation in amplifiers is usually more pronounced at lower temperatures (particularly below 0°C). Consequently, consideration must be given to the temperature of the DUT in relation to the temperature range under which the device is expected to perform in a particular operation.

A test parameter of importance is the time duration of observation. Because the frequency of burst-noise occurrence is frequently less than once every few seconds, the minimum test period should be in the range of from 15 to 30 seconds.

Pass-Fail Criteria

A test system built to accommodate the test philosophy outlined above has the ability to reject or pass a DUT on the basis of two variables: burst-amplitude and the frequency of burst occurrence. The burst-amplitude which will trip the counter can be no lower than the background l/f noise peaks of burst-free units, otherwise normal background noise will fail the DUT.

The background noise peaks depend on the source termination Rs, the wide band l/f noise figure of the DUT, and the test system bandwidth. A good estimate of the normal background noise-peak levels can be computed from the definition of noise factor and an empirically determined noise-crest factor of approximately 6:1. The crest-factor is the ratio of the maximum peak-noise voltage to the RMS noise voltage. The noise factor is defined as the ratio of the total noise power at the amplifier output to the output-noise power due to the source resistors alone. In terms of the RMS noise voltages at the input terminals of the amplifier this is equivalent to:

$$\text{Noise Factor (F)} = \frac{E^2\text{input noise total}}{E^2\text{noise source resist}} = \frac{(E_{NTi})^2}{(E_{NRS})^2} \quad (1)$$

E_{NTi} is the total input noise-voltage, i.e., the sum of noise generated in the source termination resistance and noise generated by the DUT.

E_{NRs} is that part of E_{NTi} due to Rs alone.

$$\text{Therefore,} \quad E_{NTi} = (\sqrt{F})\,(E_{NRs}). \quad (2)$$

E_{NRs} can be computed by using the well known expression for "white-noise" generated across the terminals of a resistor (R):

$$E_{NR(RMS)} = \sqrt{4kTBR} \quad (3)$$

where k = Boltzmans Constant = 1.372×10^{-23} j/°K

T = Absolute Temperature in °K
B = Noise Bandwidth in Hz
R = Value of the resistor in ohms.

Thus, at a room temperature of 290°K

$$E_{NR(RMS)} = 1.28 \times 10^{-10}\sqrt{BR}$$

For example, a 100 kΩ resistor preceding a system with a bandwidth of 1 kHz will generate a noise-voltage of

$$(1.28 \times 10^{-10}) \quad (\sqrt{10^3 \cdot 10^5}) \quad = 1.28\,\mu V_{RMS}$$

Both inputs of an Op-Amp are usually terminated in Rs, hence it is necessary to combine the effects of both resistors to determine the effective E_{NRs} at the input of the DUT. Because the noise voltages from these two resistors are uncorrelated their voltages must be added vectorally rather than algebraically.

$$E_{NRs}\,(\text{effective}) = \sqrt{(E_{NR_{s1}})^2 + (E_{NR_{s2}})^2} \quad (4)$$

because $E_{NR_{s1}} = E_{NR_{s2}}$, when $R_{s1} = R_{s2}$

$$E_{NRs}\,(\text{effective} = (\sqrt{2})\;(E_{NRs})$$

and for 1 kHz bandwidth at 290°K

$$E_{NRs}\,(\text{effective}) = (\sqrt{2})\,(1.28\,\mu V) = 1.81\,\mu V_{RMS}.$$

If in this example, the DUT has a wideband l/f noise figure of 4 dB (2.5:1 power ratio) the total RMS background noise-voltage at the input will be

$$E_{NTi} = (\sqrt{F})\;(E_{NRs}) \text{ (from eq.(2))}$$
$$= (\sqrt{2.5})\;(1.81) = 2.9\,\mu V_{RMS}$$

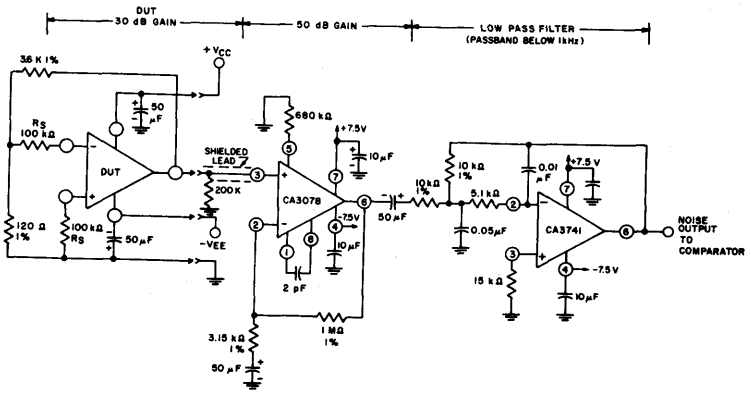


Fig. 3– Schematic diagram of high-gain amplifier/filter.

If a crest factor of 6:1 is assumed, the peaks of the background noise will be approximately (6) (2.9) = 17 µV peak. This voltage is the lower limit of the burst-amplitude rejection level. A reasonable threshold for burst detection and rejection might be 50-100% greater than this minimum value.

An alternate method used to set the burst-threshold limit involves a direct measurement (at the output of the high gain amplifier-filter combination) using a storage oscilloscope or a "true RMS" voltmeter. By this method the noise peak or RMS noise voltage of burst-free units is determined. This measurement provides a good practical check on the accuracy of the computation outlined above. Selection of the acceptable number of burst counts in the test period is arbitrary, but dependent on the type of application intended for the DUT. To be acceptable in some critical applications, the DUT may not generate even a single burst-pulse in a relatively long period of time.

Burst-Noise Test System Circuits

1. High gain Amplifier – Filter

Fig. 3 shows the schematic diagram of the high-gain amplifier-filter which provides a fixed gain of 80 dB with a 12 dB octave roll-off above 1 kHz. The gain-function is somewhat arbitrarily distributed between the DUT and post-amplifier: 30 dB and 50 dB respectively. This distribution is based on the need for sufficient gain in the DUT portion to eliminate significant noise-signal contributions from the second stage while simultaneously allowing adequate loop-gain in each stage to provide accurate gain-setting with precise external resistors. The first stage is shown as a plug-in module so that any type of DUT configuration having 30 dB gain can be tested.

The capacitive coupling employed provides a low frequency cutoff of about 1 Hz and eliminates the need for dc-offset zero-adjustments. The dc offset-voltage at the filter output is less than 5 mV which corresponds to less than 0.5 µV error when referred to the noise input (an 80 dB gain is assumed.) Several seconds must be allowed, however, for the DC operating point to stabilize after the power is applied to the DUT.

2. Bi-Polarity Comparator

Fig. 4 shows the schematic diagram for the threshold-detecting comparator. Because bursts of either polarity must be detected and converted to positive output pulses, two comparators are required: one having a positive-threshold reference and the other having a negative-threshold reference of equal magnitude. The RCA CA3060 triple OTA is convenient to use because a single package provides circuits for both comparators plus a reference inverter for the negative threshold reference. The positive feedback provided by the R_f and R_i connections produces a hysteresis effect with reference to the input switching threshold, (i.e., the comparator does not return to its quiescent state until the input noise signal drops well below the initial threshold trip-level). This feature is necessary to prevent multiple triggering by the background noise signals superimposed on top of the burst-noise pulse. By this means, multiple counting of a single burst-noise pulse is avoided.

The magnitude of the threshold reference voltage E_R determines the burst-level which trips the comparator. If a voltage gain of 80 dB is provided by the amplifiers, a 200 mV reference voltage will enable the circuit to be triggered when a burst-noise pulse (whose amplitude is equivalent to the level of 20 µV referred to the DUT input) is present.

3. Counter-Latch-Timer Control Circuits

The remaining circuits of the go-no-go burst-noise tester are shown in Fig. 5. The decade-counter is incorporated in a single COS/MOS IC (RCA CD4017AE) which has clock, reset, and enable inputs, and an output terminal for each of ten count-positions (0 to 9). A carry-out signal is available if the use of more than a single decade is desired. The clock input-signal must be positive-going and have a magnitude of at least 70% of the supply-voltage and rise-time equal to or less than 15 µs. The comparator shown in Fig. 4 provides an output signal which meets these requirements.

Selection of the reject count is made by a pin-jack connection of the latch-circuit input-lead to the appropriate output terminal of the counter. Whenever the selected count-position voltage goes "high" the latch-circuit is switched to the latched-state, and the fail-indicator lamp "on". The latch and lamp will remain "on" until the reset button of the electronic timer is switched to the "Timer On" position. This action provides a momentary reset signal (≈ 20 ms) to both the latch and counter circuits and places a continuous enable voltage on the counter for the duration of the test period.

Spurious Noise Sources and Their Suppression

The very low voltage levels and the high source impedances normally used for burst testing render the system highly susceptable to external spurious noise sources. This problem is particularly serious if a test unit is going to be rejected for as little as one or two input burst-noise pulses exceeding 20-30 µV. The major sources of spurious noise encountered in the development of this test system were:

1. 60-Hz hum pickup,
2. power supply transients,
3. electromagnetic pick up of switching transients.

60-Hz hum is introduced by capacitive or inductive coupling or as power-supply ripple. Power-supply ripple is not normally a problem when testing operational amplifiers with regulated supplies, because the Op Amps generally have good power-supply rejection. This source of noise must be considered, however, when testing devices that do not have good inherent power-supply rejection. Capacitive or inductive coupling of hum can occur when 60-Hz line cord leads are within a few inches of the input terminals of the DUT. Precautions, such as proper lead dress and twisting of the 60-Hz leads, eliminate this problem.

Power-supply transients, as distinguished from power-supply ripple, can be of sufficient amplitude to introduce detectable noise pulses at the operational amplifier input. Such transients are produced when other equipment on the same ac line is switched on or off. A typical power-supply rejection ratio for an operational amplifier is 50 µV/V (i.e. a 1 volt transient on the power-supply is equivalent to a 50 µV noise pulse at the DUT input). This example demonstrates that the test system cannot tolerate power-supply transients greater than approximately 100 mV even when testing units with good power-supply rejection. Unless the power-supply is known to be free of such transients, a battery-operated system is recommended. Even when this system is battery-

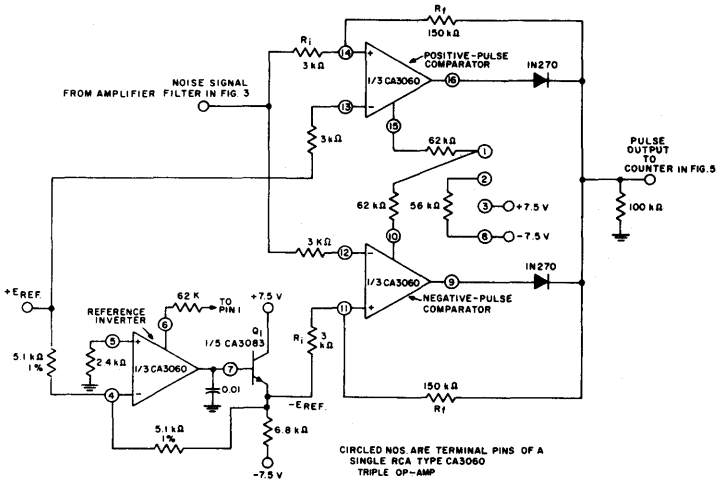


Fig. 4– Schematic diagram of threshold-detecting comparator.

operated, "On-Off" switching of nearby equipment introduces detectable transients into the system. These problems are eliminated by placing the test circuitry in a completely shielded enclosure with a hinged top for easy access to the test unit. The external noise problem is best solved by use of a shielded enclosure and by use of a battery-operated power-supply contained within the enclosure. Fig. 6 shows a photo of the circuit board layouts of the test unit.

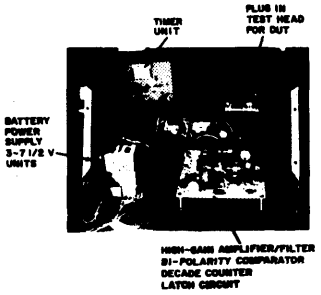


Fig. 6— Photo of circuit-board layout.

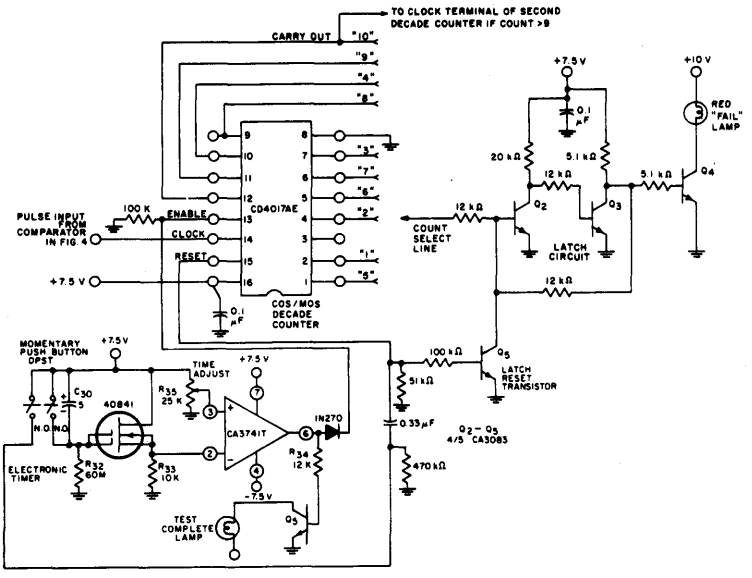


Fig. 5— Counter-latch-timer-control circuit schematic.

Abstracts of Other Application Notes

AN-3193 . . . . . 9 pages
Application Considerations for the RCA-3N128 VHF MOS Field-Effect Transistor

This Note describes applications and vhf circuit considerations for a high-frequency n-channel MOS field-effect transistor, the RCA 3N128. Biasing requirements and basic circuit configurations are discussed, and selection of the optimum operating point and methods of automatic gain control are explained. The cross-modulation and intermodulation distortion characteristics of the 3N128 MOS transistor are compared to those of bipolar transistors, and procedures are given for the design of a practical vhf amplifier that uses the 3N128.

AN-3341 . . . . . 3 pages
VHF Mixer Design Using the RCA-3N128 MOS Transistor

The 3N128 is a vhf MOS field-effect transistor suitable for use throughout the vhf band (30 to 300 MHz) as an amplifier, mixer, or oscillator. This Note discusses some of the design criteria pertinent to the construction of MOS mixers, and presents an example of a complete vhf MOS converter.

AN-3452 . . . . . 7 pages
Chopper Circuits Using RCA MOS Field-Effect Transistors

Although electromechanical relays have long been used to convert low-level dc signals into ac signals or for multiplex purposes, relays are seriously limited with respect to life, speed, and size. Conventional (bipolar) transistors overcome the inherent limitations of relays, but introduce new problems of offset voltage and leakage currents. This Note describes the use of MOS field-effect transistors in solid-state chopper and multiplex designs that have the long life, fast speed, and small size of bipolar-transistor choppers, but that eliminate their inherent offset-voltage and leakage-current problems.

AN-3453 . . . . . 6 pages
An FM Tuner Using an RCA-40468 MOS-Transistor RF Amplifier

This Note describes an FM tuner that incorporates an MOS field-effect transistor as the rf amplifier, and shows how the MOS transistor is instrumental in minimizing the spurious responses normally found in FM receivers.

AN-3535 . . . . . 6 pages
An FM Tuner Using Single-Gate MOS Field-Effect Transistors as RF Amplifier and Mixer

Selection of the transistors for use in FM-tuner stages involves consideration of such device characteristics as spurious response, dynamic range, noise immunity, gain, and feed-through capacitance. MOS field-effect transistors are especially suitable for use in FM rf-amplifier and mixer stages because of their inherent superiority for spurious-response rejection and signal-handling capability. This Note describes an FM tuner that uses an RCA-40468 MOS transistor as the rf amplifier and an RCA-40559 MOS transistor as the mixer.

AN-4018 . . . . . 5 pages
Design of Gate-Protected MOS Field-Effect Transistors

MOS (metal-oxide-semiconductor) field-effect transistors are in demand for rf-amplifier applications because their transfer characteristics make possible significantly better performance than that experienced with other solid-state devices. Unless equipped with gate protection, however, MOS transistors require careful handling to prevent static discharges from rupturing the dielectric material that separates the gate from the channel. This Note describes the design of dual-gate MOS field-effect transistors that use a built-in signal-limiting diode structure to provide an effective short circuit to static discharge and limit high potential buildup across the gate insulation.

AN-4125 . . . . . 7 pages
MOS/FET Biasing Techniques

Field-effect transistors are applied in rf amplifiers and mixers, if and audio amplifiers, electrometer and memory circuits, attenuators, and switching circuits. The dual-gate MOS/FET appears to be particularly useful in rf stages because of low feedback capacitance, high transconductance, and superior cross modulation with automatic-gain-control capability. The rules for biasing FET's vary slightly depending on type. However, most possibilities are covered in this Note through examination of the biasing of a single-gate, a junction-gate, and a dual-gate transistor. Substrate biasing and biasing to compensate for temperature variations are also discussed.

AN-4590 . . . . . 16 pages
Using MOS/FET Integrated Circuits in Linear Circuit Applications

A brief review of MOS/FET IC device theory is given, and some linear circuit applications are surveyed. Theory discussed includes gate protection and electrical requirements. Applications include choppers, attenuators, constant-current sources, general-purpose amplifier circuits, and rf amplifiers, oscillators, and mixers.

ICAN-4072 . . . . . 8 pages
Applications of the RCA-CA3048 Integrated-Circuit Amplifier Array

The RCA-CA3048 integrated circuit, an array of four identical amplifiers, each with independent inputs and outputs, all on a single monollthic silicon chip, has an operating and storage temperature range of −25°C to +85°C. Each amplifier in the low-noise array has a typical open-loop gain of 58 dB and input impedance of 90,000 ohms. The gain-frequency response, stability, output swing versus supply voltage, and noise of the device are discussed. Circuit applications include Hartley and Colpitts Oscillators, astable multivibrators, a 4-channel linear mixer, a driver for a 600-ohm balanced line, and a gain-controlled amplifier.

ICAN-5015 . . . . . 15 pages
Application of the RCA-CA3008 and CA3010 Integrated-Circuit Operational Amplifiers

This Note describes the circuit arrangement, lists the performance characteristics, explains the major design considerations, and discusses typical applications of the CA3008 and CA3010 operational amplifiers. These amplifiers are silicon monolithic integrated circuits designed to operate from two symmetrical low- or medium-level dc power supplies (at supply voltages in the range from ±3 volts to ±6 volts).

ICAN-5022 . . . . . 26 pages
Application of the RCA-CA3004, CA3005, and CA3006 Integrated-Circuit RF Amplifiers

The CA3004, CA3005, and CA3006 rf amplifiers are discussed. These silicon-epitaxial monolithic integrated circuits are designed to operate from low or medium levels of dc supply voltage, over a range of ambient temperatures from −55°C to +125°C, and at frequencies from dc to 100 MHz. They may be used with external tuned-circuit, transformer, or resistive load impedances to provide wide- or narrow-band amplification, mixing, limiting, product detection, frequency generation, and generation of pulse or digital waveforms.

ICAN-5030 . . . . . 11 pages
Application of the RCA-CA3000 Integrated-Circuit DC Amplifier

This Note describes the RCA-CA3000 dc amplifier, a stabilized and compensated differential amplifier that has push-pull outputs, high-impedance (0.1-megohm) inputs, and gain of approximately 30 dB at frequencies up to one MHz. Its useful frequency response can be increased to several tens of megahertz by the use of external resistors or coils. The CA3000 can be used as a signal switch (with pedestal), a squelchable audio amplifier (with suppressed switching transient), a modulator, a mixer or a product detector. When suitable external components are added, it can also be used as an oscillator, a one-shot multivibrator, or a trigger with controllable hysteresis.

ICAN-5036 . . . . . 9 pages
Application of the RCA-CA3002 Integrated-Circuit IF Amplifier

The RCA-CA3002 integrated-circuit if amplifier described in this Note is a balanced differential amplifier that can be used with either a single-ended or a push-pull input and can provide either a direct-coupled or a capacitance-coupled single-ended output. Its applications include RC-coupled if amplifiers that use the internal silicon output-coupling capacitor, video amplifiers that use an external coupling capacitor, envelope detectors, product detectors, and various trigger circuits.

ICAN-5037 . . . . . 4 pages
Application of the RCA-CA3007 Integrated-Circuit Audio Amplifier

This Application Note describes the RCA-CA3007 audio driver, a balanced differential configuration with either a single-ended or a differential input and two push-pull emitter-follower outputs. The circuit features all-monolithic silicon epitaxial construction, and is intended for use as a direct-coupled driver in a class B audio amplifier which exhibits both gain and operating-point stability over the temperature range from −55 to 125°C.

ICAN-5038 . . . . . 8 pages
Application of the RCA-CA3001 Integrated-Circuit Video Amplifier

The CA3001 silicon monolithic integrated circuit is designed for use in intermediate-frequency or video amplifiers at frequencies up to 20 MHz and in Schmitt-trigger applications. This integrated circuit can be gated, and gain control can be applied. The CA3001 incorporates all-monolithic silicon epitaxial construction designed for operation at ambient temperatures from −55 to 125°C, balanced differential-amplifier configuration with low-impedance double-ended input, and a built-in temperature-compensating network for gain or dc operating-point stability over the temperature range from −55 to 125°C.

ICAN-5213 . . . . . 6 pages
Application of the RCA CA3015 and CA3016 Integrated-Circuit Operational Amplifiers

The integrated-circuit operational amplifiers CA3015 and CA3016 are identical in circuit configuration to the CA3008 and CA3010, but have an improved device breakdown voltage that permits operation from ±12-volt supplies as well as from ±6 volt or ±3 volt supplies. This Note describes the operating characteristics of the CA3015 and CA3016 at ±12 volts, and discusses applications that take advantage of the higher gain-bandwidth product and increased output signal swing obtained at the higher

voltages: a 50-dB amplifier; a 10-dB, 42-MHz amplifier; a twin-T bandpass amplifier; a 20-dB, 10-MHz bandpass amplifier; and a voltage-follower.

ICAN-5269 . . . . . 7 pages
Integrated Circuits for FM Broadcast Receivers

This Note describes several approaches to FM receiver design using silicon monolithic integrated circuits. The tuner section is described first, and then the if-amplifier and detector sections. Performance characteristics are described where applicable. The FM receivers discussed are designed for use from a +9-volt supply. The key to design simplicity is the use of the RCA multifunction integrated circuits CA3005, CA3012, and CA3014. The CA3005 may be used as a cascode rf amplifier, a differential rf amplifier, a mixer-oscillator, and an if amplifier; the CA3012 and CA3014 perform if amplification, limiting, detection, and preamplification.

ICAN-5296 . . . . . 5 pages
Application of the RCA-CA3018 Integrated-Circuit Transistor Array

The CA3018 integrated circuit consists of four silicon epitaxial transistors produced by a monolithic process on a single chip mounted in a 12-lead TO-5 package. The four active devices, two isolated transistors plus two transistors with an emitter-base common connection, are especially suitable for applications in which closely matched device characteristics are required, or in which a number of active devices must be interconnected with non-integrable components such as tuned circuits, large-value resistors, variable resistors, and microfarad bypass capacitors. Such areas of application include if, rf (through 100 MHz), video, agc, audio, and dc amplifiers.

ICAN-5299 . . . . . 6 pages
Application of the RCA-CA3019 Integrated-Circuit Diode Array

The CA3019 integrated circuit diode array provides four diodes internally connected in a diode-quad arrangement plus two individual diodes. Its applications include gating, mixing, modulating, and detecting circuits. Because all the diodes are fabricated simultaneously on a single silicon chip, they have nearly identical characteristics, and their parameters track each other with temperature variations. Consequently, the CA3019 is particularly useful in circuit configurations that require either a balanced diode bridge or identical diodes.

ICAN-5337 . . . . . 10 pages
Application of the RCA-CA3028A and CA-3028B Integrated-Circuit RD Amplifiers in the HF and VHF Ranges

The CA3028A and CA3028B monolithic-silicon integrated circuits are single-stage differential amplifiers intended for service in communications systems operating at frequencies up to 100 MHz with single power supplies. This Note provides technical data and recommended circuits for use of the CA3028A and CA3028B in rf amplifiers, autodyne converters, if amplifiers, and limiters. The CA3028A and CA3028B are suitable for use in a wide range of applications in dc, audio, and pulse amplifier service; they have been used as sense amplifiers, preamplifiers for low-level transducers, and dc differential amplifiers.

ICAN-5338 . . . . . 14 pages
Application of the RCA-CA3021, CA3022, and CA3023 Integrated-Circuit, Wideband Amplifiers

The CA3021, CA3022, and CA3023 integrated circuits are multipurpose high-gain amplifiers designed for use in video and AM or FM if stages in single-power-supply systems. Specifically, they can be used in video amplifiers operating at frequencies through 30 MHz, AM and FM if amplifiers, and buffer amplifiers in which an isolation capability greater than 60 dB at 1 MHz is desired.

ICAN-5380 . . . . . 7 pages
Integrated - Circuit Frequency - Modulation if Amplifiers

The discussion in this Note shows that the simplest approach to the use of the CA3012 and CA3028 integrated circuits in FM if-amplifier strips is to replace each stage in present discrete-transistor if strips with a differential amplifier. This integrated-circuit approach requires a minimum of re-engineering because a cascade of individually tuned if stages is used. From a performance point of view, this approach results in better AM rejection than that obtained with discrete circuits because of the inherent limiting achieved with the differential-amplifier configuration.

ICAN-5766 . . . . . 8 pages
Application of the RCA-CA3020 and CA3020A Integrated-Circuit Multipurpose Wideband Power Amplifiers

The CA3020 and CA3020A integrated circuits are multipurpose, multifunction power amplifiers designed for use as power-output amplifiers and driver stages in portable and fixed communications equipment and in ac servo-control systems. The flexibility of these circuits and the high-frequency capabilities of the circuit components make these types suitable for a wide variety of applications such as broadband amplifiers, video amplifiers, and video line drivers. Voltage gains of 60 dB or more are available with a 3-dB bandwidth of 8 MHz. Applications covered include audio, wideband, and driver amplifiers.

ICAN-5831 . . . . . 5 pages
Application of the RCA-CA3044 and CA3044VI Integrated Circuits in Automatic-Fine-Tuning Systems

This Note describes the use of the CA3044 and CA3044VI integrated circuits as automatic fine-tuning (AFT) system components and discusses the advantages of integrated circuits in this application. The CA3044VI is electrically identical to the CA3044, but is supplied with formed leads for easier printed-circuit-board mounting. The construction and performance of a typical automatic-fine-tuning system for a color television system are examined.

ICAN-5841 . . . . . 4 pages
Feedback-Type Volume-Control Circuits for RCA-CA3041 and CA3042 Integrated Circuits

This Note describes feedback-type volume controls for use with RCA-CA3041 and CA3042 integrated circuits in television receivers. In television sets using these integrated circuits, the volume control is often located remote from the amplifier. The long leads required in such a configuration sometimes pick up undesirable signals that, in turn, cause the system to exhibit hum and noise at low volume levels. The proposed feedback-type volume control reduces hum and noise pick-up by reducing the gain of the system rather than the signal level, and thus eliminates the cost of shielding the leads.

ICAN-6259 . . . . . 10 pages
Integrated-Circuit Stereo Decoder Using the CA3090AQ Stereo Multiplex Demodulator

The CA3090AQ integrated-circuit provides features heretofore unavailable to the receiver designer. This device needs only a single tuning adjustment, which reduces to a minimum the manual effort during assembly; the phase-locked loop maintains performance under conditions of temperature variations, humidity, and aging. The compactness of the CA3090AQ and of the required external components, added to the other attributes, makes this stereo decoder a significant advancement in the state of the art of stereo decoder designs.

ICAN-6302 . . . . . 9 pages
Description and Application of the RCA-CA3120E Integrated-Circuit TV-Signal Processor

The CA3120E is a 16-pin, dual-in-line-monolithic-silicon integrated circuit that processes a video signal and provides the following outputs: non-inverted video output; noise-processed, inverted video output; dual-polarity, composite synchronization signals; and automatic gain-control signals (agc). The IC, which can be used in color or monochrome TV receivers, requires a single-polarity power supply (positive) and includes impulse noise inversion and delay circuits that reduce the deleterious effects of impulse noise in the receiver agc and synchronization (sync) circuits. Standard agc strobing techniques are also used. The agc and impulse-noise thresholds are automatically set and require no controls. The if maximum-gain bias and the tuner agc delay may be adjusted for optimun. TV-receiver performance; the time constant for the sync-separator input can also be optimized by the set designer.

ICAN-6724 . . . . . 8 pages
A Flexible Integrated-Circuit Color Demodulator for Color Television

This Note describes the circuit operation and application of the CA3067 in a color television receiver. The CA3067, which is supplied in a quad-in-line 16-lead plastic package, provides the following color-demodulator circuit functions: amplification, balanced chroma demodulation, dc-operated tint (phase) control, and zener-diode voltage regulation.

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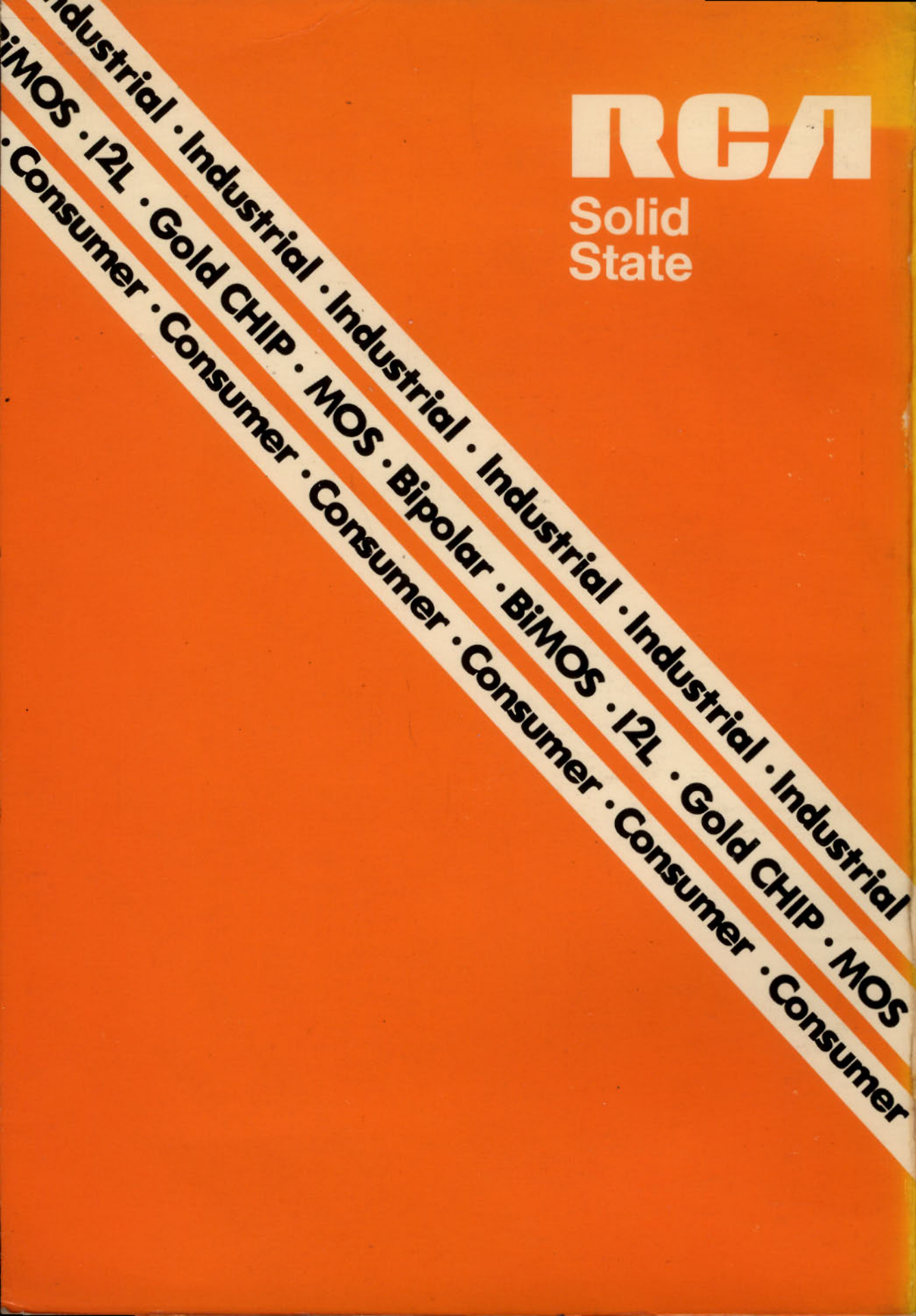
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